

THE
SEAMAN'S
Medical Instructor,

IN A COURSE OF
LECTURES
ON
ACCIDENTS AND DISEASES
INCIDENT TO SEAMEN,
IN THE VARIOUS CLIMATES
OF THE WORLD.
CALCULATED FOR
SHIPS THAT CARRY NO SURGEON.

By N. D. FALCK, M. D.

THE
SEAMAN'S
Medical Instructor,

IN A COURSE OF
LECTURES
ON
ACCIDENTS AND DISEASES
INCIDENT TO SEAMEN,
IN THE VARIOUS CLIMATES
OF THE WORLD.
CALCULATED FOR
SHIPS THAT CARRY NO SURGEON.

By N. D. FALCK, M. D.

The Project Gutenberg eBook of The Seaman's Medical Instructor

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Seaman's Medical Instructor

Author: N. D. Falck

Release date: May 3, 2019 [eBook #59425]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59425

Credits: Produced by The Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

***** START OF THE PROJECT GUTENBERG EBOOK THE SEAMAN'S
MEDICAL INSTRUCTOR *****

The cover image was created by the transcriber and is placed in the public domain.

THE
SEAMAN'S
Medical Instructor,
IN A COURSE OF
LECTURES
ON
ACCIDENTS AND DISEASES
INCIDENT TO SEAMEN,
IN THE VARIOUS CLIMATES
OF THE WORLD.
CALCULATED FOR
SHIPS THAT CARRY NO SURGEON.

The Whole delivered in a plain Language, and founded
on a long and successful Experience.

By N. D. FALCK, M. D.

LONDON:
PRINTED FOR EDWARD AND CHARLES DILLY.

M,DCC,LXXIV.

THE

PREFACE.

If we consider the many benefits that navigation contributes to commerce in general, and how much the prosperity of nations are indebted to its source, we must without hesitation declare seamen, a most respectable part of mankind; and if we reflect a moment, on the many hazards and perils they are exposed to on that precarious element, and the advantages we reap from their toil, the common feelings of human society, must awake and remind us, not only how much we are obliged as social members, even from a motive of self interest, to study their preservation, but as fellow creatures, how forcibly our duty calls us to give them every aid in our power, and to lighten the burthen of their many toils.

A ship at sea may be considered as a floating kingdom; and the subjects, however few in number, are not only liable to the same accidents as those on shore, but to many more, peculiar to that precarious and fluctuating element that surrounds them. For this reason there are many things requisite to be known by the mariners besides the art of conducting a ship from port to port.

Of all the various knowledges that distinguish the human species from the brute creation, what is of more value than that which tends to the preservation of life and health? I believe that every man who is actuated by motives of philanthropy will with me wish, that mankind in general would

advert more to it in the principles of education, than what they at present do. Indeed it is astonishing that so valuable an acquisition, as to have some knowledge of the body we possess, and its preservation, should be so totally neglected, as to be entirely excluded from education; an acquisition that not only enlivens the mind, but in itself is so very essential to our existence; yet what is more to be wondered at, is, that a seaman who launches into the main ocean, is cut off from every assistance, in case of an accident or sickness, and rendered wholly unable to give the least aid to himself, or those who are entrusted to his care. Large ships I own are exempt from this observation: but are not the lives of men on board of small ships equally as valuable, and worth preserving, as those of greater burthen?

I am not the first who has had the welfare of seamen at heart; various writers, and men of abilities too, have presented the world with observations, that undoubtedly have proved beneficial to that class of men; but then these books were neither intended, nor fit to be put into the hands of such seamen, as actually stand most in need of those very observations they have made, and only intended for the perusal of men of physical talents; nevertheless they merit for their labour public thanks, as useful members of society. But if we take an impartial view of the number of seamen, I believe the greatest part are employed in small ships, that either carry no surgeon, or (I am sorry to say it) sometimes one they had better be without; especially in war time, when they stand most in need of a good one.

To benefit these men, and at the same time to make that benefit universal, is the plan I have aimed at; and I believe this is the first attempt of the kind; at least I know of none whose steps I have followed in this design.

Should I ask physical writers why they have neglected so valuable a part of society as seamen, in giving them their friendly instructions respecting their health, I doubt not, but the majority would reply, that the education of masters of ships and seamen in general, has not enabled them to understand a subject so much above their comprehension, or sphere. But if seamen have not the advantage of an extensive education, are they divested therefore of common sense and understanding? and is it not possible to deliver them instructions, dressed in so plain a language as to adapt it to their capacity?

Nevertheless, though men of abilities have not thought seamen worth their pen, others have; but such whose pursuit in life has been stimulated by motives of a different cast. Those have furnished seamen not only with directions to cure *all* diseases, but even given them medicines to do it with, into the bargain; or to speak more seriously, given them directions how to use such medicines as would yield the most profit to the apothecary, who with pretended wisdom equipt them out: and I am apt to think, that kind of quackery has stolen away the lives of many valuable seamen.

It is universally asserted that seamen are not fond of reading; this in a great measure is true, but far from being a general rule; and though blind prejudice has given them an aversion to every thing *physical*, yet I will venture to say so much in their behalf, that hitherto nothing has been offered to them that could alter their opinion in this respect, or open their eyes, in shewing them the beauties of that useful study, and in how small a compass its true principles might be contained.

Physic like religion is rendered intricate and abstruse, the more it abounds in errors and absurdities; but truth like the brilliant sun, drives away all the phantoms of perplext mysteries, and proves that both alike are simple and beautiful, founded on reason, and that its most useful part is comprehensive to common understanding. This pleads in favour of my attempt, and with every man of sense, must overpower prejudice.

That I might the better succeed in my intention, I have begun the whole at the first principles, namely the mechanism of the human body; for I cannot see how any man can pretend to know the nature of diseases, or presume to administer any thing with certainty, without he has some knowledge of the parts that suffer: this then is the subject of my first lecture; and to make it the more pleasing, and to entice the reader to pursue, I have stript it of every thing pedantic, or the unnecessary perplexing particulars, and confined myself to such things only, as gives a concise and true idea of the whole animal system; and if I may be allowed to judge from the opinion of some of my learned friends, I flatter myself, that my readers, physical or not, will give me their sanction, as well with respect to plainness of language, as conciseness of the subject, and the utility of such a short dissertation, founded on experience, and stript of prejudice and errors.

The second lecture contains equally as copious a subject as the first; contracted in a very small compass, and which I hope will give as much satisfaction as the forementioned.

Out of the whole *Materia Medica*, I have chosen but very few medicines, and these I have confined myself to throughout the whole book. Many surgeons at sea will not, I know, be contented with so small an assortment; and I remember the time that such an observation might have been applicable to myself; but as I have by long experience learned to shake off prejudices, and been taught by the help of philosophy to pry into the nature of things in general, I solemnly declare, that now, I would venture to go to any part of the globe, not only with that little assortment of medicines I have mentioned, but even with many less. The *emetick tartar*, *sublimate mercury* and the *bark* are powerful medicines; but they should be in very skillful hands, otherwise they are rank poisons, and therefore I have omitted them here; but strictly speaking, every good medicine is the same. I have two maxims in physic, which I strictly attend to; one is, that to my friends I give the least medicines; the other is, that the principal virtue of a medicine depends on its application.

The form of the medicine box, I have constructed so as it appears to me most convenient and handy; and every kind of medicine, I would advise to have labeled with their proper names, as well as with another mark that may take the eye at first sight; and the more to prevent mistake, one label should be fixt on the pot that contains the medicine, and the other on the box: for in medicines too much caution cannot be used.

One thing with respect to the medicines, I have omitted, and that is their quantity requisite for the box, and their prices: I intended to have taken notice of this, but found that that could not with exactness be ascertained, unless entering into another subject, and that would take off from the conciseness of my plan. As I have however many other things respecting the benefit of seamen to offer, peculiarly respecting distant climates, I shall take notice of this, where it may come in, more pertinently to the subject: till then, that must be left to the management of the apothecary who is applied to for fitting out the box. If he is a man of candour and judgment he cannot be much out of his calculation, when he knows the number of hands, and the voyage intended.

With respect to the other lectures, I have, as near as it was practicable, divided them into their different classes and sections; at the same time the whole is so linked together, that one lecture must support the other. And as I have throughout the whole course of the book differed more or less from the general mode of practice, I must hereby declare, that I have in this consulted my own experience more than any other authority, without fear of censure, or hope of applause, any farther than of truth, and my own conscience.

Though I have principally written with an intent to be understood by masters of ships who carry no surgeon; yet I am well persuaded that every practitioner, whether on sea or shore, will find many things to his improvement.

With respect to my receipts, they are simple and powerful; yet I do not enjoin so strict a conformity to them, by the practitioner, who has an extensive knowledge of the *materia medica*, and is prepossessed in favour of some particular medicines; but as he will discover my indications, it may serve greatly to compare that of mine with his own, and then he may judge for himself. I have touched very slightly on the requisite diet for patients on board a ship; though this is an essential article regarding health in general: But I found that subject also of so very extensive a kind, that it would swell the book; but as I have many valuable things to offer of the same nature, I must beg, that in the mean time, the reader will be satisfied with the universal observations in general; namely, that temperance is the principal object and that patients who are deprived of exercise ought to have the most easy digesting food. Copper and lead are pernicious; but iron and earthen ware are safest either on sea or shore.

In order to render the book as useful as my zeal has intended, I recommend it a fair perusal, from the first page to the last; a book of this nature is apt to be looked into, only when advice is wanted: that method is wrong; and in such a case, the best book in the world is apt to mislead. It is not enough that we know what we are to do with such and such disorders, but we should previously know what the disorder is; and how shall we know that, without having some idea of the evil before it comes on?

A book of this size is soon perused; taking a lecture or a section at a leisure time on board, will soon send the reader through the whole; and I

flatter myself, that one perusal will lead to the second, when then it may be referred to for advice with safety; and I hope a study so essential to self-preservation, will be viewed in a more agreeable light, than it has hitherto been.

To compleat the whole, a copper-plate is annexed, wherein the most material utensils are exhibited, which was scarce capable to be explained by bare description, to men unacquainted with surgery.

To conclude, I must beg leave to observe, that the respect and esteem, with which I have been constantly honoured by every worthy seaman, whose life and health were committed to my care, have I confess, stimulated my endeavours, to offer them this book as a grateful return. And I am happy to reflect, that my experience in life enables me to make it worth their acceptance. May they reap that benefit from it, which my heart has intended them; and this is all the reward I wish to obtain.

N. D. FALCK.

ERRATA.^{[\[1\]](#)}

Page [5](#). line 22. for cataplexy, read apoplexy. Page [6](#). line 16. read, that a pressure on that part, not only occasions an apoplexy, but even stops life itself. Page [26](#). line 17. for theirs, read thus. Page [35](#). for Sect. vii. read viii. and following ix. x. Page [61](#). line 4. for five or six, read about four or five. Line 21. for iniform, read inform. Page [91](#). line 18. read, in order to get into. Page [131](#). line 29. for purify, read putrefy. Page [135](#). line 12. delete them. Page [142](#). line 2. from the bottom, read, and is a sufficient. Page [158](#). line 6. for malignitus, read malignities. Page [239](#). line 7. read, the patient when not able to crawl to the tub, so revived &c.

[1](#). These errors have been corrected for this ebook (May, 2019).

To the BINDER.

The Copper-Plate and its Explanation are to be placed at the End of the Book.

THE CONTENTS.

LECTURE I.

An anatomical and physiological description of the human body

SECT. I.	<i>Of the composition and close connection of parts in the human body</i>	<u>2</u>
SECT. II.	<i>Of the brains and nerves</i>	<u>5</u>
SECT. III.	<i>Of the heart and lungs</i>	<u>9</u>
SECT. IV.	<i>Of the circulation of the blood</i>	<u>12</u>
SECT. V.	<i>Of the joint connection of the brains, heart, lungs, &c.</i>	<u>15</u>
SECT. VI.	<i>Of chylifaction, and nutrition</i>	<u>19</u>
SECT. VII.	<i>Of the bones, muscles, ligaments, tendons and their connection.</i>	<u>28</u>
	<i>A skeleton of the human body</i>	<u>31</u>
SECT. VIII.	<i>Of the urinary organs and genital parts</i>	<u>35</u>
SECT. IX.	<i>Of the skin</i>	<u>40</u>
SECT. X.	<i>Of the external senses</i>	<u>43</u>

LECTURE II.

On the requisite means and medicines towards restoring health.

SECT. I. *Of the pulse* [50](#)

SECT. II. *Of the effects of* [55](#)
 bleeding

SECT. III. *Of the operation of* [58](#)
 bleeding

SECT. IV. *Of the requisite* [63](#)
 medicines useful at
 sea

A plan of a medicine [84](#)
box—[64](#). *Of*
utensils, and
medicines—[65](#).
Lancets, ibid.
scissars, probe,
spatula—[66](#).
Bistouries, and
incision lancets,
teeth instruments,
syringes—[67](#).
Clyster pipes and
bladders, clyster
syringe, scales and
weights—[68](#).
Splints—[69](#).
Bandages, rags,
tow, lint, tape,
needles, thread and
pins—[70](#).
Camomile flowers,

balm, sage,
sassafrass—[71](#).
Oatmeal, cinamon,
allum, chalk—[72](#).
Salts, diachylon
plaister—[73](#).
Mercurial plaister,
blistering plaister
—[74](#). Spirits of
wine and camphire
—[75](#). Liniment of
soap, sweet oil,
sweet spirit of nitre
—[76](#). Honey,
yellow basilicon;
elixir of vitriol—
[77](#). Elixir
proprietary,
lenitive electuary—
[78](#). Turner's
Cerate, fever
powders, stomachic
powder—[79](#).
Calomel—[80](#).
Precipitate—[81](#).
Mercurial ointment
—[82](#). Rhubarb,
jalap, ipecacuanha,
liquid laudanum—
[83](#). Essential oil of
peppermint,
Turlington's balsam

LECTURE III.

On accidents, and their proper method of treatment

SECT. I.	<i>Of falls</i>	<u>89</u>
SECT. II.	<i>Of wounds—91. Cut wounds</i>	<u>92</u>
	<i>bruised wounds—94. Gun-shot wounds</i>	<u>96</u>
SECT. III.	<i>Of bruises</i>	<u>98</u>
SECT. IV.	<i>Of dislocations</i>	<u>99</u>
	<i>Dislocation of the arm with the shoulder</i>	<u>101</u>
	<i>Dislocation of the elbow or the knee</i>	<u>103</u>
	<i>Of spraining the wrist, or ancle</i>	<u>103</u>
	<i>Dislocation of the thigh bone</i>	<u>104</u>
SECT. V.	<i>Of fractures</i>	<u>107</u>
	<i>A fracture of the skull</i>	<u>108</u>
	<i>Fracture of the arm</i>	<u>109</u>
	<i>A fracture of the thigh bone</i>	<u>110</u>
	<i>A fractured leg</i>	<u>111</u>
	<i>Compound fractures</i>	<u>114</u>
	<i>Conclusive observations on fractures</i>	<u>115</u>
SECT. VI.	<i>Of scalding</i>	<u>118</u>

SECT. VII.	<i>Of burning</i>	<u>119</u>
SECT. VIII.	<i>Of drowning</i>	<u>120</u>
	<i>The method to recover a drowned person</i>	<u>123</u>

LECTURE IV.

On external diseases, and their cure

SECT. I.	<i>Of inflammation</i>	<u>130</u>
	<i>Of resolution</i>	<u>132</u>
	<i>Of suppuration</i>	<u>133</u>
	<i>Of tumours</i>	<u>134</u>
	<i>Of mortification</i>	<i>ibid.</i>
SECT. II.	<i>Of ulcerations</i>	<u>137</u>
SECT. III.	<i>Of the diseases of the skin</i>	<u>142</u>
	<i>Of the itch</i>	<i>ibid.</i>
	<i>Of boils</i>	<u>144</u>
	<i>Of ring-worms, daw-worms, &c. &c.</i>	<u>145</u>
	<i>Of the lousy disease</i>	<i>ibid.</i>
SECT. IV.	<i>Of promiscuous disorders</i>	<u>148</u>
	<i>Of the tooth-ach</i>	<u>149</u>
	<i>To draw a tooth</i>	<i>ibid.</i>
	<i>Of sore eyes</i>	<u>152</u>
	<i>Of the ear-ach</i>	<u>153</u>
	<i>Of hoarseness and sore throat</i>	<u>154</u>
	<i>Of the venereal disease</i>	<u>156</u>
SECT. V.	<i>Of the gonorrhœa</i>	<u>157</u>
	<i>Of the phimosis and paraphimosis</i>	<u>158</u>

	<i>Of the chordee, and priapism</i>	<u>159</u>
	<i>Of the dysuria</i>	<i>ibid.</i>
	<i>Of shankers</i>	<u>160</u>
	<i>Of the bubo</i>	<u>161</u>
	<i>Of swelled testicles</i>	<u>162</u>
	<i>Of venereal excrescences</i>	<u>163</u>
	<i>Of the second infection</i>	<u>164</u>
SECT. VI.	<i>Of the bite of venomous animals</i>	<u>166</u>
	<i>To cure the bite of a rattle snake</i>	<u>167</u>
SECT. VII.	<i>Of the bite of a mad dog</i>	<u>170</u>
	<i>The signs of a mad dog</i>	<u>171</u>
	<i>The effect of a bite of a mad dog</i>	<u>172</u>
	<i>Of the cure of the bite of a mad dog</i>	<u>173</u>

LECTURE V.

Of fevers and inflammatory diseases, and their proper method of treatment

SECT. I.	<i>Of fevers in general</i>	<u>182</u>
SECT. II.	<i>Of accidental fevers</i>	<u>184</u>
SECT. III.	<i>Of a cold</i>	<u>192</u>
SECT. IV.	<i>Of intermitting fevers, and agues</i>	<u>193</u>
SECT. V.	<i>Of putrid and malignant fever</i>	<u>203</u>

LECTURE VI.

Of the various internal diseases, and their method of cure

SECT. I.	<i>Of diseases of the head, of the apoplexy</i>	<u>208</u>
	<i>Of the phrenzy and delirium</i>	<u>210</u>
	<i>Of a stupor and lethargy</i>	<u>211</u>
	<i>Of a violent head-ach</i>	<u>213</u>
SECT. II.	<i>Of diseases of the neck</i>	<u>214</u>
	<i>Of the malignant sore throat</i>	<i>ibid.</i>
SECT. III.	<i>Of diseases in the breast</i>	<u>219</u>
	<i>Of the cough</i>	<i>ibid.</i>
	<i>Of the pleurisy, and a peripneumony</i>	<u>221</u>
SECT. IV.	<i>Of diseases of the bowels, and the alimentary canal.</i>	<u>224</u>
	<i>Of a Depraved Appetite</i>	<u>225</u>
	<i>Of the Colic</i>	<u>228</u>
	<i>Of looseness, and bloody flux</i>	<u>230</u>
SECT. V.	<i>Of the Scurvy</i>	<u>233</u>

LECTURE I.

AN

Anatomical and Physiological Description

OF THE

HUMAN BODY.

The mechanism of the human body is so wonderfully, so curiously, and so wisely contrived, that the more we examine it, the more we must with admiration acknowledge, that nothing but an *ALMIGHTY BEING* could be its author. This incomprehensible *BEING* let us ever remember to reverence and adore, when we examine into the stupendous contrivance of our fabric; *HE* is the fountain, the benevolent upholder of nature, and all the creation; in whom we live, breathe, and have our being.

SECT. I.

Of the Composition and close Connection of Parts in the Human Body.

The human body is composed of solids and fluids; these are so closely, and so intimately combined with each other, that we find upon a physical enquiry and examination, they are scarce separable. They aid and give action to each other, and life depends not on each singly, but on the united, and on the separate spur, and on the soul of activity each gives to the other: Their substances form the body, and their action produces the operation of the mind.

Whilst every thing moves free and agreeable to the talk prescribed by nature, the automaton is in a state of health; body and mind enjoy peace and tranquility. So closely are body and mind in the most perfect bond of friendship, that they share pleasure and pain sympathetically with each other; and reciprocally contribute to each other's welfare.

With what satisfaction is the mind continually engaged to give pleasure to the body! and with what alacrity are for ever the Members ready to obey the will! Happy harmony, by which we so wonderfully exist!

The *solids* are the canals and springs of life, and the *fluids*, which circulate in those canals, nourish them, and set the springs in action.

By the *solids* we understand,

1st The *bones*; the frame and support of our wonderful structure.

2dly. *Gristles* or *cartilages*; which cover the ends of the bones, in order to make them move with ease and agility.

3dly. *Muscles*; the fleshy parts, which by a nervous sensation, are induced, either voluntarily or involuntarily, to contract or dilate themselves, in order to give motion to the various parts of our body, to which they are fastened or destined to move.

4thly. *Tendons*; the chords as it were, by which the muscles are fastened to the bones, so as to make them follow the contraction of the muscles.

5thly. *Ligaments*; the chords by which the bones are connected with each other, and which give a firmness to their contact and articulation.

6thly. *Arteries*; the canals which transport the blood from the heart, to all the parts of the body.

7thly. *Veins*; the canals which carry the blood from the remotest arteries (in the most minute parts of the body) back again to the heart, for a new rotation of the blood.

8thly. *Lymphatic Vessels*; which are small cellular canals, that convey the lymph in different parts of the body.

9thly. *Glands*; which are kernels formed to *secrete* or separate various humours from the blood.

10thly. *Nerves*; these are medullary branches springing from the brain, and distributed in the most minute parts of the body; ordained to give the act of sensation, as the soul of animal activity.

11thly. *Hair* and *nails*; these are a horny substance, and appropriated either for defence or warmth of the different parts where they are placed.

The *fluids*; are, 1st. *Blood*; which is circulated in the arteries and veins, and is the nutriment and source of life.

2dly. The different humours separated by the glands from the blood; of which there are many, as the *lymph*, *serum*, *spittle*, *seed*, *chyle*, *urine*, *gall*, &c. appropriated to different uses, as hereafter will be described.

3dly. *Fat*, and *marrow* in the bones; which are designed not only as a store of nourishment, but to lubricate and warm the parts where they are lodged.

These are the principal materials which compose our wonderful machine. There are various other parts mentioned by Anatomists, both solids and fluids; but as this is designed only to give a universal idea of the most material parts of the structure of this stupendous fabric, we will let this definition suffice, and now consider with what order the principal organs of life act, either in consort, or in opposition to each other.

SECT. II.

Of the Brains and Nerves.

The brains are by all Physiologists esteemed the most mysterious organs of life, being the fountain of the nerves, the organs of sensation, and hence allowed by all to be the seat of the soul, or the *sanctum sanctorum* of the human understanding.

They are distinguished into the common or great brain called *cerebrum*; the small part called *cerebellum*; and the *spinal marrow*, which projects down the back-bone. The great brain is divided into two great lobes, seated in the upper and fore part of the skull, of which it takes in the greatest cavity. It is so exceedingly sensible, that the least pressure puts an immediate stop to the sensation of the animal; and which is generally termed a *apoplexy*.^[2]

2. Hence in a fracture of the skull, or any other heavy contusion, where the brain is compressed either by the fractured bone, or the extravasated blood, the patient will remain in a state of stupefaction, till either the fracture is reduced, or the extravasated blood removed by trepanation; after which he immediately comes to himself, as if suddenly revived from a trance.

This part of the brain, though it is so exquisitely sensible, seems to be designed by nature as preparatory to the *cerebellum*, the immediate root of the nerves.

The *cerebellum* is the lesser, but the more material part of the brains. It is seated in the hinder cavity of the skull, (which for the sake of protecting this exquisite sensible organ, is remarkably strong and firm) and from thence as the fountain of life itself, the nerves, the instruments of sensation in the whole animal frame, take their commencement: For whereas the other is so sensible, as that the least pressure will cause a stupor; this is so far superior in point of exquisite sensibility, that a pressure on that part, not only occasions an apoplexy, but even stops life itself.

From this *cerebellum* prolongates the marrow in the back-bone, called *medulla spinalis*, which is but a continuation of the same, and is equally sensible, and of the same substance with the former.

The whole brain, that is, the *great brain*, the *small brain*, *spinal marrow*, together with all the nerves which spring from thence, are covered with two coats, called the external stronger coat, *dura mater*; and the thinner and inner coat, *pia mater*; which continue with each nerve, even to its smallest ramification. The whole in a living subject is in a continual motion peculiar to itself.^[3]

3. This motion in the brain is not unlike the peristaltic motion; this is an observation too seldom taken notice of by anatomists, though evidently plain in living animals, and is of singular service in physiological enquiries.

From the *cerebellum*, and *medulla spinalis*, all the nerves take their commencement and origin. Ten pair spring forth from the part in the skull; the principal are those which form the various senses, tasting, smelling, hearing, and seeing, and that common to all, feeling. From the *medulla spinalis*, or marrow in the back-bone, proceed thirty pair of nerves, which spread themselves variously over the whole system.

All nerves have this in common with each other, namely, exquisite sensibility; but in other respects they differ however widely; particularly the nerves destined for some of the external senses.

It is from that common quality *feeling*, that they become the subtile springs of life; and give an edge to all the animal functions.

There is nothing so plentifully, and nothing more curiously interwoven with every other substance, as the nerves; and so amazingly quick is their sensation, that the very instant a single nerve, even the smallest and remotest in the system, is affected, the soul, or the common *sensorium*, is sensible of the identical part so affected.

It has been asserted, that this amazing quick conveyance of the sensation of the nerves, is owing to a subtile nervous fluid; but on a rational reflection, and a just enquiry into the animal œconomy, it must appear, that such conveyance of a fluid is inconsistent with probability and experience: For, neither are the nerves tubulous, nor has there ever been discovered any circulating humour in them. And secondly, is it consistent with reason, that a conveyance of a fluid from the toe to the finger, should be so momentarily

quick, as is sometimes experienced in the gout, and other the like ailments? That the quick transposition of sensation of the nerves, is rather effected on a principle of *vibration*, appears more probable, and will admit of a more convincing experiment; which may be exemplified in the following manner: Suppose (in the language of a seaman) a rope runs from the top-mast-head, or any other part of the rigging, to the deck, and is kept tought and free from any interception; then it will be evident, that the least touch at one end will instantaneously be conveyed to the other end; in like manner the nervous sensation may be communicated, perhaps not unlike electricity; which similarly will in an instant be conveyed to any part, by a direct communication, but equally subject to be intercepted.

Hence, a paralytic limb will lose, together with its irritable sensibility, its circulation of fluids, and its nourishment; and recover again when the pressure or interruption is ceased. Hence also, spasmodic contractions will be affected at the irritation of some principal parts, and from this instantaneous irritability, the motion of every part is thought to obey the will.

SECT. III.

Of the Heart and Lungs.

The heart is a muscular body, situated in the breast or chest, somewhat between the lobes of the lungs, and formed by nature to be the principal organ for the circulation of the blood and other fluids.

Its structure is wonderful, and of a most curious piece of mechanism. It has two principal cavities, called *ventricles*, which are separated lengthways by a wall, and distinguished into the right and left.^[4]

4. They might with equal propriety, be termed *anterior*, or foremost, and *posterior* or hindermost.

Each of these cavities or *ventricles*, have particular valves, call'd ears, or *auricles*; which perform the function somewhat similar to the suckers of a pump: By these contrivances the heart receives the blood from the veins, and expels it again by the arteries, to every part of the human body; in a manner that I shall hereafter describe.

The lungs are organs immediately concerned in the circulation of the blood also; their function is to receive the air, to purify it, to circulate, and to distribute it in due proportion in the sanguineous mass, and to extract and expel that part of air already made use of; and thus act the part of ventilators in the animal œconomy.

They are divided into two lobes. These lobes are a continuation of little bladders, that have an immediate continuation with the wind-pipe, *aspera arteria*. The wind-pipe from the larynx down to the very lungs, is protected externally with semicircular gristles, in order to keep it always open, and so shelter it from external injury; when it enters the lungs it spreads itself into numberless branches like a tree, and sends ramifications into every little air-bladder in the lungs.

All along these ramifications and the air-bladders, are arteries, veins, and lymphatic vessels, which have an immediate communication with the heart, by means of the pulmonary arteries and veins.

Now at the instant of *inspiration*, these little bladders in the lungs are filled with air, which infuses itself by the small arteries into the whole mass of blood; and at the *expiration* again, the air which has been made use of, and drawn from the neighbouring little veins, is again expelled by the wind-pipe through the mouth: And thus is *respiration* performed by alternate motions of the lungs; receiving the fresh, and alternately expelling the foul air again; thence justly may they be deemed the ventilators of the human *automaton*, and a principal organ of life, in the animal creation in general.

The upper part of the wind-pipe, which is in the *fauces* of the mouth, is called the *larynx*. This is the organ of voice.

In this animal mechanism is contained, all the variety of formation requisite to every kind of sound that can be performed upon any instrument whatever. Its parts are chiefly of a gristly substance, at the same time exquisitely delicate, with nerves and glands; from whence it is subject to diseases on the slightest indisposition.

In men it is larger than in women; hence their voice is rougher; which generally increases by exercise of the voice also. Its formation is very complex, and too tedious here to enter upon; we will therefore pass it over, and hereafter say as much as relates to its preservation, and giving it relief when disordered.

SECT. IV.

Of the Circulation of the Blood.

We shall now consider the circulation of the blood; the grand source of our existence. At the contemplation of which we must be lost in admiration, at the wonderful wisdom of its contrivance; and with uplifted eyes adore the cause of its motion!

The Heart, as I before observed, has two chambers or *ventricles*, distinguished by the right and left; each of these chambers have also a valvular cavity call'd *auricle*, or ear. Each *ventricle* opens itself into an *artery* and *auricle*; and each *auricle* opens itself into a *ventricle* and *vein*. These openings have valves, which open and shut alternately, and by this, prevent any confusion or hindrance in the influx and efflux of the blood.

The right *ventricle* is the longest and greatest, but weakest: From this ventricle springs forth the *pulmonary artery*, which spreads itself throughout the Lungs.

The right *auricle* is also the largest; in which opens itself the large vein, called *vena cava*; as the common trunk that receives the blood from all its branches spread over the whole human body.

The left *ventricle* is shorter, but the walls are considerably stronger than that of the right *ventricle*. Here begins the *aorta*, or the great artery, by which the blood is sent to all the parts of the human body.

The left *auricle* is also somewhat smaller than the right; and in that opens itself the *pulmonary vein*, which returns the blood from the lungs, after it has received its necessary assistance from the vesicles of air in the lungs.

This premised and understood, the circulation is performed in the following manner:

The right *ventricle* being full of blood, contracts itself. This contraction is called the *systole*, by which it forces the blood through the *pulmonary*

artery into every part of the lungs, even into every vesicle; where it receives a portion of the air, drawn in by the wind-pipe.

At the end of these *arteries* join the ramifications of the *pulmonary vein*, which receives the blood from the ramifications of the arteries, by *anastomosition*; which, by little quantities, these veins gather gradually into little branches, and at last become one common trunk, which empties the blood into the *left auricle* of the heart.

From the *left auricle* it is let, by means of a peculiar constructed *valve*, into the *left ventricle*; which, by a dilation called the *diastole*, receives it from its *auricle*.

Immediately upon this reception, the *systole* or contraction takes place, and the blood is forced into the great universal artery, *aorta*; by which it is farther transported into every part of the human body.

This *artery* spreads as it goes, and continually decreases in magnitude, as it becomes numerous in branches; until it terminates into the smallest ramification in the remotest part of the body.

The *systolic contraction*, and the *diastolic dilation*, which alternately take place in the heart, continue throughout the arteries from the great trunk to the remotest capillary ramifications,^[5] by which means the blood is transported with more certainty and facility to the minutest part.

5. Concerning this great act of continued systole and diastole, Anatomists have greatly varied; but from the structure of the arteries, and the continued pulsation, this way of transporting the blood is confirmed both by reason and experience.

At the extremity of these arteries the veins take their commencement, in ramifications equally small with the former; these take up the blood from the arteries, by many infinite small quantities, and carry it gradually back from those extremities to the heart again, for a new rotation.

These veins commencing infinitely small, but gradually gathering, become branches, and at length form one general trunk, called the *vena cava*, (or the great universal vein); and this empties itself again into the *right auricle*; from whence it is let again into the *right ventricle*, whence it came; then again forced into the lungs; thence back again into the left ventricle; from thence all over the whole body, and continues the Circulation.

SECT. V.

Of the Joint Connection of the Brain, Heart, Lungs, &c.

These are the principal actions in the grand movement of the human automaton.

The nerves are the subtile springs of the sensation, by which the whole becomes sensible of irritation, and agitated to its functions. Their ramifications are extended *ad infinitum*, and so curiously distributed, that though there is not a part in the whole system exempt from their spreading, yet they are distributed with such regularity, that they don't interfere with each other. A regularity absolutely requisite to the order in the animal œconomy: For wherever the vibration of the nerves is obstructed, life or motion must in consequence cease.

The lungs are the ventilators in the system: they receive and let in the air; they sift it from all its gross particles, and extract the useful parts from this universal element of life; thus they give from every little vesicle, a due proportion of extracted air to the blood by the adjacent veins; and by the neighbouring arteries, again extract the foul air from the circulated blood, which by the mouth and nostrils is expelled again; at that instant a fresh quantity of air is drawn in again for a new supply.

Thus by intervals, the lungs, like a perfect ventilatory machine, repeatedly, and without intermission, keep time with *systolic* and *diastolic* motions of the heart, in exchanging fresh and foul air for the support of this wonderful animal structure. This is the function of breathing *respiration*, namely *inspiration*, or the drawing in the fresh air; and *expiration*, the expelling the foul air.

The heart may be considered as the grand perpetual pumping engine, constructed on principles, so as to have the least friction possible.

This curious pump, *the heart*, sends the fluid of life, *the blood*, to all the parts, even the remotest in the whole machine. The *arteries* are the canals of

the conveyance; they are strong elastic tubes, whose fibres are of spiral direction, and receive from the heart a power of contraction; which is continued from the heart in the moment of the *systole*, even to the remotest and most minute ramifications; by which the blood is pushed along to the minutest extremities. This is immediately succeeded with a *diastolic* reception of a fresh quantity of blood; so that on succession, the *diastole* and *systole* is perpetuated not only in the heart, but throughout all the arteries, in a friendly correspondence over the whole system: And this constitutes the *pulse*.

The veins, however, are somewhat of a different texture from the arteries; whilst the arteries transport the blood by an elastic and pulse-like contraction, the veins on the contrary, receive it from them at the extremities, in an easy and imperceptible gradation; for whereas the fibres in the arteries are in a spiral direction, elastic and strong, the coat of the veins are tender, and the fibres more longitudinal.

Hence, the wounds in the arteries and veins differ greatly; for whilst the arteries are for ever in a strong agitation, the veins perform their functions with less exercise; and therefore the wounds in arteries are always attended with danger, whilst the veins heal easy. From the same reason also, the arterial blood differs from the venal; for as the arterial blood is sent to all the Parts, as the grand nutriment, from whence the glands secrete their peculiar humours; the veins only transport it back again, to prepare it for a new rotation. Whence the blood in the arteries is of a high colour, and of a more alkalascent nature, than that in the veins.

Collateral with the minute ramifications of the veins and arteries, over the whole system, are the *lymphatic vessels*, which separate from the blood as it passes along, a transparent lymphatic humour, adapted for the different parts, which they are connected with.^[6] These vessels are materially different, from either arteries and veins in their structure, and justly to be compared to a continuation of little cells; so constructed, as to transport the humours, without admitting a return.

6. An opinion has lately been broached, that the lymphatics are all absorbing vessels, of the same nature with the lacteals, and that they all join to empty themselves in the subclavian vein to the common mass of the blood; but this to me, seems too general, and contradicts experience.

From hence we see plainly how these three organs, the *brains*, the *lungs*, and the *heart*, are concerned, and mutually assist each other in the actions, and circulation of the fluid of life; nothing could move without the sensation of the nerves; these could not exist without the blood, from which they receive their support; the blood could not circulate without the heart, arteries, and veins; and in them it could not move with a necessary freedom, except its being supplied with a sufficiency of air from the lungs: So that evidently all three are so interested in this grand movement, that the stopping of the one, must needs be the stopping of the whole.

SECT. VI.

Of Chylifaction and Nutrition.

In the preceding we have briefly considered the state of the fluid of life, and mechanical instruments of their motion in the human machine. In this we will consider how the whole is supported, and the vital lamp, from time to time, becomes supplied.

Under this consideration we will examine the canal of food, and the various changes the morsel undergoes from the time we take it in our mouths, till the nutriment is reduced to blood, and the remaining dross is expelled the body as useless.

The animal functions towards *chylifaction* and *nutrition*, are *mastication*, or chewing the *food*; *deglutition*, or swallowing; *digestion*; *chylifaction*; *nutrition*; and the *excretion* of the *fæces*.

In order therefore to understand this clearly, it becomes necessary to describe the organs which nature has formed for the requisite performances thereof.

In the first place then, *smelling* is that sensation which nature has given to every animal, that has a choice of food, as the first safe-guard to inform it of any thing agreeable or disagreeable, useful or pernicious to its body; its seat is in the nose chiefly, but we find by experience, that it has a friendly connection with our palate and stomach; for the effluvia of any thing will either create a desire, or give us the greatest aversion to every substance that throws out a flavour.

It is performed by means of a subtile spreading of nerves, peculiarly delicate; which continues through the membrane of the nose, the roof of the mouth, gullet, and the very stomach.

Taste is the next sensation, which nature has given us, not only as a distinguisher of proper food, but a sensation from which we receive many

luxurious pleasures; and to the indulgence of this sensation, most evils and plagues to mankind, take their origin.

The tongue is the principal instrument of that peculiar quality; but if we examine somewhat closer into this affair, we shall find that the soul of pleasure and pain of that sensation, as well as that of smelling, has its seat in the stomach; for that which will taste pleasing and good at the first approach, will soon lose its relish when the stomach is gratified: and, if any thing tastes disagreeable, the stomach receives it with reluctance, and will ever incline to discharge it again.

The tongue is an instrument (if I may be allowed the term) very curiously constructed; it is moved by a variety of muscles, and serves not only for tasting, but also as a labourer, to shovel and to turn our meat between our grinders; so that nothing may escape being well masticated, and intermixed with that fine digesting balsam, the spittle, in order that it may be easily swallowed.

Besides this, it makes the most requisite instrument for the noble and excellent faculty of speaking; which forms one of the principal characteristics that distinguishes man from the brute creation.

The gullet or *oesophagus*, is the canal which conveys drink and food from the mouth to the stomach; this canal is a muscular, tendinous, and vascular tunic.

The commencement of it is in the mouth, and is called the *pharynx*; a curious structure, that receives the food, and by its contractive motion, and the help of the tongue, forces the aliment into the stomach.

The stomach is much like the bag of a Scotch bag-pipe; it lies immediately under the *diaphragm* or *midriff*, covered partly on the right side with the liver, and on the left side with the spleen. The left and superior part, is continued with the *oesophagus*; and the right and inferior part, or orifice, commences the *intestines*.

The first orifice is called the mouth of the stomach; and the second the *pylorus*, or *porter*: At the porter there is a curious valve which lets the aliment out by small parcels into the intestines, where it undergoes its various other changes.

The stomach has three teguments, a *muscular*, a *tendinous*, and *nervous coat*; this *nervous coat* has another slimy one, but this in reality, is a delicate lining, interwoven with nerves, and the ramifications of fine blood vessels.

The *intestines* or guts, are a continuation of the stomach, they are a canal which is generally reckoned six times as long as the subject it is taken from; it is distinguished in small *tenuia*, and wide *crassa*.

Each again is divided into three parts: the *tenuia*, or small narrow intestines, are the *duodenum*, or twelve finger-gut; the *jejunum*; the *ileum*: The wide or *crassa*, is divided into the *cæcum*; the *colon*; and the *rectum*.

Throughout the whole canal of intestines are numbers of little vessels, called *lacteals*, which lead the chyle, extracted from the aliment, into a *receptacle*, which is lodged in the *mesentery*, and from thence, by another duct call'd the *thoracic duct*, is carried along the back-bone upwards, and joins to the left subclavian vein, where the *chyle* gradually commences to be blood.

This short description we will let suffice, and now enter upon the action itself.

The morsel now, which is designed for food, is taken into the mouth, masticated with the teeth, turned about with the tongue; and as the mouth is at work, the *saliva* or spittle is squeezed from the salival glands, and thus intermixed with the aliment; when enough chewed and moistened with this saliva, it is conveyed to the *pharynx*, or swallow, which receives it, and, by its contraction, forces it into the *oesophagus*; and by a repeated contraction, is carried down into the stomach.

There it is again moistened with *saponaceous liquid*, or *pancreatic juice*; by which and by a perpetual motion of the stomach, it is brought into a state of digestion; then by small degrees entered through the *pylorus* or porter, into the first division of the gut, the *duodenum*.

This gut is about twelve fingers long; and whilst the aliment is there, it is intermixed with the gall, which is a liquor separated by the liver, and contained in the gall bladder; this liquor, the gall, is carried into the *duodenum*, by a small duct, called the *ductus cysticus*; where also enters another kind of liquor called the *pancreatic juice*.

When the aliment is thus prepared, and fit for a particular state of dissolution, it is carried into, and through the *jejunum*. This gut is in length about twelve or thirteen hands breadth, and its motion somewhat brisk; through which the aliment passes pretty quick, and hence, generally is somewhat empty.

As it passes through this part, the chyle is separated from it by the lacteals, which are small vessels that separate the chyle from the aliment, and abound there more than in any other part of the gut.

From thence it comes into the *ileum*; that is the longest of all the divisions of the guts, and is in length about twenty-one hands breadth; it has a great many circumvolutions, and next to the *jejunum*, has many lacteals to separate the chyle.

Now the aliment comes into the wide gut, and gradually becomes fæces; and first, the *cæcum*: This part is rather an appendix only, and hangs from the main part like a finger to a glove. The use of this gut has been much controverted by anatomists; it seems however, very probable, that this appendix is designed to keep the aliment in for further digestion, as it now begins to putrefy, and becomes fæces or excrement.

From thence it enters the *colon*, which is a long, and very winding intestine; it runs up along and about the liver, touches the gall bladder, and the spleen; from thence it descends again to the *os sacrum*. It has but few lacteals, and is, as it were, the last drainer of the fæces: It is this intestine which is the seat of the Colic, and of most other complaints of the belly.

Next to this comes the last and straightest, the *rectum*; this gut is closely adherent to the *sacrum*, and ends in the fundament; which is provided with muscles to open and shut the *anus*, in order to contain the fæces, and discharge it.

The mechanism of chylifaction in the human body differs from the brute creation in general, except that most contemptible of the whole, the hog; to which it bears a very near resemblance, insomuch that there is very little distinction.

Both have that advantage over the generality of terrestrial animals, that they are confined to no particular food; which favours greatly the luxury of the one, and the beastiality of the other.

The brute creation are generally distinguished into *carnivorous* and *granivorous*: The first is that kind which feeds upon flesh; and the latter upon grain and vegetables. Upon examination however, we find, that the stomach and guts are peculiarly adapted to their food; and that grass agrees no more with the dog, than mutton does with the horse.

But man is so happily made, that any thing which is food, is proper for him, and he may become used to it; and thus is either carnivorous or granivorous.

The whole canals, from the stomach to the anus, is in a continual vermicular motion, which is called the *peristaltic motion*; by this the aliments are dissolved, and disunited; and as they pass along, are drained by the lacteals, of their *nutriment* or *chyle*.

These lacteals are, by means of a membrane (with which they are surprisingly interwoven, and connected to the whole canal) called the *mesentery*, lead regularly into one common cistern, lodged almost in the middle of the intestines, in that membrane, called the *receptacle of chyle*; and from thence the *chyle* is carried by a duct up along the back-bone, called the *ductus thoracicus*, into the left *subclavian vein*, where it gradually commences to be blood.

By this mechanism we are nourished, and the substance of our food converted into blood, and transported through the whole animal machine, for the support of every part of its wonderful composition.

As all animals which feed upon flesh, are more subject to diseases, nature has provided them with these advantages: that when any thing is obnoxious to their nature, and received into their stomach, or their being over loaded, it can discharge itself of so troublesome a burthen, by vomiting, which is effected thus: when the inner coat of the stomach, which is irritable and nervous, is stimulated by whatever is obnoxious, it will cause in the whole stomach, a contraction; and by that, force its contents to the shortest direction of evacuation, namely, by the canal of the oesophagus, through the mouth. This expulsion is peculiar to carnivorous animals only.

Purging or discharging by the fundament is common to all animals of whatever kind; and is performed by an irritation in the intestines, by which the peristaltic motion is increased; to this I must add a reversion of the

secretion of the lacteal vessels, by which the humours are increased, the motion accelerated, and the fæces discharged, without giving any nourishment to the body, and consequently the system diminished.

Digestion is that act by which the aliment or food is prepared, so as to produce a good chyle, and consequently good blood, for the nourishment of the body. Though no animal has a more delicate stomach than man, yet it must be observed, that none has a stomach better adapted for all kinds of food.

Whence therefore in the common course of life, temperance and gentle exercise is what nature requires to maintain health. But nothing becomes more obnoxious to that blessing, than gluttony, voluptuousness, and idleness.

SECT. VII.

Of the Bones, Muscles, Ligaments, Tendons, and their Connections.

The bones may be considered as the timber-work of the human frame; by which this wonderful fabric is supported, and kept in its due form, that the whole may be brought into its various movements, without confusion or obstruction to each other.

The bones are the most solid parts of the human body, composed of hard and indurated fibres, striated over each other, in a manner peculiar to that substance. The bony fibres are in themselves insensible, yet as the parts are variously distributed with arteries and veins, and that the nerves must necessarily have a share in their formation, they have a peculiar sensibility, which is perceptible in some parts more than others; they cannot strictly speaking, be deemed quite insensible.

The whole bony frame is covered with a tendinous and nervous tegument, called *periosteum*; except such parts of the teeth as are designed for mastication, which are provided with a peculiar enamel, that is harder than the rest of the bony substance.

The *periosteum* is exquisitely sensible, and is the safe-guard to the substance of the bone, which is delicately tender notwithstanding it is not so sensible as the skin that covers it; insomuch that it will become *carious* on the least exposure to the air, or the attack of any foreign body of matter whatever; whence in wounds and fractures in general, great attention should be paid to the substance of the bone, being very subject to become *carious* and to *exfoliation*, which is of the greatest consequence; but of this I shall say more in another place.

The marrow is principally designed for the nourishment of the bones; which is evident from its being plentiest in young people, when the bones are strongest; and that when it is deficient, they become brittle, and lose their tenacity.

The marrow is contained in a cellular substance, partly in vesicles of a nervous texture, and partly bony cells. At the ends of the long bones the texture is more spungy than in the middle, where the cavity is less, but the substance is most compact. Though anatomically there is no perceptible circulation in the bones, yet, that a circulation is actually existing, is evident, from a liquor oozing out from the ends of a fractured bone in the living animal; by which a fractured bone again unites, and this is called the *callus*; and whilst in its liquid state, resembles the white of an egg, which gradually ossifies, and becomes as hard as the main substance of the bone.

The number of bones differ somewhat in various subjects; ordinarily they amount to two hundred and fifty-two: In the head sixty-three; in the trunk seventy; in the arms and hands sixty; and in the legs and feet sixty.

As the bones are the support of the animal fabric, I have inserted in the next page, a catalogue of the human skeleton, which occasionally may be referred to:

A Skeleton of the Human Body.

BONES in the HEAD.

The SKULL.

The Forehead.

Os Frontis	1
------------	---

The Hindhead.

Os Occipitis	1
--------------	---

The Sides of the Head.

Ossa Parietalia	2
-----------------	---

The Temples.

Ossa Temporum	2
------------------	---

The Basis of the Skull.

Os Ethmoides	1
Os Sphenoides	1

The Bones of Hearing.

Ossicula Auditus	8
---------------------	---

The FACE.

The Upper-Jaw.

Ossa Malæ	2
——Maxillare	2
——Unguis	2
——Nasi	2
——Palati	2
Os Vomer	1

The Under-Jaw.

Maxilla Inferior	1
------------------	---

The Teeth.

Dentes Incisivi	8
——Canini	4
——Molares	20

The Tongue-Bone.

Os Hyoides, is composed of	3
	—
	63

BONES in the TRUNK.

The Spine.

Vertebræ Cervicis	7
----------------------	---

——Dorsi	12
——	5
Lumborum	

The Ribs.

Costæ Vera	14
——Spuria	10

The Shoulders.

Scapula	2
Claviculæ	2

The Hip and Bason.

Os Sacrum, is composed of	6
Os Coxygis, is composed of	3
Ossa Innomenata, composed of	
——Ischium	2
——Ilium	2
——Pubis	2

The Breast.

Sternum, is composed of	3
	—
	70

UPPER EXTREMITIES.

The Upper Arm.

Os Humerus	1
------------	---

The Under Arm.

Ulna	1
Radius	1

The Wrist.

Ossa Carpi	8
------------	---

The Hand.

Ossa Metacarpi	4
----------------	---

The Fingers.

Ossa Digitorum	15
	—
	30

UNDER EXTREMITIES.

The Thigh.

Os Femoris	1
------------	---

The Knee Pan.

Patella	1
---------	---

The Leg.

Tibia	1
Fibula	1

Ankle.

Ossa Tarsi	7
------------	---

Foot.

Ossa Metatarsi	5
----------------	---

Toes.

Digitorum	14
	—
	30

The connection of the bones are in various ways; those connections that are designed for rest, are by close contact of parts, and are called *sutures* or *seams*; such are the bones of the skull with themselves and the face. Those connections which are designed for motion are called *articulation*. Some of the articulations have but an obscure motion, as the ribs with the back-bone, and the back with itself, &c. Others have an angular motion, as the elbows and knees; and others again have an universal motion, as the arm-bone with the shoulder, and the thigh bone with the hip. The *sutures*, and such connections as have no motion, are merely dovetailed into one another in close connection; but those articulations that are designed for motion are connected by cartilages, either in close contact, or so as to move slippery over one another.

The ligaments are those tough tendinous parts, by which the articulations designed for motion are joined together; at some parts they cover the joints only, and at others, they are immediately fastened to each other, besides the external coverings; those ligaments are very strong and elastic, and have a close connection with the nerves that pass by them.

The surrounding ligaments of all moveable joints, form a *capsular-bag*, which contains a slippery liquor, called *synovia*, that lubricates the ends of the bones covered with *cartilages*, that they may move with ease and agility over one another.

Muscles are the fleshy parts on the human body, appointed for motion. They are a composition of arteries, veins, nerves, and tendons.

Muscles are generally divided into two kinds; those for necessary or involuntary motion, and those for voluntary. Some again are fastened at each end to some of the bones at their extremity, and contract themselves in a straight direction; others again are annular, or in the form of a ring; and by this manner open and shut. The first kind are generally divided into three parts, the head, belly, and tail. The whole muscle begins and ends in a tendon, by which it is fastened to the part it is designed to move; which, when the belly of the muscle contracts itself, must consequently draw both bones nearer together, fastened to the head and tail; and thus performs the motion. Some of these muscles act in consort, and others in opposition to each other.

The tendons are the principal parts of the muscle; and that part thereof, as before observed, which is fastened in the bones, namely, at the head and tail of the muscle, and is the chord, as it were, by which the limb is brought into motion.

A tendon is a hard, substantial, flexible, and elastic body, which, at the head of the muscle, is very compact; but as it enters the muscle, spreads its fibres over the whole body, gathers again at the tail, in as compact a manner as it began, and continues in this chord-like texture till it fastens itself at its appropriated place.^[7]

7. It must be observed however, that tendons take on many various forms, as different as the muscles; which would be too tedious here to particularize.

The tendons and ligaments are of equal substance, and differ in nothing but their various uses; namely, the ligaments to connect the articulation; and the tendons, or the ends of the muscles, to give motion. The muscles are a composition of the tendinous fibres, plentifully intermixed with small ramifications of blood vessels; all which compose the fleshy part thereof. Hence, the tendons being white and hard, the muscles gradually grow softer and redder in the middle of them; and they verge gradually to become tendinous at the tail, the same as the head.

Hence, from the close texture of the tendons, they are very painful when wounded, or otherwise hurt; but as the muscles are softer and more pliable, their wounds and other accidents are of less consequence, and not so dangerous as the former.

The number of muscles are undetermined, for though the principal muscles are so distinct that they cannot be mistaken as to their form and use, yet there is so great a number of small ones, or so many of the great ones capable of being subdivided again, that anatomists have not, as yet, agreed about their number.

SECT. VIII.

Of the Urinary Organs and Genital Parts.

The *kidnies*, the *ureters*, the *bladder*, and the *urethra*, are the principal urinary organs. The kidneys receive two arteries called the *emulgent arteries*, from the great arterial trunk, called *aorta*, before described; they each send also two veins back again, accompanying the arteries to the *vena cava*. Whilst the blood is circulated in the kidneys, the urine is secreted in many small quantities, and carried from each kidney by a tendinous tube, the bigness generally of a small goose quill, called the *ureter* to the bladder. The *ureters* enter the bladder between its *lamillas*, so as not to admit of a return. Hence, a bladder in its natural position is both wind and water tight, and if reversed, is neither.

The bladder of urine is of a tendinous and nervous texture, and capable of great extension, but at the same time exquisitely sensible. The neck of the bladder is very muscular, and by this forms a *sphincter*, or annular muscle, by which the urine is retained; and as it is of a very sensible nature, this part is subject to inflammatory constrictions; great pain, and many evils have there been occasioned, either from venereal cases, or gravelous complaints.

From the neck of the bladder to the end of the *penis*, is the *urethra*; which canal serves for the emission of the urine, as well as that of the seed.

The whole is lined with a number of very little glans, to lubricate and protect the sensitive lining from the sharpness of the urine.

These are the urinary organs. We will in the next place, proceed to the organs of generation.

The *testicles* are the principal instruments for forming the human seed; they are generally two in number. The testicles receive the blood, of which they secrete the seed, from the *aorta*, about the same parts, whence the emulgent arteries project. These arteries run down from the aorta to the testicles, in a very contorted and winding manner, accompanied by the veins

which return back again from the testicles, and fix themselves in the ascending *vena cava*. These contorted and interwoven vessels are called the *spermatic vessels*, which carry the blood from the arteries in little quantities, and slowly return it again into the great vein.

Whilst the blood is circulated in the testicles, the seed is secreted in them; but as it is secreted, it is transported back again from thence by tubes, called *vasa deferentia*, towards the neck of the bladder, where there are two irregular bladder-like vessels, wherein the seed is contained, and kept for use, called the *vesiculæ seminales*.

In these seed bladders the seed is kept for use, till it is wanted. They are remarkably delicate and nervous; and when they are filled with seed, they stimulate all the organs of generation with a desire for venery.

The *penis*, or the manly member of generation, is partly covered with the common skin; the foremost part of which, that covers the glans, is called the *præpuce*. This *præpuce* is tied underneath to the substance of the *penis*, by what is called the *frænum*.

The inner part of the penis is composed of two kinds of bodies, the *cavernous bodies* of the *penis*, and the *cavernous body* of the *urethra*. The first are bodies enveloped in their peculiar teguments, and make the greatest part of the *penis*. Each of these bodies takes its origin from the erecting muscle of the *penis*, which is near the *pubis*; through each of them goes an artery and a nerve, which spread themselves through these bodies.

These bodies are full of cavities, which have all a communication with one another. The cavernous body of the *urethra* continues from the *bulb*, near the neck of the bladder all along to the end of the *penis*, and includes the *glans*, or the nut of the yard; and is much of the same substance with the former. At the back of the *penis*, goes along the large vein, called the *vena penis*, which spreads itself all along into numberless branches that connect themselves with the ramifications of the arteries throughout all the cavernous bodies; and from thence is caused the erection, as I shall presently describe. The *glans*, or the nut of the yard, is remarkably delicate and sensible, as being the seat of pleasure in coition; it is also of a cellular substance, and in one continuation with the cavernous body of the urethra.

Its sensation is greatly heightened by the *frænum* being tied underneath, from the *præpuce* to that part which keeps it, as it were, in an agreeable constriction when erect.

The penis is allowed two pair, and a single muscle. The first pair is the *erectores*, which take their origin from the fleshy protuberances of the *ischium*, and lose themselves in the cavernous body of the penis. The second pair of muscles is the *transversales*; these take their rise near the forementioned, and fasten themselves at the root of the penis. The fifth is the *accelerator*, or the odd muscle; this muscle takes its commencement from the cavernous body of the *urethra*, and grasps the whole penis, and adheres to the *sphincter* of the anus; the muscles aid and assist each other in the erection of the penis, which happens in the following manner:

At the influx of the animal spirits, the *vena penis*, or the great vein in the penis is somewhat contracted, whence the blood is forced into the cavernous bodies of the penis and that of the urethra, as the reflux of the circulating blood from the arteries is hindered; hence, the cellulæ of the cavernous bodies are obliged to swell up and distend all the parts, and thus make the penis turgid and erect; when the penis is further titillated by the friction of coition, the parts become generally irritated to an agreeable constriction, whereby the seed rushes from the seed-bladders into the urethra, and is thus forcibly ejected in the time of ejaculation; then the whole fabric slackens, the constriction of the great vein relaxes, the blood regains its free circulation, and the penis becomes slack again.

These are the faint out-lines of the genital parts; but to have a full description and comprehension of the peculiar properties of the genitals of both sexes, and also the sacred act of generation, I recommend to a serious perusal of the anatomical and physiological description of the genital parts of both sexes, in my *Treatise on the Venereal Disease*, where they will find a full and satisfactory account.

SECT. VIII.

Of the Skin.

The whole human automaton is covered with a skin, the mechanism of which is not less curious than what we have already described.

It is a texture curiously interwoven with an infinite number of tendons, arteries, veins, nerves, and lymphatic vessels. The tendons form the network of the whole, and give it an elasticity, from whence it requires a muscular motion; and the rest add each their parts, as I shall hereafter describe.

The skin, universally over the whole body, is divided into three parts; but in most parts it is found to have four.

The external, is the *cuticula*, or scarf skin; this is a horny and insensible substance, formed by nature as a shelter for the next part of the skin, which is exceedingly sensible; on examination it appears to be a horny substance, and as it were, little scales continued from the immediate skin, which on any external injury, such as repeated friction, fire, scalding, or blistering, separates from the main skin. The next is the real skin, called *cutis*; This is strongly interwoven with tendinous and nervous fibres, interspersed with numberless ramifications of arteries, veins, and lymphatic vessels; whence it is not only exquisitely sensible, but elastic, and on the least injury, subject to bleed. On the external surface is a very thin *plexus*, called the *reticulare*; which has the tinge of the natural hue of the skin, and distinguishes the complexion of the *negro*, the *tawny*, and *white*, from each other.

The structure of the *cutis* is peculiarly curious, and may justly be considered as a continued *secretory* and *excretory gland*; or more properly, like the bark of a tree; for whilst there is continual transpiration from the body, it, at the same time, is capable to absorb the subtile particles of whatever surrounds it, and exclude the grosser. Over the whole surface are innumerable little *pores*, which are nothing but the little openings of the

various lymphatics, and of the blood vessels corresponding with their peculiar *glandules*, called the *milliary glans*; between those on the hairy parts, the hair is planted as it were, with their distinct roots, corresponding with the hue of the subject they belong to. These *pores* widen or contract according to the internal or external warmth of the body; which when they contract, cause a kind of roughness, or continuation of little warts; and on the contrary, when dilated, smooth the skin. Next to this follows the fat skin, or *membrana adiposa*; which is a continuation of cells, wherein the fat is contained, and has an immediate communication with the neighbouring blood and lymphatic vessels. This part is not universal, for some parts of the face, the genitals, &c. are without it. The last of all is the *membranosa*, a thin parchment-like texture, which adheres spontaneously to whatever part it covers, either *periosteum*, the *tendons*, *ligaments*, or *muscles*.

These are the four principal divisions of the skin; but that which is the principal, is the second mentioned, namely, the *cutis*; on which I shall have occasion to offer something hereafter, that may prove of the greatest advantage to the sea-faring people, for whose benefit I have particularly intended these Lectures.

SECT. IX.

Of the External Senses.

Feeling is the universal sensation of the nerves, on which all other senses depend, as the subtile spring of life itself. What it is, and whence it proceeds, has already in the foregoing been explained; namely, a sensation of the nerves, by which we perceive an idea of *hard* or *soft*, *wet* or *dry*, *hot* or *cold*, or in general terms, *pleasure* or *pain*. The peculiar seat thereof seems to be placed at the ends of our fingers; whether this is the absolute seat ordained by nature, or rendered so by custom, is not my business here to enquire; perhaps, both conjoin to make it there most perfect, as best suited for that purpose.

Taste is the next sensation of the nerves, regarding an immediate approach and action of substances themselves, in order to distinguish betwixt the nature of foods, by which we are enabled to form an idea of *salt*, *sweet*, *sour*, *bitter*, *sharp*, or *mild*; which according to their mixture and proportion, render *taste* agreeable or disagreeable. Its seat is principally in the tongue; but that it extends to the very stomach itself, is evident, from the immediate consent of parts; namely, that what is disagreeable to the palate, will ever be nauseous to the stomach: Of which I shall say more hereafter.

Smelling is that sensation given to the animal creation, as a friendly messenger of taste, in order to enable us the better to chuse our food. It is actuated by means of the *olfactory nerves* spread in the *pituitary membrane* of the nose, and has an immediate communication with the brain; which nervous plexus is actuated by odorous particles, exhaling from substances, by which the olfactory nerves are stimulated agreeably, or disagreeably, as the first intelligence to the palate, or to the whole consent of the nervous system.

Hearing is that sensation of the nerves which is occasioned by the vibration of the air, and forms the sound, which enters the cavity of the ear,

and tremulates and puts in action, the organs of hearing. The ear is perhaps as curious a piece of mechanism to our conception, as any part whatever we shall meet with in anatomy.

To the organs of hearing we must reckon the external and internal *auditory*, divided by a thin membrane, which is a thin spreading of nerves, called the *drum*; within this, or in the internal cavity, we observe, 1st. Four little bones called, from their forms, the hammer, *malleus*; the anvil, *incus*; the stirrup, *stapes*; and roundling, *orbicularis*. 2dly. Their peculiar muscles. And, 3dly. Two considerable cavities, the *labyrinth* and *aqueduct*. How hearing is actually brought about, Anatomists have not as yet agreed; and which would be too tedious here to enquire into. That the little hammer is perpetually moving on the *tympanum*, according to the different sounds, may be one part; which, however, is greatly assisted by the other organs.

The *labyrinth*, with many turnings and windings, penetrates through the skull and forms part of the aqueduct, by which the sound is modulated and magnified, so as to cause the distinction between them, and convey the idea of their sonorousness to the common receptacle of sensation.

Seeing is that great faculty by which the form, colour, and motion of objects is represented to our ideas. The principal conveyer of this wonderful sensation, is the *optic nerve*; the eye is a telescope finished to the greatest perfection; or like a *camera obscura*, where objects are pictured to the utmost nicety, and a due proportion observed in their nature. It is a globe enrolled with a strong tendinous coat, and filled with three kinds of humours perfectly clear and transparent, which differ in nothing but their various degrees of liquidity and form. Each of these humours is enclosed in a delicate thin transparent membrane. The external coat of the eye is called the *cornea*, from its resemblance to horn, being a continuation of the *dura mater* and *pia mater*; and is every where opaque, except at the entrance of the light, which is called *pupilla*, and projects somewhat beyond the spherical form of the eye, similar to the crystal of a watch. Inside the *cornea* it is lined with a spreading of the optic nerves, called the *retina*, where objects are reflected; by which the ideas are immediately conveyed to the brain.

The humours are the following: The *aqueous humour*, the most external, which is designed for collecting the rays of light; the second is the

crystalline humour, which is in the form of a magnifying lens, and invested with a subtle transparent tegument, and so curiously fixed, as to be contracted and dilated, according as the objects are near, or removed from the eye; and accordingly the crystalline humour requires to be flattened or thickened; the last is the *vitrous humour*, which fills the remaining cavity of the eye; this humour serves to spread the objects again on the *retina*, and arrange them in their natural order, so that the picture of the objects viewed, may be preserved in its due proportion and regularity.

Now the act of vision is performed thus: The convex transparent part of the *cornea* faces at once every object which comes within the direct lines thereof, and this is termed a *full view*.^[8] This view enters through the aqueous humour, where the whole is contracted according to its convex and concave form, so as to make it fit to enter the *pupilla*, which is a muscular continuation of the *sclerotis*, and called the *uvea*; and from thence the whole view passes through the crystalline lens, by which the view is again magnified, and passed through the vitrous, and the whole depicted on the *retina*, in the same concave curve, and at equal angles in which it was first received, and apparently the same magnitude; thence representing the objects in their natural perfection, figure, colour, magnitude, and motion.

8. It is the general received opinion, that objects are reversed in the representation on the *retina* of the eye. This is however a mistake; I hope on another subject to prove this in a full and satisfactory manner.

When, however, an object is singled out of the whole view for particular attention, the *uvea* then contracts or dilates, which, together with the muscular action of the whole ball, proportions all the humours suitable to the magnitude and distance of the object under examination, and the quantity of light; and to make the eye more capable, the lids are frequently moving, in order to moisten and varnish, as it were, with a liquid pressed from the innermost gland, called *canthus major*, to keep the surface smooth, and assist its transparency. If this intense looking is continued beyond the strength of this noble organ, it relaxes from its proper destined expansion, and the objects are rendered obscure.

Darkness is ease and rest to the eye, because it is then out of action; whilst light always keeps it in exercise; And in diseases of the eyes, light should be kept away as much as possible.

Here then I shall beg leave to conclude this short description of the wonderful structure of our frame; which when we duly consider, will never fail of filling our breasts with a sense of admiration; and induce us to lift up our souls with reverence and gratitude, to that incomprehensible GREAT CAUSE, by whose wisdom we are formed; and by whose goodness we exist.

LECTURE II.

ON THE NECESSARY

MEANS AND MEDICINES

REQUISITE TOWARDS

RESTORING HEALTH.

Having, in the preceding lecture, given a general view of our wonderful structure, I shall, in this, endeavour to lay a foundation of the healing art, in order to assist nature when impaired by the attack of disease, or any unforeseen accidents.

But that I may become useful, and yet concise, I will, without loss of time, lead the reader to the principal distinction between health and disease; for without forming a proper judgment of an ailment, there can be no reasonable intention of cure. I shall therefore beg leave to solicit a serious attention to the following important observations, particularly, as they are founded upon the purest principles of *physic*, authenticated by happy success in an extensive practice, and communicated by the dictates of a candid and well disposed heart, that wishes Existence only as a useful member to society.

SECT. I.

Of the Pulse.

When we consider the admirably curious structure of the human mechanism, it must clearly appear, how easily the animal œconomy may become disordered. But so kind has the ALLWISE AUTHOR of our existence been to his creatures, that on the slightest indisposition, the whole system is sensible of the least disagreeable sensation; and every part, with a social and sympathetic friendship, is ready to relieve the fabric from whatever is obnoxious, and causes the disturbance.

As nothing has a greater share in life than the blood, so nothing likewise on the least indisposition, is sooner set in commotion, and gives the alarm, than this very liquor, in which life itself is contained; and the circulation of this fluid in the arteries, furnishes us with the greatest prognostic, namely, the *pulse*, by which we may, with any degree of certainty, judge of the state of the body, and foretell good or bad events. How beneficial it is then to be acquainted with this knowledge, I shall leave to every rational being to determine.

Every artery in the living body constitutes a *pulse*, as has been explained in Sect. V. in the foregoing lecture.

And it is demonstrative from simple reason, that the pulsation of such arteries must be in proportion to their magnitude, greatest in the heart, whence they take their origin; till the vibration must gradually be lost to the touch, as they become insensibly small, and lose themselves in their ramificatory capillary tubes.

The artery which is most convenient to our examination, is that which lays immediately to the touch on our wrist, below the thumb, on the radius, just below the knuckle of that bone; and it is there we are best able to distinguish the pulsation of the arteries, as being most exposed to the

external touch; because it is immediately between the skin and the flat of that bone.

Thus much premised, we will first consider the *pulse* in the sound state; and next make a comparison with that in people labouring under diseases.

In a full grown man, in the vigour of health, the pulse ordinarily beats once to every second, that is, 60 times in a minute, 3600 in an hour, and 31,536,000 times in a year. It varies however in different subjects, according to the stature, temperament, condition, time, and action of the same; insomuch, that two persons equally in a good state of health, shall, however, differ widely in their pulses. Yet, notwithstanding, there are peculiar signs, which will ever make a true distinction between health and sickness; practice however is requisite to form a proper judgment of such distinction.

In order to lay a proper foundation to understand the nature of this great prognostic, I shall divide the pulse into the following classes:

1st. A *full, slow, and regular* pulse, denotes a perfect state of health; for whilst the blood moves regularly in the arteries, without pain to the nervous system, or disturbance to the animal œconomy, the pulse will beat precisely regular, without trembling, fluttering, hurry, distinction, or feebleness, and discover to the touch, a peculiar easiness; and this, every one, who wishes to become master of that subject, should carefully attend to.

2dly. A *full and quick* pulse, indicates a fever; for by the irritation of the nerves, the vibration of the arteries are accelerated. If a gentle perspiration attends it, it may be deemed salutary; and with this pulse it appears, that nature makes a vigorous effort to expel what is obnoxious: therefore in such a pulse, a gentle perspiration generally gives relief.

3dly. A *full, quick, and hard* pulse, denotes great irritability in the nerves, and a redundancy of blood; this is the pulse of a fever; and here bleeding, if timely applied, is very requisite, as also medicines which promote perspiration.

4thly. A *small and quick* pulse, denotes an irritability in the nervous system, and a contraction of the arteries; this is generally the pulse which attends sudden shocks and frights, as also in the cold fits of agues. The causes should carefully be attended to: if, from a sudden shock, bleeding is

useful; but in the latter case, it is hurtful. It seldom lasts long, and a full feverish pulse generally follows, and the state of the patient ought here to be the guide: for, the same pulse may proceed from too great a plenty of blood, as well as from a deficiency; consequently requires different treatment; and this shall be pointed out more fully hereafter.

5thly. *An irregular full pulse*, is at all times a bad one; many may be the causes, and generally denotes a great confusion in the nervous system, and in the circulation of the blood. This pulse is generally the attendant on violent inflammations, phrenzy, delirium, &c. and if it continues any time, forebodes great danger: bleeding in time is greatly requisite, and other remedies, in order to assist the nerves, and bring the circulation to its proper regulation again.

6thly. *An irregular small pulse*, is frequently the attendant of an emaciated constitution, putrid fevers, and consumptions; and this dangerous pulse too generally forebodes the approach of death. The more irregular the pulse grows, *small, weak, trembling*, and leaves off by irregular intervals, the sooner dissolution is at hand; till at last, the pulse entirely disappears, and life ceases. But dangerous as this pulse is, yet it must be observed, that it often appears in the strongest constitutions, as in sudden emotions of the mind, falls, &c. or in a swooning. This is also the last and the first pulse in suffocations; or in other words, the struggling pulse between life and death.

These few distinctions of the pulse we will let suffice, as they, when properly attended to, will in all cases enable us to make a true distinction between health and the different stages of sickness, and accordingly direct us to a true method of cure.

SECT. II.

Of the Effects of Bleeding

There is, perhaps, not a greater remedy in medicine than bleeding; but good and effectual as it is, yet I will venture to say, there is none more pernicious than this, if injudiciously applied. As I value the life of my fellow creatures, I would at all times caution them against the injudicious application of this operation, being persuaded, that more have lost their lives by the lancet, than by the sword, and pestilence itself. Not to be tedious on this important subject, I shall give here some very few rules, whereby a patient may be judged in a proper state for bleeding; and likewise, where that operation ought to be avoided: for all the service we can possibly hope and receive from bleeding, is to lessen the quantity of blood, by which the whole body becomes relaxed, consequently less liable to the inflammatory irritation, that a too great quantity of blood might occasion; but on the contrary, if the solids are but weak, and the blood poor, we only augment the evil by bleeding, which we would wish to remove, as thereby the system becomes weakened, and not able to disengage itself from the obnoxiousness it is incumbered with; and thus proves the very destruction it was intended to remedy. The notion of drawing off the bad blood by bleeding, is very erroneous and absurd; the whole is too intimately mixed, and will ever generate the same again till the disease is removed.

Bleeding is requisite

1st. When a strong robust person, of a full and sanguine complexion, by accident receives a heavy contusion; a broken limb; or wound, whereby the parts become inflamed; a fever like to ensue; and that the pulse becomes hard and full, the veins distended, &c. &c.

2dly. When such a person receives a great shock, fall, terror, or any other strong emotion of the mind; whereby the blood becomes rarefied, or threatens a fever.

3dly. In the beginning of pleurisy, peripneumony, sudden and violent fevers, great colds, &c. as also scalding, burning, apoplexies, convulsions, palpitations, suffocations, and all such dangerous violent disorders; in these cases only bleeding is useful, and then it ought to be performed immediately, and with great moderation: six or eight ounces is enough from the strongest man; if requisite, it may be easily repeated, but if overdone, it is not so easily replenished.

Bleeding in general is hurtful

1st. In agues, notwithstanding the violent paroxysm of the hot fit, because the solids are here too much relaxed.

2dly. In all contagions distempers; this is a circumstance worthy the greatest attention. Reason and experience prove that bleeding here is very improper; for by bleeding, the contagious miasmata is drawn only the more into the whole mass of blood, and this is the fatal stumbling block, by which thousands have lost their lives. The manner of treating such distempers with success, I shall shew in its proper place.

3dly. In all old standing diseases, where there is a low, weak, though quick pulse; for in such cases bleeding is very improper, as the system is already too much reduced.

4thly. In all dropsies, scurvies, lentors, consumptions, &c. for here instead of bleeding and lessening the power of the solids, the patient wants strengthening, and has no blood to spare.

5thly. In all rheumatic and gouty complaints; for here nature must be assisted in order to throw off what is painful, by such means as will strengthen the solids, expell what is obnoxious, and prevent it from returning to the blood, or falling upon some more important part.

6thly. In all venereal cases, particularly if recent; as by bleeding, the pox will unavoidably be the consequence; as the venereal virus will be absorbed in the whole mass of blood.

7thly. In all paralytic cases, and such weaknesses where the strength of the body is already impaired.

8thly. *and lastly*, In the time of other evacuations of the body; as purging, fluxes, bloody fluxes, &c. and particularly when the body is in a sweat, or perspiration. From this, I hope, the good and bad effects of bleeding will be understood in many other cases, which would be too tedious to mention here; I shall therefore proceed to the operation of bleeding itself.

SECT. III.

Of the Operation of Bleeding.

Bleeding is thought in general to be a very simple performance; but in fact, there is not an operation in all surgery, requires more care, or is liable to more dangerous consequences, if ill performed.

It is not only the opening a vein, but it is to open it properly, (taking care not to prick an artery or tendon) to take away a sufficient quantity of blood, and to heal up the orifice again.

In the bend of the arm are three distinct veins, the *head*, *median*, and *basilican*. The ancients, before the discovery of the circulation of the blood, fancied the *head*, or upper vein, carried the blood from the head; the *median*, or middle, from the breast; and the *basilican*, from the liver, &c. but this idle conjecture is laid aside, as being highly absurd. When it is remembered what was said in the first Lecture, in Sect. V. of the circulation of the blood, it will then plainly appear, that all these veins become one before they enter the body; so that there can be no difference as to their quality in being opened: either therefore, which lays most conspicuously easy, is the vein that should be chosen; for the arms of people differ in this respect, as much as their features.

In order to chuse therefore properly, examine with the finger how the veins lie; if upon a flat hard substance, it is a tendon; if a pulsation is perceived, there is an artery; both which ought to be avoided if there is a vein that lays more free and conspicuous to the sight and touch. If however it cannot be avoided, the vein ought to be opened with the greatest caution, for the pricking of a tendon would not only be exquisitely painful, but endanger the arm by inflammation, and even life itself; and the pricking of the artery would endanger the patient's bleeding to death, or else form an *anauresma*, (which is a bag of blood protruding from the artery) equally dangerous to the life of the patient. In general the middle vein is the safest and most convenient, provided it does not cross the tendon: for the upper

vein lies frequently very deep, and is but small, and the lower frequently crosses a tendon, or lies immediately over an artery.

However, one or the other will frequently suit, and, with caution, the operation may be performed with ease.

As to the choice of the arm, it can make no difference in regard to the effect, whether it is the right or left, for the veins of both arms end at last in the great trunk of the *vena cava*, before they enter the heart. To the operator, however, the right is the handiest, and to the patient, the left is most convenient. In that arm however where the best vein offers, that should be chosen.

When the arm is fixed on, then tie it up. The place should be on the upper arm, about two inches from the bend, and the vein fixed on to be bled, first drawing up the skin a little, and laving on the ligature twice round the arm, drawn together with a draw knot; the reason of laying on the ligature above the intended orifice of the vein, is plain, as it stops the reflux of the blood, which is transfused from the heart by the arteries towards the fingers, from thence taken up by the veins, and by the ligature prevented from flowing back again, and consequently distends the veins, and swells them up.

When the ligature is laid on, (which may be a garter, or a piece of tape, about four or five feet long) and the veins are swelled up, then open the vein. Bend the lancet so that the blade and handle may make an acute angle; take the blade between the thumb and fore finger in the right hand, if the right arm is bled, but if the left arm, the lancet ought to be held in the left hand,^[9] and open the vein obliquely, so that the lancet cuts all the while, stretching the vein a little downwards with the opposite thumb. The lancet should cut a little slantingly, for if perpendicular, it would not only cause much pain, but be apt to slip through the vein, prick a tendon or artery, and occasion dangerous mischief.

9. This is the proper method of bleeding; for a bleeder ought to be equally dexterous with the left hand as with the right, which custom soon makes familiar; but those who are awkward with the left hand, must stand behind the left arm to bleed, which method may be adapted rather than to risk danger.

The orifice should be somewhat obliquely in the vein, neither directly length-ways, as the veins would be apt to slip, and the orifice not sufficiently opened to let the blood pass freely; neither should the opening

be right across, as thereby the vein might be divided, and the blood not be easily stopt, or the orifice heal up. When now the vein is well opened, let the blood run, (the arm a little bent) till it changes to a higher colour, the surest sign of being enough, and this will be in about four, six, or eight ounces, according to the constitution of the patient. This simple rule I would advise to be attended to, for whatever may be pretended about presaging the quantity of blood beforehand, certain it is, it cannot be ascertained but in the time of bleeding.

When it is time to leave off, slacken the ligature, and gradually untie it, when generally the blood ceases to run; draw close the orifice, and wipe it clean all round. Let the patient bend the arm a little, and close the orifice very close, then lay a four-folded dry compress, the bigness of a crown, of clean tolerable fine linen rag, and tie the arm up: lay the ligature (a garter or a piece of tape about four feet long) at the middle of the compress, the one half round the upper arm, the other on the lower arm, alternately, so that they always cross each other in the bend of the arm on the compress, till at last it is either tied or pinned; care being taken that it neither be too tight or too slack, so that the arm may be bent or extended with tolerable ease. Let the ligature remain on twenty-four hours, when all will be closed up.

What is to be done if unfortunately a tendon or an artery should be pricked, I shall mention hereafter.

SECT. IV.

On the requisite Medicines to be used at Sea

As my greatest ambition in this life tends only in becoming useful, I shall studiously avoid that false pride of appearing learned.

There is perhaps not a medicine in the modern *materia medica*, but what I have examined into, and opportunely experienced the virtue of; but so often have I been deceived in my expectations, that out of the many hundreds, with which I am acquainted, there are not above a score, or at farthest two, which in any case whatever I should expect any benefit from. Yet I mean not to throw out prejudices, for every physical man has his favourite medicine, and we are all apt to praise the bridge that carries us well over.

The little assortment of medicines however, which I here offer, I know from experience to be good, and as such I recommend them. I have plowed the ocean myself, visited most climates in the known world, and in my practice on that precarious element, drawn such observations, as makes me happy to think, that they may prove useful to a class of men, who daily experience the uncertainty of human life.

In the first place I shall draw the plan of a sea medicine box, as may suit a ship that carries no surgeon; in which I shall endeavour to be so plain as I hope will prevent any mistake.

The MEDICINE CHEST.

1 { Lancets Scissars Probe Spatula Teeth Instruments 2 Syringes 3 Clyster Pipes & Bladder 4 Scales and Weights	A. Spt. of Wine and Cam- phor.	B. Liniment of Soap.		C. Sweet Oil.	D. Sweet Spirit of Nitre.	11 Camomile 12 Balm 13 Sage 14 Sassafras 15 Oatmeal
	E. Honey.	N. Calom.	O. Fine Prec.	P. Me. Oi.	G. Elixir of Vitriol.	
5 Splints 6 Bandages 7 Rags 8 Tow 9 Lint 10 Tapes, Thread, Nec- cles and Pins	F. Yellow Bafi- licon.	Q. Rhub.	R. Jalap.	S. Ipecac.	H. Elixir Pro- prietat.	16 Cinnamon, &c. 17 Allum 18 Chalk 19 Salts
	L. Iinitive Electuary.	T. Liq. L.	U. E. O. Pep.	W. T's. Ba.	M. Stomach Powder.	20 Diachylon Plaifter 21 Mercurial Plaifter 22 Bliftering Plaifter

Of the Utensils.

First then, I shall speak of the utensils and the instruments which must necessarily belong to a medicine box, be it ever so small.

In the box, No. 1.

Lancets: There ought to be three or four, and they should always be kept clean and bright in a case, free, as much as may best from the dampness of the sea air. The best method for this is, to wipe them often with a dry clean linen cloth, warmed by the fire, (taking care in the wiping of them, that the blade of the lancet lays upon one of the handles, by which the point is preserved) and when properly cleaned, wrap the whole case up in a large piece of dry paper. Let it be a constant rule, to clean them well every time they are made use of.

And here I must remark once for all, that all iron or steel instruments are ever best preserved clean and bright, by rubbing them clean with a warm dry cloth, and then wrapping them up in clean dry paper; by which method I have preserved my capital instruments with the same brightness, as they came out of the workman's hands throughout my voyages, notwithstanding they have been made use of, and frequently too.

The notion of oiling or greasing instruments is very erroneous, and so far from preserving them, that it will never fail of making them more rusty. The wiping them quite clean and dry, and wrapping them up in warm dry paper,

is certainly the best secret for keeping all iron or steel bright at sea, and deserves strictly to be attended to.

Scissars: There ought at least to be two pair, one for common use, and the other for particular purposes; and if the last mentioned are crooked, they are certainly much the better, being very handy on many occasions. The best pair should be carefully preserved as already directed.

Probe: A probe is very requisite for searching of wounds, &c. It should be of good silver, and the smaller it is in thickness, even like a small wire, the better for use, as it will bend and twist agreeable to the place it is to search. At one end it should have a neat smooth button, and at the other, be triangularly pointed, like a sail needle, in order to wrap lint round it to clean sores with; its length should be six or eight inches.

Spatula: 'Tis not much matter what the spatula is made of, whether silver or iron, be it what it will, (provided it is neither brass nor copper) one end should be flexible, so as to be handy in spreading salves upon lint, or for other the like purposes; whilst the other should be more stiff, being very useful for pressing the tongue down, in looking and examining the throat. It should be about six or eight inches long, and an inch broad.

Bistouries and Incision Lancets: A bistoury, or neat sharp cutting knife, is very useful indeed in many cases. And a large incision lancet, in the form of a common lancet, except that the point inclines more one way, is very necessary for opening aposthumes, and where a collection of matter has been formed.

Teeth Instruments: Of the whole tribe there is certainly none more universal than the key; this ought to have three different hooks, and so that they may shift and unshift, according as the operation requires. The gum lancet I have no notion of, I never saw any occasion for it, and am apt to think it rather an hindrance to the operation. The instrument should be kept clean in the same manner as I have said before.

2. *Syringe:* Each box ought at least to have two. Ivory is the best substance they can be made of; they ought to be pretty thick and stout, as they are liable to warp. A small one with a neat pipe, slender, and a button to it, will be very useful for the clap, or other cases, where injections are

requisite. A larger one for the ulcerated sore throat, or other the like cases, where quantities are required to be injected.

3. *Clyster Pipes* and *Bladders*: There ought to be sundry. I would have them previously mounted; and to prevent the bladder from being decayed, or eaten up by vermin, or moths, fill them up with wormwood; a method I have found very effectual.

When they are to be made use of, they are easily softened by a little lukewarm water.

The general method of mounting the bladder on the pipe, is, by the assistance of a cork in the pipe, which is to be drawn by a string, when the clyster is injected; for my part, I found it always a hindrance; and therefore instead of the cork, I always chose to secure the whole with a string, tied with a slip knot, which when the pipe is introduced, is much more easily undrawn.

A *clyster syringe* of pewter, that holds at least a pint, or a pint and an half, is far more preferable at sea, and might easily be made with an additional pipe, by which a man may be the operator himself.

4. *Scales* and *Weights*: This is a requisite article, in order to proportion the doses of medicines. The scales need scarce any description, as they require no more than their being of an equal length, and the whole in equilibrio.

The *Apothecaries weights* come in the following order, and signified by these characters:

- 1℔—A Pound is 12 Ounces.—℥xii.
- ℥i.—An Ounce is 8 Drachms.—ʒviii.
- ʒi.—A Drachm is 3 Scruples.—℥iii.
- ℥i.—A Scruple is 20 Grains.—gr.xx.

These marks are generally stampd upon the pieces. As a pound and ounce are not generally used in the proportioning of doses, they are omitted in the common little boxes prepared for that purpose. The rest, drachms, scruples, and grains, are marked upon the requisite pieces; and which at the first inspection may easily be understood.

I shall next proceed to another division of the chest or box.

5. *Splints*: Of those there should be at least four pair; two pair for broken legs, and two pair for broken arms. They are generally made of very thin deal boards, pasted on leather, and slit, so as to bend to the parts applied, sideways, but remain stiff length-ways.

Paste-boards may however be made use of with good success; and if the paste-boards are kept in sheets, they have the advantage, that splints may be made of them occasionally; but by fomentation they are apt to become soft and flabby, which ought to be guarded against.

6. *Bandages*: There ought to be three or four long ones, two or three yards each, torn length-ways off an old sheet, not too much worn, stitched together so as not to make a seam, three fingers broad, and neatly rolled up, so that they always may be ready for use.

7. *Rags*: are a most requisite article at sea, and I can only say, the more there are the better. They should by all means be clean, and neither too coarse nor too fine.

8. *Tow*: is also a very requisite article. A good large bundle should be provided, and it should be cleared from the shaggins, as they render it stiff and unpliant: the grey is generally the best for medicinal purposes.

9. *Lint*: should also be somewhat plenty; but as this may be made at leisure, the quantity of that is not so material. Lint in the common method is best made from rags torn length-ways, about three fingers broad, and the transverse thread drawn by the help of a knife.

10. *Tape*: is very useful, particularly that of two fingers breadth, and ought to be pretty fine. At least a dozen of yards should be provided.

Needles, threads, and pins should by no means be omitted, being highly useful on many occasions.

Secondly. I shall now proceed to the herbs, which need but be few, as they are very subject to decay.

11. *Camomile flowers* are much used by some, and supposed to strengthen the stomach; I could never think them of that effect. When however a vomit is given, camomile tea is very useful, and promotes greatly the operation; but it should not be made too strong: as much as may

be taken between three fingers, pouring on it a quart of scalding water, and thus made into a tea. But that kind of tea should not be too often repeated, unless it particularly agrees with the stomach.

12. *Balm*: This is an excellent herb on many occasions; it is cooling, and there is not a tea I have more to recommend to sea-faring people, let their ailment be what it will; it is of a balsamic and healing nature, and will not easily pall on the stomach, except where there is an universal weakness in the system; where then what is more warming to the body than balm, may with propriety be added.

13. *Sage*: This is an herb of a more hot nature; and in fevers should be used with caution, unless where a transpiration is required. It promotes urine and perspiration, in many cases it may be used with the balm, and then makes a very good tea, both for the sick as well as those in health.

14. *Sassafras*: This is one of the warming and balsamic woods, which, if it is mixed with *lignum vitæ chips*, makes an excellent decoction for all rheumatic complaints, and where the blood wants to be diluted or purified.

15. *Oatmeal*: This article cannot strictly be deemed a medicine, yet, as water gruel is frequently wanted, that of oatmeal I think the best; besides, it is excellent for making poultices, being of a softening and rich quality. But if it should be wanting, *pounded biscuit* will answer the purposes. In regard to diet, I shall say more in its proper place.

16. *cinamon*: This is a necessary article at sea, as it is of a gentle astringent nature, together being very warming, and is undoubtedly the best of all the spices, where the bowels are weak. Its peculiar uses I shall speak more of hereafter, when I come to treat on loosenesses and fluxes, and such diseases where it is useful.

17. *Allum*: This astringent medicine is very useful for gargles, for scurvy gums, and various other uses. Burned allum (which may be done by exposing a piece on a fire shovel over the fire till it is bubbled up and become of a white cake) is an excellent medicine, for gently checking proud flesh in ulcers; besides which, it makes an excellent tooth powder.

18. *Chalk*, is as necessary an article as any medicine, and so useful is it in long voyages, that a ship should be well stored with it; for it is not only an excellent absorbent, and will stop fluxes, when made use of in decoctions

for that purpose, but, makes also a great purifier of water, and thereby contributes greatly to the preservation of health; as I shall hereafter further take under a stricter consideration.

19. *Salts*: Glauber's purging salts, is a medicine which has had many virtues ascribed to it; as a cooling purge it is very well, but further I cannot recommend it: An ounce, or an ounce and an half is the dose; and it will be best to dissolve it in a tea cup of water over night, so that it may be taken early in the morning, and worked off with drinking gradually some tea after every motion.

20. *Diachylon plaister*:

Take lytharge fine prepared one pound, oil of olive one quart, boil them over a gentle fire, putting into the pan a little water, stir it all the while, and take care it does not burn; continue boiling, till it becomes of a consistency of a plaister, and make it into rolls; which, according to art, is best done before it is quite cold, upon a wet marble slab, and wet hands, and then put up in paper, previously rubbed over with some soap, to prevent it from sticking.

There is not a medicine so generally useful for all hands on board, as this very plaister, in case of cuts, bruises, or sores of any kind; it is not only the medicine box that should be well provided with it, but every man on board, I advise to have a roll, and some spread on cloth in his chest; for in every little accident, a cut, a broken nose, or broken shin, &c. to which a seaman is liable, nothing is handier, and nothing is more beneficial than this plaister, as it will prevent inflammation and festering, heal it up, and prevent small sores from becoming great ones.

21. *Mercurial plaister*:

Take of the diachylon one pound, purified mercury half a pound, triturate the mercury with a little hog's lard or turpentine, then melt the diachylon, and incorporate the triturated mercury into it.

This is principally fit for venereal cases, for buboes, and other hard tumours; it may also be put to shankers and very bad ulcers; but it should never be used unless necessity requires it.

22. *Blistering plaister*:

Take Spanish flies eight ounces finely powdered, common drawing plaister one pound, melt the plaister and sprinkle the powdered Spanish flies till all is well incorporated, then make it into rolls according to art.

This is intended for rising blisters. In regard to the use of this, I have only to observe, that it should be made use of only, when the greatest necessity requires it; that is, in bringing on a crisis of a fever, which I shall point out in its proper place. At present I shall only speak of its application. Whatever place is intended to be blistered, either the neck, between the shoulders, or the calf of the leg, let the place first be cleaned with a little vinegar, and rubb'd till it becomes red, then let the blistering plaister, the bigness of the palm of the hand, pretty thickly spread, (best upon leather) be laid on the part, and remain there for about twelve hours, and be tied on, that it may not shift: at that time the blister will rise; let it be cut to let the water out, (but care should be taken not to pull off the skin, as is frequently practiced) dress it with *Turner's Cerate*, spread upon a thin rag, (or what is still better, dry lint) and let that dressing be shifted once every twelve hours, till all is healed up.

I shall now proceed to the inner part of the chest.

A. Spirit of wine camphorated.

Take spirit of wine rectified one quart, camphor two ounces, mix it, and let the camphor be dissolved in it.

This is an external application, for bruises and other inflammations, where discussion, or dispersion of humours is necessary, its chief benefit is in sprains, dislocations, and fractures. I have this however to observe with the camphorated spirit, that though it is exceedingly useful in the forementioned, and in some rheumatic cases, yet it should be used with caution, and not too plentifully, as it is liable to dry up the vessels, and might incline the limb to wasting, if too freely made use of.

B. Liniment of soap.

Take spirit of wine rectified one quart, distilled oil of rosemary one drachm, camphor one ounce, castile soap half a pound; cut the soap small, and let the whole be mixed and dissolved.

This is an excellent external medicine for sprains, rheumatism, bruises, &c. This medicine is commonly called opideldock, and in all such cases is

far superior to the spirit of camphor singly; unless where the constitution is sluggish, and requires a greater stimulus; in general, however, this excellent composition will supply the place of both.^[10]

10. Since these compositions are so easily made, they might occasionally be prepared on board; Instead thereof, I would advise the seaman to furnish the box with camphor and soap; and instead of rectified spirit, good rum or brandy will answer equally as well.

C. *Sweet oil*: This ought to be pure and good; it is of many uses in medicine, which I shall treat of as occasion requires.

D. *Sweet spirit of nitre*.

This is a spirit distilled from spirit of wine one quart, and spirit of nitre half a pound, according to the art of chymistry.

In regard to its virtue, it is an excellent medicine in most acrimonious cases, and acts principally as a gentle diuretic; besides this, it is of an antispasmodic and antiputrescent quality, and therefore useful in all kinds of inflammatory disorders, particularly of the putrid kind of fevers; which I shall, in the course of this work, take notice of.

E. *Honey*: This is a useful article, which ought to be plenty, and of the best kind; it is particularly useful for making gargles, clysters, poultices, &c. &c. as shall be farther specified.

F. *Yellow basilicon*:

Take olive oil, yellow bees wax, yellow rosin, Burgundy pitch, of each half a pound, turpentine two ounces; melt the whole, and mix them properly.

This is an external digestive application to cleanse ulcers, and to make them discharge good matter, by which the inflammation will abate.

G. *Elixir of Vitriol*:

Take stomachic elixir one pint, oil of vitriol four ounces; mix them, and filter it through paper.

This is an excellent elixir for weak stomachs, that proceed from bile and other causes of indigestion; it is also a great specific against the scurvy, and the like habits; as also in feverish complaints. The dose is from 20 to 40 or 60 drops, in a little water.

H. *Elixir proprietatis*:

Take of myrrh in powder one ounce and an half, succotrine aloes in powder an ounce, saffron four ounces, of dulcified spirit of vitriol six ounces, of rectified spirit of wine twenty-four ounces; digest them in a sand heat for four days, and then pour off the elixir from the dregs.

This elixir is an excellent stomachic, it will promote digestion, strengthen the stomach, and create an appetite; it will also keep the body gently open, and on that account be of great benefit against many ailments in the bowels. The dose is about a drachm or two, or a middling spoonful; to be taken either by itself, or in a glass of wine, and makes a most agreeable bitter.

I. *Lenitive Electuary*:

Take dried figs one pound, tamarinds, cassia, french prunes, each half a pound, sena leaves eight ounces, coriander seed four ounces, liquorice root three ounces, double refined sugar two pounds and an half; reduce the sena and coriander to a powder, and sift it through a sieve, boil the figs, cassia, prunes, and liquorice into a pulp, and strain these also, then mix the powder gradually amongst it, and make it, with the help of the sugar, into an electuary.

This electuary is a gentle cathartic, keeping the body coolly open, and therefore very useful in fevers, and weak constitutions when openness of the body is required. The bigness of a large nutmeg is a dose, which may occasionally be repeated.

K. *Turner's Cerate*:

Take olive oil a pint, yellow bees wax and prepared calamine stone, of each half a pound; melt the wax and oil, and mix the calamine stone: keep stirring till it is cool.

This is a healing cerate, very useful in scalds and burns; as also to dress blisters with; likewise for the external dressing of sores, when near healing, or to keep other dressings on.

L. *Fever powders*:

Take purified nitre powdered half a pound, crabs eyes prepared four ounces, cinnabar of antimony finely prepared two ounces, calomel one drachm, and mix them.

This is the celebrated antispasmodic powder of the great *Stahl*, with some considerable amendment by means of the calomel; which, in the course of a long practice, I have found the most beneficial for fevers. There is not a kind of fever to which mankind are subject, either at shore or at sea, but what this powder may with safety and with great benefit be applied. A scruple, or half a drachm is a dose, and which may be repeated, as the ardency of the fever shall require.

M. Stomach powder:

Take purified nitre prepared and chalk, of each eight ounces, nutmeg two ounces; reduce all to a fine powder, and let them be properly mixed.

This is an excellent powder against the heart-burn, or other crudities of the stomach and bowels; for whilst it absorbs the crudities and accidities in the first passage, it is at the same time a gentle digestive, and withal a carminative: Hence in heart-burn, belching, and other disagreeable symptoms of depraved appetite it is highly useful.

N. Calomel:

This is a dulcified mercury, made of sublimate mercury four ounces, pure quicksilver three ounces; intimately mixed, and sublimed six times, according to the art in chymistry.

This is the safest and gentlest of all mercurial preparations, and if judiciously administered, may be deemed one of the greatest of all medicines. I shall therefore be somewhat particular on this head. In the first place, it should be faithfully prepared; and secondly, levigated as fine as is possible; to do this therefore, I have in my Treatise on the Venereal Disease, recommended it to be ground with a little water on a flat marble, to the finest degree possible, and then dried up for use. The fineness of this excellent medicine is of the greatest importance, and ought to be strictly attended to, for not only much mischief has been done by its rough particles in the bowels, but it must follow, that the finer it is, the farther it goes. And with all mercurials, it should be strictly observed, the less quantity we can make do, the more we may hope for success.

There is not an acrimony in the human body but what may be corrected by this universal antacid medicine, if properly administered. In venereal cases, it should be one of the first, and the principal remedy throughout the

cure; but care should be taken that the quantity be small, and adapted to the strength of the patient, and to prevent it from falling on the salivary glands, and thence to produce salivation. The dose should never exceed one grain, a quantity sufficient for twenty-four hours, except where there is evacuation, or other discharges of the body required; in which case it may be occasionally increased, especially in malignant fluxes, or when purges are given, or in other evacuations.

It may be made up in many forms; the best, however, is in pills, with a little flower and water, &c. or it may occasionally be mixed up with other medicines, as I shall frequently have occasion to speak of.

O. Precipitate:

This is made up of equal weight of Quicksilver and compound Aqua Fortis, well mixed, evaporated to dryness, in a broad bottomed vessel, by a sand heat, according to the art of chymistry; in which operation it turns red.

This is an excellent medicine, applied to all kinds of ulcers, as it will correct the malignity of the matter, take off fungous flesh, bring on a kind suppuration.

It will also clear the skin from all kinds of breaking-out, and perfectly destroy lice, nits, &c. It should, however, like the former, be levigated very fine, for thereby it loses greatly its sharpness, and becomes more efficacious. It should also be mixed with some cerate, ointment, or other dressing, and never, or very seldom, be used by itself, as I shall hereafter point out.

P. Mercurial Ointment:

Take hogs lard eight ounces, purified quicksilver one ounce, Venice turpentine two drachms; rub the mercury into the turpentine in a mortar, till the quicksilver disappears, then mix the whole together.

This is popularly called Unction, and peculiarly useful in venereal cases, as also to destroy a particular vermin called *Crab-Lice*, and other foulnesses. But I must here give a caution against the too free use of mercurial ointment, as it is not only apt to bring on salivation, but if the mercury is adulterated with lead, (as sometimes is the case) it is liable to occasion many incurable pains in the limbs; and by this the remedy may prove worse than the disease.

Q. *Rhubarb*. This is a necessary article in all kind of fluxes, and relaxations in the bowels; for, besides that it purges gently, it has withal such an astringency as to brace up the weakened vessels, and to restore their proper tone again.

The dose of rhubarb is one scruple; which should be mixed with some fine chalk, or *Stomachic Powder*. (M.) about equal quantity, particularly in the beginning of fluxes.

R. *Jalap*. This of all purging medicines is undoubtedly the best, when purging is required. The dose is from a scruple to half a drachm, mixed up into a draught, with a little syrup and water, or any other form. A grain or two of calomel is ever a proper addition, let the case be what it will, that requires purging.

S. *Ipecacuanha*. This is the best of vomiting medicines; to be given about a scruple, either in form of a little draught, or in a bolus.

Vomits are often of more benefit than purges; not so much on account of cleansing the stomach, as by the agitation it gives to the whole body; whence it is brought into perspiration, and thereby promotes the circulation of the fluids that was obstructed.

T. *Liquid Laudanum*:

Take purified opium one ounce; cinamon and cloves, of each one drachm; white wine one pint, steep them for a week without heat, and filter it for use.

This is a medicine of much use in painful and restless diseases, but should never be had recourse to, unless the greatest necessity require it. For, whilst it lulls the nerves from their irritation, it is apt to weaken their tone, and by that, greatly lessen the strength of the whole body; but where really requisite, it is a happiness that ease can be given, by which to relieve nature in pain that it may be restored again to tranquillity. The dose of this tincture is from 10 drops upwards to 30.

U. *Essential Oil of Peppermint*: This is one of the most grateful nervins in the whole materia medica; and at the same time the least stimulant. In all cases where there is weakness and feebleness, I would advise this in preference of spirit of hartshorn, or any other stimulant whatever. It warms the stomach, comforts the bowels, and infuses a liveliness over the whole

nervous system. The dose is from 3 or 4 to 6 drops, on a little lump of sugar, and dissolved either in a glass of wine, or water; which may be repeated occasionally, without the least hurt to the constitution.

W. Turlington's Balsam:

Take St. Johns worth a small handful, rectified Spirit of wine two quarts; digest them for a few days, then strain and filter the extracts; then take Angelica root, Olebanum, Myrrh, and Socotorine Aloes, of each one ounce; Styrax, Benzoin, and Peruvian balsam, of each an ounce and a half; digest the whole in a slow heat for some days, and filter the balsam off for use.

This is the genuine receipt of the celebrated *Turlington's Balsam*, and an excellent medicine it is, for many purposes. Externally, it is an efficacious remedy for all green wounds, cuts, bruises, &c. if immediately applied.

But particularly for internal uses, its virtues are great: It warms the stomach, promotes good digestion, enriches the blood, warms the whole system, and, above all, is a sovereign remedy in all rheumatic complaints. The dose is about a tea spoon full, in a glass of wine, or any other convenient vehicle; and on all occasions, where bitters are made use of, this ought to have the preference, being one of the best stomachics. But at the same time it should be observed, that where there is a fever in the system, it ought to be avoided, as it is remarkably heating in its nature, and therefore agrees best with cold constitutions, and where the habit is weak.

Here then I shall end this lecture; and only observe, that though there are various other medicines of the greatest importance, as well for external as for internal uses, I have purposely avoided them, as their application requires more care, than what we may expect from men who have not made medicine their peculiar study. At the same time I must also confess, that the forementioned will be sufficient, if managed as I have already intimated. It is a maxim with me, 'tis not so much the medicines themselves whereon the success depends, as the judgment requisite to know how and when properly to make use of them.

I must likewise take notice, that I have purposely adhered as close to the recipes of the Dispensatory as I could with propriety, because it may the better suit the apothecarys' shops in all parts of the world; and that when a medicine is wanted it may the easier be supplied. It would have been an

easy matter for me to alter every medicine in its composition; but I conceive this would have tended but to raise difficulties in procuring them, and taken away from the general utility of the book.

LECTURE III.

OF

ACCIDENTS,

AND THEIR

Proper Method of Treatment.

I Shall now proceed to the actual assistances that the human body stands in need of, when the order of health is impaired. This is the noblest office, in which mankind can possibly give aid to each other.

Health is certainly the greatest blessing this side of the grave; and what service can we render to our fellow creature, so valuable, as to assist him in the hour of distress.

From this consideration it naturally follows, what caution it also requires, and how circumspect every one ought to be, who takes upon him that noble office.

Before therefore I begin this important lecture, let me request of the reader the strictest attention, and never to act before the ailment, as well as

the method of treatment is thoroughly understood; since nothing less than health, and even life itself, is liable to be the forfeit of a fatal error.

Mankind are ever liable to accidents, how cautious soever their conduct in life may be: But seamen, particularly are more so; not only from their occupation, but from the precariousness of the element on which they are dependant, and which so often exposes them to the most imminent danger.

I shall divide the accidents of bodily dangers, into falls, wounds, bruises, dislocations, fractures, scalding, burning, and drowning; and treat on each part separately, in as plain a manner as is in my power.

SECT. I.

Of Falls

The man who falls, is partly bereft of his senses, or his presence of mind, and particularly so the more timorous he is of his own preservation; add to this, because he is unaccustomed to it; for custom makes us familiar in every thing: Thence we perceive the different effect in different persons in their falling; for some will fall with a good deal of judgment, prevent some unlucky blow or other; whilst another shall fall as heavy as a dead log, or like one drunk, without having the least chance of saving himself in any shape. I remember once I fell from the puttock shrowds of the fore-top, but providentially escaped without the least injury, my watch chain having caught some how, which I endeavoured to save. I confess this in a great measure was owing to good fortune; at the same time I was naturally very active, and having acquired a flight of jumping off the main-deck into the hold of a light ship, I thereby had habituated myself to stop my breath, and a presence of mind which principally saved me when I thus fell in good earnest. I only mention this to shew, that the greatest mischief in falling proceeds from timorousness and surprise. I knew a wag, who would play pranks that way to astonishment; he could let himself fall from any part of the rigging, catching as he came down like a cat, make all hands come round to his assistance, and then laugh at their credulous good nature.

A man who falls is apt to lose his breath, particularly if he is not very careful, and has not presence enough to stop it the moment he finds himself fall; and the instant he loses his breath, he loses also his mental faculties; consequently comes motionless and exposed to every fatality. Thence we generally find a man, who has fallen from a considerable height, lays motionless like death on the spot, even though he has not received the least injury otherwise.

The Method.

When I have been called to such an accident, and found my patient motionless; I have untied his neckcloth taken him by the collar of his jacket, and shaken him heartily; which in the space of half a minute has brought him to, with a heavy sigh. Next I have bled him without loss of time, but not too copiously; after which I have examined him, and acted according to circumstances.

A person falling is liable to a number of dangerous consequences, not mentioning immediate death. The consternation, the shock, fright, terror &c. are as alarming and dangerous as the external hurt itself. The brain, and the blood vessels in that part are very delicate, and by the shock, as well as the want of respiration, they often burst; thence an apoplexy is liable to hurry the patient from the stage.

A fever generally attends a fall, which ought to be strictly attended to; after, therefore, the patient is bled, and seems to have recovered his reason, he should have every six hours a dose of the *Fever Powders*, (L.) and drink some balm tea, in order to get into a gentle perspiration, which will in this respect soon recover his health again, and enable him to do his duty.

SECT. II.

Of Wounds

A Wound is an unnatural separation of any part of our body, whereby the order of circulation is hindered, the solids divided, and the fluids let out; suddenly caused by some violent means or other. But when such a separation becomes ulcerated, it loses the appellation of wound, and is called a sore, or ulcer.—This distinction is requisite to be taken notice of; for wounds and sores are often confounded, which shews not only ignorance, but as their nature is materially different, their method of treatment is not less so; and ought therefore previously to be understood.

Wounds are best distinguished into three different kinds; cut wounds, bruised wounds, and gun-shot wounds. I shall briefly shew how to treat each in particular.

Cut Wounds.

A Cut wound is barely a separation of parts; and the sharper the instrument with which the wound was made, the easier will the wound be to heal up again.

It is however a natural thing to observe what parts of the body is wounded, for tho' nature is kind in uniting the parts, yet some parts are more easier healed than others.—Observe therefore

The different Methods.

If an artery is wounded (which is known by the blood being of a florid colour and spinning out by starts) care should be taken to have it stopt to prevent bleeding to death. But in this you should not be too scrupulous. Two or three ounces from a wound makes a great shew; and very often a little blood lost is of great benefit to the healing. Should, however, the blood

gush from the wound violently, especially from an artery, make a compress of lint, strewed with *Powder'd Allum*, (17) which secure well on the parts with a roller, laying over it a compress, with *Spirit of Wine and Camphor*; (A) if that should not be yet sufficient, put a piece of money in a compress, and secure it on the wound, that is, provided the part of the body will admit of it, such as the arms or the legs, &c.

Let the ligature, or the first dressing, lie on for two or three days, and, if no hindrance to the patient, longer. If all these things are not sufficient, recourse must be had to the needle.

In every other kind of cut wounds, care should be taken to have it first cleaned with some water; (a small matter of vinegar, and brandy may be mixed with it) next, let the lips of the wound be well closed, dressed with dry lint, and *Turlington's Balsam* (W), and over which, a compress with *Camphor Spirits* (A).

Plaisters, salves, &c. in green cut wounds, are altogether needless. All that is required to healing, is to keep the wound clean, to give it rest, and not to aggravate it with any thing acrimonious. *Turlington's Balsam* (W) is admirable, but yet it suits small wounds better than large ones.

As a finger is very liable to be cut, and sometimes gives a great deal of trouble, I will here give a simple and expeditious remedy:—If you have cut your finger, wash it in a little salt or fresh water; tie it round with a good long thread, not too tight,—take care that your woolling may keep on without being removed; and a day or two will heal it up, without any farther plaistering. If inflammation attends the wound, methods should be taken accordingly, by bleeding, fever powders, &c.

Of Bruised Wounds.

These kinds of wounds are of a worse nature, and are not so kind to heal as the former; for here the parts seem to be torn to pieces, so that there is not only a separation of parts, but a destruction of a great many tender fibres all round the wound.

Hence we see in such cases a great deal of swelling, inflammation, and pain attend it; for the blood being hindered, the circulation naturally swells up the parts; and the nerves being lacerated also, must occasion great pain.

Hence it must follow, that nature must first get rid of the obnoxious and torn part before she will consent to unite the wound again; and this must be performed by suppuration.

In small cases, nothing is more beneficial than *Diachylon Plaister*, (20) which assuages the inflammation, corrects the acrimony, and brings the wound to a good state.

Broken shins often occur on board a ship, which is of that species of wounds. And as from such a case, (though in general slighted) many evils have arisen, I would caution the mariner to apply immediately *Diachylon Plaister* (20) to a broken shin; by which he will prevent having an ulcerated leg; a thing very troublesome.

If the bruised wound is on other parts, and the parts all round are bruised, care should be taken that it is cleaned as before directed, and a pledget of lint, with *Yellow Basilicon*, (F) will be necessary; over which a compress with *Spirits of Wine and Camphor*, (A) or rum, may be applied; and afterwards, when it begins to heal, it may be dressed with dry lint.

As these kind of wounds are liable of becoming ulcers, great care should be taken in the beginning to treat them properly.—Wounds in general should not be dress'd or look'd to too often; in general it aggravates the parts, and retards healing. A wound should not be opened the first three days after the first dressing: nature is very kind, and requires care only to replenish what is deficient.

When an ulcer has commenced, see the treatment under that head.

Gun-shot Wounds.

These are the most terrible of all sorts of wounds; for it is not only the contusion that attends them, but frequently the ball forces strange things, as cloaths, &c. into the wound with it.

Gun-shot wounds at best are tedious in their healing, but very often, from their nature, liable to mortification, and thence become dangerous. Besides, it is seldom that the fleshy parts alone are wounded, but they frequently penetrate into the very bones themselves.

The first care in gun-shot wounds, is to extract the ball, or whatever is forced into the wound; in the next place, the wound should be dressed with *Yellow Basilicon*, (F) mixed with fine *Precipitate* (O).

No. I

Take Basilicon one ounce, fine Precipitate one scruple, and mix them.

The parts all round should be well fomented with warm claret, intermixed with *Spirits of Wine and Camphor*.

If the patient is full-bloody, and no blood lost by the wound, he should be blooded, and immediately treated as a patient in a fever.

If the pain in the parts become violent, eight or ten drops of *liquid laudanum* (T), together with thirty or forty drops of *sweet spirit of nitre* (D), should be given once, twice, or thrice a day. If the wound begins to discharge plenty of matter, it should be dressed with the same dressing often. But if the parts become black, very foetid, the patient in great pain, and grows faint, a mortification is to be feared; which I shall treat of in its proper place.

When the suppuration is unkind, and the parts are much inflamed, a poultice will be very proper.

No. II

Take oatmeal (15) or (if that is not on board) biscuit pounded about three spoonfuls, honey one spoonful, water a sufficient quantity to boil it into soft poultice; in which put of sweet oil about half a spoonful. This poultice should be renewed twice a day at least, and put on as warm as possible it can be borne.

As gun-shot wounds are apt to degenerate into malignant ulcers, I shall refer the reader to the treatment of ulcers in general.

SECT. III.

Of Bruises.

Bruises, in whatever degree, have more or less some blood and other vessels torn and injured, and may properly be considered as wounds under whole skin. It is from the tearing to pieces blood vessels, whereby the blood extravasates from their natural channels, that occasions the swelling. If the bruise is but slight, the vessels are soon enabled to circulate the humours; but if the contusion is great, the extravasated humours are apt to corrupt, and make a collection of matter; thence an aposthume and ulcer must naturally follow.

Bruises on fleshy parts are not so bad by far, as on the joints, because the fibres in the muscles are more pliable, and sooner unite; the ligaments on the joints are very rigid, and extremely sensible; for which reason also, they are generally more painful and tedious in their cure.

The first intention is to disperse the extravasated humour, and to strengthen the tone of the injured fibres. To this intention the parts should immediately be bathed with *spirit of wine and camphor* (A). But if the bruise is immediately on the joints, the *linament of soap* (B) is still superior. A bandage over the part, so as to keep it in rest as much as possible, is also of great service. If, however, the part is so very much bruised, that there appears a collection of crumous blood, and that an aposthume is actually the consequence, recourse may be had to the above poultice in order to ripen it the more, and bring it the sooner to a head; and then treat it as a common ulcer. But that seldom is the case.

If the contusion is great, and the patient is full of blood, bleeding is necessary; for in such a case a fever is apt to ensue, which should be guarded against, and the patient treated accordingly.

SECT. IV.

Of Dislocations

There are many accidents by which a limb may become dislocated; the reduction of which is a material point in surgery. But if a surgeon is not at hand, a man ought nevertheless, not to remain in that painful and deplorable state, that endangers not only the limb of becoming useless, but even the loss of life itself.

I shall proceed therefore in my directions as plain as possible, avoiding every expression that may render this operation prolix.

It must first of all be remembered what was said of joints, articulations, and ligaments in the first lecture under that head. We must thence observe, that a dislocation of the upper arm with the shoulder is the most common that happens, because of the great variety of motion, and the flatness of the head of the arm-bone, and the cavity of the shoulder blade. Next to this is the thigh bone, with the hip. But this is done by much greater force, and therefore more troublesome in reducing. All other joints that are angular are seldom really dislocated, being stronger secured, and cannot be separated from their articulation, without first tearing the ligamentous capsula that surround them; yet nevertheless, some are subjected to be partly dislocated; and this is what is commonly called sprains.

Dislocations may be from two causes; first from violence, secondly from weakness of the ligament. In the latter case the reduction is the easiest, but the limb is apt to slip out again by the least accident.^[11]

¹¹. I remember a man who frequently had his shoulder dislocated, and was his own operator in reducing it again. The first time the accident happened in the ship I was surgeon of, I was naturally called to his assistance: "Stop Doctor," (said Tom) "I have got a tackle in my chest, and I will soon bouse it in again;" and sure enough, he had a pully, with which he immediately reduced it.

I shall briefly consider each kind of dislocation, that an accident makes a man liable to.

Dislocation of the Arm with the Shoulder.

The dislocation of that, may be two ways; (viz) downwards, and forwards; but it cannot be backwards nor upwards; except some part of the shoulder-blade be broke.—The best method to examine the patient what kind of dislocation it is, (especially to one unacquainted with these branches of surgery) is strictly to compare both shoulders, and the difference will be very plain, both to sight and touch.

If the dislocation is downwards that is, the head of the arm-bone, slipt in the arm-pit, there will be a cavity on the top of the shoulder, different from the sound arm, and the elbow will be drawn upwards, without being able to be moved either towards the body; or so, as the patient to put his hand out without the greatest pain.

When this is found to be the case, then, in order to reduce it, set the patient low, on a steady seat. Take a napkin or towel, of which make a girt; put one part over your neck, and the other part under the arm-pit. Let an assistant hold the patient steady, another take hold of the patient's arm, with the elbow bent, and extend it gradually; then with both your hands take the arm, and whilst you lift up the arm by help of the napkin resting on your neck, you press gently down the arm with your hands like a lever; in the mean time the assistant extends, and thus you reduce it; which you will hear by its snapping: upon which the patient will be able to move his arm in all kind of direction, without much pain.

If the head of the arm-bone is dislocated forwards, there will be a cavity backwards on the shoulder, and the elbow will stick backwards, and remain motionless, with great pain. In this case, as before, let the patient sit low, and with the help of the napkin, and assistances, first extend the arm forwards; then like a lever bring the head of the bone in its cavity; which, like the former, will be perceived by a snapping, and the voluntary motion of the patient's arm.

When the shoulder is reduced, secure the whole with a long double-headed bandage, cross-wide over the shoulders, and round the body. Foment it night and morning with *spirits of wine and camphor* (A). Keep the arm close to the body, with the elbow bent, and wear it in a sling, that it may be free from motion till it is well.

Dislocation of the Elbow, or the Knee.

The elbow, as I said before, is very seldom dislocated, unless fractured, and the ligament tore at the same time; and when so, the case is terrible. But very often those joints receive a wrench, which, though nothing out of its place, is often very painful, on account of the many tendons that surround the parts. The only thing in such wrenching, is to foment it with *spirit of wine and camphor* (A), or else the *saponaceous linament* (B). The same method should be observed with the knee, securing it well with a bandage, in order to give it rest to recover: For rest is a principal remedy.

Of spraining the Wrist, or Ankle.

The wrist and ankle are complicated joints, and thence capable of a great number of motions, and subject to a variety of sprainings, but not a perfect dislocation; nevertheless, they are very painful, and sometimes tedious in getting well. Whatever be the matter, the sick part should be compared with the sound, and the difference is soon seen.

If it so happens, that there appears a perceivable difference, the parts should be gently extended, and reduced; and all that is to be done, is the securing it well with a bandage, and fomenting it with *Spirits of Wine and Camphor*, (A) or, what is yet better, *Linament of Soap*. (B)

The thumb, however, is liable to be dislocated, which is easily perceived; which should be immediately reduced, and secured with a bandage, and treated as already directed.

Dislocation of the Thigh Bone.

The articulation of the thigh bone with the hip is similar to the shoulder, with regard to the head of the bone, and the pan in which it moves, except that it is much deeper. It must be a great force that drives it out of its articulation, and consequently very difficult to be reduced. It may either be dislocated inwards, or backwards.

If the dislocation is inwards, which is the most common, the toes will be turned outwards, and the whole leg and thigh will be somewhat shorter than the other; so likewise, if it is dislocated backwards, the leg and thigh will appear shorter, but the toes will be bent inwards. The best way of

determining, as I have said before, is to make a just comparison of the sick and sound leg.

The reduction is somewhat difficult, and will sometimes baffle the most experienced surgeon, because the acting muscles are very strong; and when they have first begun to contract the parts, they will not easily suffer themselves to be relaxed. Nevertheless, lay the patient upon his back or on the sick side, upon a steady table; let an assistant hold him fast, another extend the thigh, so as to have the knee bent; the operator having a napkin resting on his neck, and under the thigh of the patient, similar to the method directed before by the shoulder; and then with both hands bend the thigh, as with a lever, into its socket again; when it will give a loud snap, coming into its proper place.

It very frequently happens in this heavy dislocation, that either the articulation is not reduced, or that it immediately slips out again when reduced; so that the patient will never recover the former use of that limb more. In this case nature is very kind, and the part where the bone slips into will become callous, and form as it were a new acetabulum; by which the patient in time comes to walk, though (as it may be supposed) but lamely. In the course of my practice I have met with but two perfect dislocations of the thigh, both of them perfectly recovered; but it must be remembered, they had youth on their side, and the greatest care joined to favourable circumstances.

At best however, it is an ugly accident, for it is not only that the ligaments that surround the joint are generally torn, but very often the thigh bone is fractured at the same time. It requires more skill than one would suppose, to determine which is the case, whether a fracture, dislocation, or both; and am apt to believe they are often confounded with each other.

When the dislocation is reduced it should be well secured with a long double-headed bandage; the patient kept still, the part well fomented, and if requisite, some blood taken away: a fever generally attends, which should be duely taken care of.

On the whole, dislocations when they proceed from violence, are sometimes attended with severe symptoms, the parts being very sensible, and it will require sometime for the patient to recover his strength again.—

Care therefore should be taken to keep the parts warm, and give them as much rest as possible.

A strengthening plaister spread upon leather, and put for a constancy round the parts after the inflammation is gone of, will also prove of singular service.

SECT. V.

Of Fractures

We come now to fractures. There is something terrible in the sound of *broken bones*; and yet in common they are by far more easier managed and healed than a dislocated joint: nature is very kind; from the fractured parts of a broken bone, there exsudes a gelatenous matter called *Callus*, that very soon glews them together again, and makes them stronger than ever. As easy as a fracture is ruined, as easily it may be well treated. I shall therefore lay down some few rules, which if properly observed, will ever be attended with success, as they are founded upon happy experience, in my own practice.

In the first place, a fracture should be reduced as soon as possible. Secondly, the part should have as much rest as possible till the callus is formed, or the bone knitted together again. Thirdly, bandages should not be laid on too tight. Fourthly, camphorated fomentations should not be too plentifully made use of. Fifthly, the habit of body should be kept in a slate of health; and Sixthly, the diet during the time should be wholesome and sparing.

With regard to the reduction, it was formerly a great error for a limb to be stretched out in an extensive posture; for the extension of the muscles contracted the limb, and made the reduction both painful and troublesome.

I shall here recommend a better method; namely to have the muscles in a flexible position, during the operation and the cure. But to proceed to particulars.

A Fracture of the Skull.

This accident is frequently attended with many bad symptoms, and often requires the operation of the trepan; which I shall here pass over, being too prolix for this place. I have only to observe, that we have frequent instances

of the patient doing well without any other help than bleeding; and where there is no surgeon on board, this is the only remedy first to fly to, and gives the greatest chance for the patient's life: next to this, shave the head, and treat him as circumstances shall direct, without meddling with the fracture, any farther than applying gentle fomentations, with *flowers of chamomile* (11) boil'd in claret, and some *linament of soap* (B) added to it; and if there is any wound, have it properly dressed as directed in the article of wounds.

Fracture of the Arm.

If the upper arm is fractured, proceed in the following manner:

First of all be sure it is a fracture; to know this, put one hand on the middle of the arm, and the other on the elbow, move it gently to and fro, and if there is a fracture, you will find it crackle by the touch, which sometimes is so distinct as to be heard. The arm at the same time is somewhat swelled, and either the patient cannot move it, or else it is attended with very great pain.

When the fracture is certain, then proceed in the following manner; Let the patient sit upon a low chair, or chest, let an assistant hold him steady, and let another lift up the arm, and extend it gradually, horizontal to the shoulder, keeping the elbow bent somewhat all the while.

Then with both your hands gently press the fractured bone, so that it may feel perfectly joined again. Sometimes it happens that the ends of the bones are somewhat slipt over each other; in which case, extension is the more requisite.

When this is done, then take a bandage of about two yards long, first rolled up, and wrung out of *spirit of wine and camphor* (A.); begin near the elbow, and lay it smoothly on, laping over a little; and proceed gradually till it comes to the shoulder, and then return with it back again till it is done.

This bandage should not be too tight, nor too slack, but so as immediately to suit the arm exactly. Over this bandage again, lay a double folded rag, called a compress, so as to cover the bandage; and over this again two splints, one above, and another below; which tie gently together with some tape in three places. Then put the arm into a sling; cover up the whole arm and hand, and let the patient go to rest.

If the fracture is upon the under arm, the reduction is a little more difficult; but only in the care it requires of reduction. As there are two bones in the under arm, it should well be examined which it is, or if both. In reducing of it, let the arm be bent as before; one assistant to hold the patient, one to hold the elbow, and another the wrist; gently extending, till the fracture is reduced, as before directed, and the bandage, compress, and splints, put on according to the same Method.

In this fracture the wrist and hand should be kept from moving; as the motion of their muscles will be apt in a great measure to hinder the bones from uniting again.

A fractured Thigh bone.

This is of very great importance; which, from its situation, and the strength of the muscles, in general, even by the best method is very difficult to reduce.

Let the patient lay on the opposite side, and with the knee bent; let the limb be extended by assistances, and carefully set it according to the situation, similar to the beforementioned method.

This limb requires a very long bandage, and very large splints. The method of treatment must be as before, and the patient must be well secured in bed. He should not lie constantly upon his back in a straight posture, as formerly was practised; but most part upon either side, with his knee bent, which should be kept bent as much as possible; if he can lay chiefly upon his sick side, so much the better; he may also move a little his joints, in order to prevent them from growing stiff.

A fractured Leg.

This accident is more common, and therefore ought to be paid the greatest attention to.

To know for certain whether the leg is fractured, let the patient lay on his back or on either side, with his leg and thigh bent, so that the muscles are entirely at ease, and out of action. Let an assistant hold up the leg by the knee; then take with one hand the ancle, with the other the middle of the

leg, and move the ankle to and fro, holding the other hand steady, and if you feel a crackling, and the patient feels great pain in those parts, the leg is certainly fractured; particularly if the patient could not stand upon it before. This being fully discovered, we must next examine which bone is fractured, the *Tibia* or *Fibula*; and then proceed in the following manner.

Let the patient lie as before directed; one assistant holding the knee in a bent position, another holding the leg near the foot, both gradually extending.

Then with both your hands reduce the fracture, by gently pressing it with the palm of your hand while it is extending. Then lay on a long roller; or, what is still better, a many-tailed bandage, which is made thus: Take nine or ten slips of linen, about two inches broad, increasing from twelve to twenty inches in length, laying over each other about half an inch. The whole bandage must be secured with a slip of linen behind, so as to make the breadth of the bandage the length of the leg; either of these must be previously wrung out in camphor spirits.

If you make use of the roller, begin laying it on from the ankle, going gradually upwards, folding each round over half an inch at a time; and under the calf of the leg, give the bandage a single twist, by which it will always suit. Continue this till the whole is finished.

Over this again put a compress, and then put on the splints, (previously bolstered with some tow) which secure with tape, as before observed with the fracture of the arm.

The many-tailed bandage is however superior to the roller, in the fracture of the Leg. The method of laying it on is this: the patient laying upon his back, with his thigh lifted upwards by the assistant, put the bandage under the calf of the leg; then begin from the ankle upwards, folding over alternately the tails of the bandage, so that they secure each other. When this is compleated, put over the whole a *Compress*, as before directed, as also the *Splints*; or in the room of common splints, put on the new invented splints of Mr. Sharp; of which a ship should have three or four pair of different sizes, both for the right and left leg.

When this is performed, the patient should be put to rest as soon as possible; and he will lay easiest upon the side where the fracture is, with his

thigh and leg bent forwards. This position suits not only best on board a ship where the motion is continually apt to disturb the sick part, but is also most natural both for ease, and for healing; as the muscles are all at rest, and adds firmness and ease to the body.

Compound Fractures.

When a fracture is attended with a wound, it is needless to observe, that the case is of a more dangerous nature. If the contusion has been so great, as to splinter the bones, such splints as are loose, and seem to prick through the flesh, should by all means be removed. Violence must however be avoided; for by a little patience, nature will of her own accord separate and discharge them.

Gentleness should ever be observed in surgery, though not a timid weakness, by being defective in dressing properly. The reduction of the fracture should be conducted the same as before, with this difference, that always a many-tailed bandage should be chosen, so that it may be opened when requisite.

The dressing of the wound should be the simplest possible; for the nature of bone is so delicate, that it will neither bear to be much exposed to the air, nor suffer any greasy salve or ointment to come near it; both are liable to corrode it, and bring on a carious, which is of a most dangerous tendency. Dry lint therefore is the best, particularly near the fractured bone.

In short, it is to be treated cautiously, like an ulcer, and the greatest care to be taken, that the arm, leg, or whatever fractured part it is, be kept steady, and no oftener dressed than what is absolutely necessary.

Conclusive Observations on Fractures.

In the obscure times of surgery, various medicines were applied to broken bones, with a notion to unite them the sooner. This, however, is entirely exploded from modern practice.

The inflammation that generally attends fractures requires however a fomentation of *Camphor Spirits* (A), in order to disperse the obstructed humour; but if there is no inflammation, the part undoubtedly is the better

without such applications; as then there will be nothing to hinder a free circulation; for which reason, fomentation should be used as sparingly as possible.

It is well known too, with what difficulty the fractured bones were set formerly, owing principally to the limb being stretched out in a position that excited all the muscles to contraction. This is also here obviated.

In reducing therefore a dislocation or fracture, the muscles should be as much at rest as possible, both in the operation, as well as during the whole time of the cure. It is for that very reason my direction in this respect differs from what has been delivered to us by authors on that subject, particularly such, as might be expected to fall into the hands of a seaman.

Another error has been practised, namely, the laying on the bandage very tight. It was supposed, that that would strengthen the limb, and make the bone smooth; but it has sometimes prevented the bones from growing together at any rate. The bandage to be sure should not be too slack, for then we might as well lay it aside entirely. A bandage certainly is necessary, but merely as a support, and it should never be tighter than what the patient can well bear.

In simple fractures, the first dressing should be so permanent, as to have no need for being removed, at least for six, seven, eight, nine days, or more; if the patient don't feel uneasy. For my own part, having ever been successful in fractures, I have not opened the first dressing for a fortnight, and sometimes three weeks.

The laying stretched upon the back when a leg is fractured, is altogether needless, troublesome, and even a hindrance to the healing. The patient should have a good wide cot, and so that it may swing pretty easy; or else entirely confined, when the ship has much motion; and having the leg well secured with splints, he may safely lay on what side he pleases, provided he does not sit up in the bed (though he may out of it) for the first three weeks; for that strains the muscles very much.

The diet should be sparing, the body kept open, and a temperate warmth should be preserved, in order to promote perspiration. Drinking of strong liquors should by all means be avoided, and the diet should be as fresh as the circumstances will permit.

Six weeks is generally the time allowed for the cure; that however is no certain rule. If the fracture is simple, and the constitution good, then at a month's time a small trial may be made by the help of crutches, or a trusty mess-mate, in a calm day; but if it occasions the least pain, it should be postponed.

This is then all I have of consequence to observe in simple fractures; which holds good also with compound ones, except that the wound requires to be dressed oftener; but the greatest care should be taken, that it is done in a manner, so as not to disturb the knitting of the bone.

SECT. VI.

Of Scalding

Scalding is another accident of much consequence, which ought to be remedied by times, in order to prevent the many evil consequences of inflammation it is liable to produce.

I must first of all refer the reader to what I have said on the skin, Lect. I. Sect. VII. for farther than this we will not suppose a scalding proceeds, unless it is very dreadful indeed.

The nature of scalding then, is this: the hot water coming upon the body, separates immediately the scarf skin from the real skin, thereby opens the lymphatic vessels, so that they discharge their lymph under the cuticula, and thence arise blisters. But withal, the heat of the water irritating the nerves, they are excited to contract and constrict the delicate vessels, and hinder the circulation; thence ensues the fiery colour and inflammation, and create acrimonious tendencies. Thence we learn, that a relaxation of the solids is to be relaxed, and the humours to be rectified.

The moment a man has scalded himself, immediately apply *Turner's Cerate* (K), spread upon a rag, to the part. What is equally as efficacious, and sometimes better, is the *Linament of Soap* (B). If notwithstanding this, blisters arise, open them not, but let them remain with the application on them, whereby a cure will soon be facilitated; but if they break open on their own accord, or that the scarf skin was scalded off, then apply on the sore, *Turner's Cerate* (K), intermixed with a little *Calomel* (N), and levigated *Precipitate* (O), which will prove very beneficial.

No. III

Take Turner's Cerate (K) one Ounce, Calomel, and fine Precipitate (O) of each ten grains, and mix it.

This spread upon a little lint will prove a fine healer. If the *Diachylon Plaister* (20) can be applied, it will also prove so powerful an *antiphlogistic*, that any other will be unnecessary.

SECT. VII.

Of Burning

This is of the same nature with scalding, but is liable to become more pernicious, in proportion to the bulk and heat of the body that does the mischief. If it is slight, the treatment is equal with the former; but if it has penetrated deeper, it follows, that the inflammation is greater, and consequently the treatment requires more care.

If the sore so burned is deep, and thence an eschar in the parts, digestive medicines should be applied. It will not be amiss to put a little Mercurial ointment (P) in the poultice. If the inflammation is very great, *Bleeding*, and other methods requisite to prevent a fever, should be observed. In general, the ointment No. [III](#). will ever be found Efficacious.

SECT. VIII.

Of Drowning

A Man who unfortunately falls over board, and is taken up for drowned (immediately) should not be given up for lost. As this is a circumstance of the utmost consequence to a sea-faring man, I shall think my time well rewarded in explaining the method of recovery contributing to the preserving the life of my fellow creature.

There are different ways of drowning; and according to the nature of the accident, the probability of recovery is founded.

A man, before he comes into the water, may receive an unlucky blow, that will not only stun him, but make him expire his breath (which is generally the case in high falls) as before observed. In this case his senses are benumm'd, and by the inspiration apt to draw in a quantity of water into his lungs instead of air; thence, both from the blow, and from the total stoppage of respiration, he may have some blood vessels burst, both in the lungs and in the brains, whereby he is instantly irrecoverably dead, beyond all hopes. In this case it may happen, that either way, separately, is enough to kill him.

A man may also, in the very act of drowning, swallow a prodigious quantity of water; not only filling his stomach and intestines; but thence forcibly infuse the water into the lacteals, thereby overload them, and force it into the blood vessels that they burst. The great quantity of cold water will also be apt to chill the stomach and intestines; thereby destroy the sensibility, and prevent them from performing the peristaltic motion; and by the universal consent, destroy the whole nervous system, which is the spring of life, that promotes the blood to circulation, and consequently stops the animal motion, and life itself.

But the most common way of drowning is by suffocation; namely, a sudden constriction of the respirative organs; whereby the supply of the air

is cut off, and consequently the circulation of the blood must stop. This is verified by the frequent instances we have of people drowned, who have scarce any water, either in their bellies or in their lungs.

To understand this properly, we must first observe, that a continual supply of fresh air is requisite for the circulation of the blood; partly and principally for its motion, and partly from the nourishment the blood actually receives from the air. In the next place, we must also consider, that the lungs, appropriated barely for respiration, are so very delicate in the irritability throughout the larynx, aspera, bronchea, and vesicles themselves, that the least heterogeneous particle stimulates them to a convulsive expulsion, of what seems obnoxious to them, and thence excites a cough. But, when the parts are too irritating, the lungs are excited to a universal constriction, and occasions strangling. This we find is the case when in drinking or swallowing our aliment, that the least morsel happens by mischance coming the wrong way (as it is called) that is, into the larynx, occasions a heavy cough, or even a suffocation; to obviate which, nature has formed the epiglottis in the larynx, like a flap, that opens in respiration, but is always shut in the act of deglutition, except by some mischance or other, that it becomes lame, and unable to do its office.

These circumstances considered, we shall not only be able to account for the act of drowning, but discover the principles that furnishes us with probabilities of restoring life again.

I have mentioned three kinds of drowning that at present occurs to my memory; and unless that some blood vessels, either in the lungs or in the brains, or in some other principal viscera are burst; or, that the lungs are so much filled with water, as to be unable to recover their respirative functions again, (which is seldom the case) or, that the whole nervous system has received a universal paralytic shock by the chill of the Water in the stomach, so as to be robbed of its irritability, either way of drowning is recoverable. For the whole mystery consists, in setting the animal automaton or clock-work in motion, again; to stimulate the nerves to their sensation; to set the heart a pumping; and the lungs, in order to push forward the fluids, in respiration. I could say much on the subject; but philosophy aside, let us endeavour to recover the man to life again.

The Method to recover a drowned Person.

Three things are to be the intention to restore life in a person taken up for drowned.

1. To liquify the fluids. For the moment the circulation of the fluids is stopt, the blood and lymph congeals; and after this, putrefaction gradually commences.

2. To stimulate the nerves. For on the irritability of the nerves life itself depends; and which sets every thing in the animal fabric in motion.

3. To restore respiration. For without this, neither can the blood circulate nor the nerves have sensibility.

The moment therefore you receive the person drowned, strip him of his wet cloaths, and wrap him up in three or four dry blankets; if his belly is swelled with water lay him across a chair or barrel upon his belly, in order to make him void the water; but in this spend not much time, but be as quick as possible. Take a good quantity of salt, the more the better, with which employ three or four handy men to rub him all over;—his temples, his wrist, his arms, his breast, his back, his groins, his knees, his ancles, &c. let this friction either be by the hands or rough woollen cloaths pretty briskly; and, whilst this is doing, let one apply his mouth to that of the patient, stop his nostrils, and with all effort blow gradually into his lungs, and try to make him breathe. If the lungs are open to this, there is a good prospect of recovery, and let the breathing into the lungs be briskly repeated. In the mean time, try to open a vein in the arm; if the head is bloated, as it frequently is, open the jugular vein, or whatever vein that appears distended in and about the head.

On signs of life, continue the friction briskly, rub some *essential oil of peppermint* (U) in and about his nostrils. If he recovers, at first he will open his eyes, heave his breast, his heart will beat, and gradually he will move some of his limbs. If he opens his mouth put a little salt in it at first, next give him the following draught.

No. IV

Take essential oil of peppermint (U) six drops, upon a lump of sugar, which dissolve in a table spoonful of elixir proprietatis (H).

If he inclines to vomiting, promote it with decoction of *camomile* (11) or a vomit of *ippecacuanha* (S); besides all this, blankets should continually be warming, so as to preserve a warmth of the body.

If he is fortunately brought to life, put him directly into a warm bed, and give him a dose of the *fever powder*, and let him drink freely of good sage punch, in order to promote a perspiration, which is here very requisite.

No. V

Take sage about a pugil (that is as much as you can well take betwixt three fingers) on this pour boiling water, one quart; to which add sugar and good old Jamaica rum, sufficient to make it agreeable, but not too strong. Or instead of this give him some mulled wine.

A fever always attends a patient after the recovery of Drowning; therefore care should be taken when he is brought to himself, that he is not over heated by liquor; as this is liable to throw him into a disease as dangerous as the narrow chance he has just escaped.

Should however the above methods not succeed, you must not despair. If the lungs, or the larynx seem so contracted that it appears impossible to blow breath into him by the mouth, recourse must be had to *bronchotomy*, that is, opening the wind-pipe betwixt the third and fourth cartilage; to which a man should put his mouth and inflate the lungs; the throat should be tickled with a feather, a little pepper blown up the nostrils and the body here and there scratched with pins, or small wounds made in the legs and arms, and plentifully rubbed with salt. If there is salt enough to do it, the patient ought to be covered all over.

We have instances that patients, after two hours application, have been brought to life again; therefore every means possible should be tried. Of all the various stimulants, salt, or potashes, is the best; spirits must not be made use of, nor indeed vinegar, in the time of recovery; as that is liable to congeal the blood, and prevent it from circulation, till the recovery is procured; when the sage punch, or mulled wine, or some other cordial may be proper. Spirits of hartshorn, if it is on board may also, with propriety, be

held to the nose, or poured a little down the throat, as that is stimulating, but not coagulating.

When at length the patient has recovered, let him take the forementioned draught No. [IV](#). as also the warm sage punch No. [V](#). and take all possible care for his recovery.—If Bronchotomy was obliged to be performed, let the wound carefully be washed, closed, and dress'd with dry lint; over which lay a compress with warm claret intermixed with *camphor spirits*; and the more to favour the healing, let his head constantly be inclined; rest should also be promoted as much as possible, for which purpose 8 or 10 drops of *liquid laudanum* will be very proper; likewise silence should be strictly imposed on him; and that should be a material object to be observed, by those who tend him. In every respect else, let him be treated as is directed under the head of fevers and other circumstances, as they shall occur.

Here then I shall conclude this lecture, and beg leave to observe, that as mankind are ever liable to accidents, particularly that class I have here devoted myself to serve, it behoves us ever to be provided with preconsiderations, that we may be always ready to give our friendly assistance in the hour of distress. In time of need, the mind is too much hurried to read with any attention; and dangers particularly from accidents admits of no delay.

The prudent seaman, especially the father of his Crew will therefore make himself acquainted with all these circumstances in his hours of tranquillity, by which he may ever be ready with his friendly aid, and receive the blessing of him that is benefited by his Humanity and generous care.

LECTURE IV.

ON

EXTERNAL DISEASES,

AND THEIR

CURE.

In this lecture I shall treat of external disorders, which may take their origin either from injuries, or from internal acrimony of habit; for whatever disturbs the animal œconomy, prevents the natural order of the fibres, and those again when in disorder, corrupt the fluids.

Be the causes what they will, certain it is, that there is a strict connexion between the external and the internal parts, to which due regard ought ever to be paid. I shall without delay proceed to their various kinds.

SECT. I.

Of Inflammation

When either the solids are so injured that they will not transmit the circulation with their natural freedom; or when the humours are become so acrimonious, as to constrict the small tubes, that part of the fluids becomes confined in them, there must then of course be an obstruction of the circulation of the fluids, thence accumulation must follow and consequently swell the parts. And this is the nature of an inflammation.

The blood being obstructed, the parts must naturally swell; the blood being forced into the small lymphatic vessels, thence proceed the redness; and lastly, from the vessels being distended and irritated by the stagnated humours, proceeds the pain.

An inflammation terminates three different ways. When the solids relax, or the fluids become attenuated, so that the accumulated humours disperse, and are admitted a free circulation again, it is termed *resolution*; and this is the best and most favourable way for an inflamed and tumified part to terminate.

When some of the small vessels by the over-stretching of their contained humours break, together with the lymphatic vessels, and the extravasated blood intermixed with lymph is converted into matter, it is termed a *suppuration*; this is also a favourable way of termination, provided the matter is properly concocted and discharged, and the parts properly healed up again. And this ought to be the second indication when the first does not succeed.

When however the imprisoned humours enclosed by their tubulæ are inspissated, and mix irregularly with the solids that contain them, obstinate tumours are apt to ensue. This is the third kind of termination, and should be avoided if possible; for it either becomes tedious, or occasions malignant ulcers; which in glandulous parts are the first foundation of cancers.

But, when the humours that are obstructed get malignantly corrupted into a pernicious acrimony, and stimulate the distended solids, so that they stretch beyond their natural tone, without admitting any circulation, the parts become insufferably painful; the patient is restless, anxious, and attended with an acute fever; the humours become offensive and fœtid, and the parts at length turn of a livid colour, and this is then called a *gangrene*. Gradually the parts begin to putrefy and grow black, the humours acquire a cadaverous fœtor, the solids lose their natural texture, and die. This is called a *sphacelus*, or mortification; and unless the mortified parts are removed from the sound, death must inevitably be the consequence.

These are the different terminations of inflammation; and whether the cause is from external injuries, or from internal acrimony, the indication must ever be the same, I shall consider each separately.

Of Resolution.

Inflammation that attends wounds, contusions, &c. should be dispersed by the application I have already mentioned in the foregoing lectures, viz. *camphorated spirit* (A), or *liniment of soap* (B), the first suits best the fleshy part, and the latter the inflammation on the joints and tendinous parts. But when with the inflammation a wound is compounded, and is obstructed in its suppuration, recourse should be had to poultices; as thereby the fibres may be the more relaxed, so that the ulcer may the freer discharge the matter, thence drain the inflamed part of the obstructed humour, and consequently remove the cause of the inflammation.

Of Suppuration.

When an inflammation arises from an internal cause, there generally is some acrimonious humour that wants to be discharged; in this case resolvents is not always safe, as it is ever apt to increase the pain and inflammation. Of that kind are the boils and swellings under the armpits, hams, and other parts. In this case, poultices should be applied as warm as possible; the ingredients may be oatmeal, or biscuit powdered, honey and a little butter, warm and largely applied. When the swelling increases, grows red, and acquires a top, together with constant throbbing, it is a sign that matter is forming, which is the more confirmed by its growing soft. When

the fluctuation of matter is actually felt, and that the top becomes somewhat of a black and yellow colour, it then is mature for being opened, in order to let out the matter, which is best performed with a lancet: this being done, apply the poultices again. And when the matter is fully discharged, lay on the diachylon plaister, in order to heal up the part. Should however an ulcer be the result, it must then be treated accordingly, as will be directed under that head.

Of Tumours.

When inflammations will neither disperse nor come to suppuration, it is apt to form hard tumours; sometimes they are indolent, at other times very painful. In this case, nothing is better applied, than *mercurial plaister* (21), and especially if there is any venereal taint in the habit. At the same time, whenever tumours are dispersing, care should be taken to use internal medicines, in order to correct the acrimony, and a purge now and then, in order to incline the humours to be carried off. If, however, they incline to suppuration, poultices should be again applied; and to bring them sooner to a head, purges should be omitted, till after the matter is discharged.

Of Mortification.

A Mortification has two degrees; gangrene, the beginning, and sphacelus the compleated mortification. When an inflammation will neither disperse, suppurate, nor become indurated, but, on the contrary, increase with a burning inflammation, and is prodigious painful; when the swelling continually increases, the parts become uncommonly tender, attended with a numbness; the colour changing from a fiery red into a livid ash colour, and the cuticula seems to separate on the touch, a gangrene has commenced. In this melancholy case there is no time to lose.—Apply the following powerful medicine, which in a gangrene, and even a sphacelus, has proved beneficial to a miracle.

No. VI

Take Mercurial Ointment (P) one ounce; Calomel (N) and Powder'd Precipitate (O) of each one drachm; Oil of Peppermint

(U), one drachm; Spread this upon a rag, and apply it to the gangrenous part.

If a sphacelus has commenced, the skin turned livid, attended with a cadaverous foeture, and seems to the touch to be rotten, it must be scarified to the quick with the lancet, and then the above remedy applied. But if it has already reached the bone, (which a mortification soon will) and seems to make a rapid progress, all external remedies are of little avail, and nothing but amputating the limb can save life. But as that operation ought to be performed by a skilful surgeon, I shall therefore omit it here; only observe, that there are very few cases, but that this dreadful catastrophe may be prevented, if timely assistance is given.

To conclude this section, I think it requisite to observe, that a gangrene and sphacelus, or mortification, is far from being so often the case as many from the pain they undergo, are apt to persuade themselves. A common boil, whilst it is in an inflamed state, is sometimes so very painful as to put the patient in a fever, and will be excessively tender. But, notwithstanding, a suppuration is the result.

A gangrene and mortification is only to be dreaded, when a whole limb is inflamed, such as an arm, a leg, &c. When the inflammation is dreadful, that the blood can find no passage at all, but is obstructed on all sides, then the part must naturally die, and become cadaverous. The cause of this misfortune may be either a heavy contusion, a compound fracture, a gunshot wound, or an internal malignity of the bone itself; and as a caution, I must also add, that pricking a tendon in bleeding may occasion this dreadful malady.

There is another kind of mortification which proceeds from a bad habit of body, a poorness of constitution, or when the whole mass of blood is corrupted by the scurvy. This shall be taken notice of in its proper place.

SECT. II.

Of Ulcerations

This is a part of surgery that many pretend to, but what very few really understand. There are however laws in the animal œconomy, which when they are violated, leads us to trace its many evil consequences.

An ulcer is a dissolution of solids, occasioned by acrimonious humours, which not only act spontaneously upon each other, but prevent the remaining circulating part from the formation of unnatural substances.

Exulceration, strictly speaking, is what every part of the animal fabric is subject to; from the external surface, to the very substance of the bone itself. In this sense, every eruption, as well as the caries of the bones come under that denomination; but more particularly, the corruption of the softer parts on the surface of our body, is here the subject.

I shall not enter into a detail of the many kinds of ulcers generally enumerated; they have all in common an corruptive acrimony, and a depraved accretion of excrement in their nature; which differ only, according to the parts ulcerated, and the habit of body; and I have ever found in my practice, that one proper method serves for all; I shall therefore in a simple manner give such directions as will ever be attended with success.

The principal intention is to correct the malignity of the humour peculiar to ulcers; to bring them into a good suppuration, and nature, all kind NATURE! will perform the rest.

When an ulcer is inflamed all round, filled up with fungous substance, of various colours, the matter thin, ferous, and is very fœtid; then first wash the sore gently with some good soap suds, or let the part ulcerated be steeped in it for some little time, which will draw out the malignity; then apply the following dressing:

Take the powder of burn'd *Allum* (17), and with a little lint dab it all over the fungous part, then spread a pledget of the salve mentioned in gun-shot wounds, No. [I](#). and over which apply a good warm poultice there also mentioned. Let this dressing be repeated twice or thrice a day, till the inflammation ceases, and the fungous substance becomes reduced; when then, the washing, the allum and poultice may be left off, but the mentioned salve continued till the sore looks of a pale red, and begins all round the edge to heal up.

Cleanliness is very commendable; but in this case, when a sore looks well, the matter becomes of a pale yellow; wiping should be very sparingly; and then, dressing once a day (the morning) is full sufficient; for by wiping, and often meddling with it, the small granulations of the flesh is hindered from growing, and healing is retarded.

A sore leg is very troublesome, and very often attended with pain and expence, to no purpose. It generally swells, is inflamed and troublesome at night. Let the leg be bathed every night, (with the open sore) in some good warm soap suds; or instead of that, take what you can have plenty of, namely, sea water. If you can have it warm, it is well, if not, make use of it cold, rather half a dozen times of the day, than omitting it altogether; and by dressing the leg as before directed you'll have reason to thank me for my good advice.

Sometimes the bone is affected; in which case, the cure with the best dressing in the world will frequently be tedious. A sore of this kind will not heal till the bone has discharged a scale, (termed *exfoliation*.) But tedious as it is, very often it is made more so, by unnecessary trouble. Without intending to give offence to my brother surgeons, I have frequently seen bad consequences attending methods taught with diligence in hospitals; take my word for it, you will do no good with your scraping tools, throw them over board, and you'll succeed better than using them in this case.

The bones, though the hardest, are the most delicate parts in the whole body; but when it is affected, nature forms a new periosteum over the sound part, and gradually pushes forward the decayed scale. This wonderful work is performed by nature only, and must not be disturbed. By scraping, and boring of it, the tender periosteum will only be molested, the sound bone will become foul again, and so you may go on for ever; and I never saw a

sore of that kind healed up by this improper method, till nature overpowered art, or the surgeon was tired with scraping, perforating, &c. but I have known many amputations in consequence of it, and many lives lost in consequence of that again; yet there is no rule without exception. There are cases where operations of that kind on the bones is required; but that ought to be referred to a judicious and experienced surgeon.

When the sore remains fungous, the proud flesh sprouts out in little bunches, and the matter is muddy and stinking, it is almost a sure sign the bone is foul; and if you examine with a probe, you may generally feel it, or even see it, and the bone, is sometimes black, brown, &c. and generally very rough on the surface. In such a case, continue the dressing as before observed, and frequently examine whether the scale loosens; at the same time examine (not too roughly) which way the scale inclines, also the bigness of it; and if there is a probability of extracting it, so that the fungosity only stops it, then freely use a bistoury to make way for it; but in this be not in too great a hurry, for care should be taken (as I have said above) not to extract the scale, till the separation from the fresh bone is mature.

If an ulcer is sinous, inject in the sinus, a mixture of burnt allum and water, viz.

No. VII

Take water two ounces; burnt allum (17) two drachms; mix them well for injection. After two or three dressings with this, inject the antacid injection.

No. VIII

Take two ounces of water, with one scruple of calomel (N), as I have described in my treatise on the venereal disease.

This is the proper management of ulcers in general, and will ever be attended with success, even in the most inveterate kind. In venereal ulcers, the mercurial ointment with the precipitate, may be applicable, especially in shankers. I shall therefore end this section, for having directed the

management of the worst of ulcers, the inferior kind will of course yield to the same method.

SECT. III.

Of Diseases of the Skin

Under this head I shall consider all kinds of breaking out, exclusive of that, which attends inflammatory and acute fevers, as they are all of one nature, and will all yield to one method of cure.—All proceed from a corrupted humour, which nature endeavours to expell from the body, by driving them to the surface, as that is the least resisting part.

Of the Itch.

The itch may either proceed from original acrimony, or it may be caught from a mess-mate who is infested with the disease. In itself it is of different kinds, but the diversity in this respect depends principally on the habit of body afflicted with it. Sometimes it is dry, scaly, and attended with very great itching; in that case, it proceeds generally from a scorbutic habit, and is attended with a slow inward fever. Sometimes it is more moist, and utters itself with bladders all about the hands, particularly betwixt the fingers, and is a sufficient testimony of the disorder. The principal indication is to purify the blood; and in the mean time recourse may be had to external means. A purge is first requisite.

No. IX

Take powder of jalap (R) one scruple; calomel (N) three grains, mix it with a little sugar and water, and let the patient take it early in the morning, working it off with balm or any other tea; after this some pills may be made of calomel.

No. X

Take stomach powder (M) two scruples, calomel (N) one scruple; sugar about one scruple; mix them, and with a few drops of water make it into a mass, and divide it into twenty pills. Of which let him take one every morning and night.

Externally make use of the following:

No. XI

Take Turner's Cerate (K) two ounces; Precipitate (O) two drachms; mix them.

Let this be sparingly rubb'd on the itchy part; if it should be too hard, mix with it a little sweet oil. When the pills are all used, another purge like the former may be given; after which if requisite, a few of the same pills, one every night, will not be amiss. Should it be very obstinate, a little *mercurial ointment* (P) about one drachm to the above quantity may be added; particularly if the ship is in a warm climate. Should these medicines seem to affect the mouth, as people vary much in this respect, a purge will set all to rights again. This is a safe and expeditious method of curing the rankest itch, even of the worst kind. If however there is a fever attending, a few *fever powders* (L) should be given when going to rest, in order to promote perspiration; at the same time using the before mentioned medicines. In regard to the scorbutic itch, that I shall take notice of, under the title of that disorder. During the cure of the itch, a decoction of the *sassafras* (14) should be drank, as that is a great purifier of the blood.

Of Boils.

Boils are sometimes very troublesome, and at times exquisitely painful; especially when they are on tendinous parts: they are always a sign of foul blood, and nature very often that way frees the body from diseases. They may frequently be dispersed in the beginning of their coming, by pressing, and gently pinching them: but unless they are very numerous, such a method is unsafe; because it forces into the blood again what nature endeavours to discharge. The best method is to bring them to a head as soon

as possible, to which purpose I shall here recommend a paste that is very powerful.

No. XII

Take oatmeal and make it into a paste with honey (E); this will speedily ripen them, and facilitate the collection of matter.

Should the boil not open of its own accord, and matter should be observed to fluctuate in it, open it with the incision lancet; and continue with your honey paste, till the core is discharged; when the *diachylon plaister* will heal it up.

Of Ring Worms, Daw Worms, &c. &c.

These are very troublesome, and though at first they seem but insignificant, very often they lay the foundation to stubborn evils. In fact they are but a peculiar kind of itch; and the very same remedy that is recommended above for the itch, is equally powerful to remove this: what necessity is there then, of increasing words with a tedious description of their various appearances; since they will all yield to one substantial remedy?

Of the Morbis Pedicularis, or, Lousy Disease.

It is an unhappy circumstance on board a ship, when a man is afflicted with this filthy distemper; and such a man is as pernicious to a ship's crew, as a rotten sheep in a whole herd. If it proceeds from a lazy filthiness of body, a gentle washing with a good scrubbing-brush, and rubbing the back with the boatswain's towel,^[12] is a pretty amusement enough to circulate the blood and to excite cleanliness. But sometimes this disorder has its seat in the very blood itself, and that filthy vermin will breed under the skin, and eat their way through, by clusters.—I have opened boils as big as pigeons eggs, from which large bunches of lice have followed the lancet.

¹². A cat of nine tails.

These patients are in every respect wretched, and are actually objects of pity. They generally have a pale, sullen, heavy, half-starved, cowardly, in short a lousy, look; habitually lazy, and seem to be void of every manly passion; neither good nor ill-natured; and are stimulated by nothing but pusillanimity; and as they seem neither sick nor well, I say they are objects both of pity and contempt; though the latter is unjust, because they cannot help it.

The reason of all this wretchedness is because the blood is degenerated into a lifeless mass of pituitous corruption; and wants that spirit which gives a relish to life. Hence we see in this one instance, how much the mental faculties depend upon the state of the body.

In order to cure this unhappy patient, the blood must be purified, and the whole system stimulated to a brisk circulation. Nothing less than mercury will do the one, and stimulants together with exercise, effect the other.

In the first place give him the following medicine.

No. XIII

Take ipecacuanha (S) one scruple; jalap (R) ten grains; calomel, (N) four grains; make it into a draught or bolus.

Let him take this in a morning upon a fasting stomach, and let him drink plenty of chamomile tea after every vomiting; then give him the following electuary.

No. XIV

Take lenitive electuary (I.) two ounces; black pepper powdered, two drachms; calomel (N), one scruple; mix them.

Of this let him take every morning one tea spoonful; upon which let him rub himself all over with the mercurial cloth as follows, till he comes by this exercise into a breathing sweat, at least for an hour; and for the rest of the day keep him in continual exercise.

No. XV

The mercurial cloth.

Take a cloth or rag as much as half a yard square, upon which spread one ounce of mercurial ointment; let the patient work this into the cloth with his hands, so that it is equally all over alike.

With this cloth let him rub his wrists, arms, shoulders, top of his head, his back, knees, waist, thighs, and legs; and continue this rubbing once every morning with the same cloth for a whole month, putting the cloth in his pocket when he has done.

His sores may be dressed with the above ointment No. [XI](#). The decoction of sassafras, and a dram of spirits, with a little *Turlington's Balsam* after his morning's exercise, will prove greatly to facilitate the cure. This is the method I have pursued with this filthy disorder; and I have succeeded to my most sanguine expectations. And by this, I have been fortunate enough to be the means of making men again, of wretches that were a burden to themselves, and a nuisance to the people that were obliged to associate with them.

SECT. IV.

Of Promiscuous Disorders.

Having thus treated on the most material ailments that want immediate assistance, I shall in this section consider promiscuous disorders, that have no relation to each other; but set them down as they occur to my memory.

Of the Tooth-Ach.

Those who have experienced the tooth-ach, best know how much they have wished for ease. No man can preserve attention to his affairs, who has bodily pain; and a severe tooth-ach is enough to disturb both body and mind.—But what is to be done?

Tooth-achs are of two kinds;—It may either proceed from cold, or from a decayed tooth, or from both. If the tooth is decayed, the sooner you have it out the better; but if your tooth is sound, then it is as unreasonable to draw it, as it would be to sling you over board, because you feel the pain. In regard to drawing the tooth, take this direction.

To draw a Tooth.

Observe first, what tooth it is; and shift the hook on the key instrument accordingly. Ordinarily the hook is to be inside, and the comb outside. When this is adjusted, wrap a little rag on the comb of the key, so that it may bear a little softly on the gums.—Then seat the patient steady on a chair or chest, (and if he is a coward, let somebody hold his hands, but a man of spirit will not want this officiousness) place yourself right before him—then fix the instrument on the tooth, the hook inwards, and the comb outwards; the hook on the tooth close to the gum, pushed a little towards the jaw bone, and be careful that you have hold but of one tooth, and the right one.

When your instrument is fixed as gentle as possible, without giving pain, (for if you fumble too long, and give pain, you intimidate the patient) I say then, when you have fixed your instrument properly on all sides, so that you have the condemned tooth, and no more; then shift your hand so, that you may have a good and firm hold, allowing for the turning of your hand.—Thus being ready, give your hand a gradual, and steadfast turn, and wrench the tooth out of its socket. In this you must be calmly resolute, neither spare your strength if requisite, nor let an untimely compassion get the better of your duty; but keep your hold till you have performed the operation. When the tooth is out of its socket, it will sometimes hang a little by the gum; introduce your fingers, and they may nimbly perform the rest if requisite.—It will sometimes happen, especially by the grinders, that a little of the jaw bone follows, sticking to the tooth;—let this not frighten you, it is sometimes impossible to avoid it; but not to frighten the patient with it, pick it off before he perceives it.

The moment the tooth is out, introduce your thumb and fore finger, and compress the socket of the drawn tooth; by which you replace every thing in order again; and after which, one dram of good old stingo, for washing the mouth, and another for comforting the heart, will make the patient forget all the pain he felt in the operation.

This method is certainly the most substantial for the tooth-ach. But sometimes this is not practicable, either because that the stump is so rotted as not to suffer a hold; or that none on board are intrepid or capable enough to perform the operation; in this case a Palliative will be very acceptable.—*Turlington's Balsam* (W), upon a little lint put into the hollow tooth will frequently give ease, a *little Spirits of Wine and Camphor* (A), or if very raging, a *little Liquid Laudanum* (T). But as I have said, they are only palliatives; and neither spirits, or any thing else can be depended upon for a certainty. Sometimes stopping up the hollow tooth with wax, with lead, or sometimes putting in a little saltpetre will give ease—each method may do, and each method may fail: so that sometimes if the tooth is not extracted, all powerful *Patience* at last must step in and prove the best doctor, till nature recovers, and comforts the patient with the balm of Ease.

Very frequently a cold occasions this troublesome pain; in which case the head should be well muffled up, in order to bring on a perspiration, and a blister behind the ear also of singular service—rubbing the face also with

some *spirits of wine and camphor* is of great benefit, and attention should be had, if there be any fever in the system.

Of sore Eyes.

Bad weather sometimes will effect the eyes, whence they will become raw and blood shot; which ought to be remedied in time, in order to prevent further inflammation. In this case make the following eye water.

No. XVI

Take Alum, One Scruple. Fine Sugar, Two Scruples. Spirit of Wine, and Camphor (A) Ten Drops. Fresh Water, Three Ounces.

Mix them in a vial, and hang it up so that it may settle; then pour off the clear into another vial, which keep for use.

Of this excellent eye-water, drop a few drops into the blood-shot eye, which in a little time, will bring it to health again.—If the eye lids are raw and inflamed, *Turner's Cerate*, or what is yet better, No. [XI](#). which prove very beneficial.—Should the inflammation be very heavy, apply the following poultice, from which you may expect great benefit.

No. XVII

Take Oatmeal, and Sugar powdered an equal quantity, and make it into a poultice with the yolk of eggs; upon which pour a few drops of the spirit of Camphor.

This kind of poultice you will find very cooling and agreeable to the eye, in order to assuage the pain; or take the following.

No. XVIII

Take the white of an egg upon a stone, or pewter plate, which rub with a piece of alum, till you get a fine white cream-like substance.

This is a fine cooling application. The Camphor agrees very well with some people, but with others it will not, in which case the last is preferable; or the *spirits of camphor* may be rejected from the other receipts.

Of the Ear-Ach.

The causes of that complaint are many. I shall only here observe two kinds; namely that proceeding from cold, and that which is occasioned from hardened ear wax.

In the first case, be very cautious how you tamper with your ears—a few drops of the spirit of Camphor upon a bit of lint, and that gently put into the ear, will sometimes perform great benefit; but never drop any thing considerable into the ear itself; it is frequently dangerous; from more reasons than what I here have room to enter into.

In cases where the ear wax is hardened, put in a little slice of *rusty bacon*, or a *bit of fat ham*, which you will find very powerful in softening the ear wax; and then with gentleness you may pick them; take care however not to go too deep.—Ear-ach frequently follows a heavy fit of sickness; in which case you can't do any thing better with safety, than keeping them warm.

Of Hoarseness and Sore Throat.

There are many degrees of this ailment. I shall first mention that of a common cold; which is either without a fever, or only a slight one.

Immediately on perceiving symptoms of a sore throat, a gargle should be applied. This, make of the following ingredients.

No. XIX

Take vinegar, and water, each four ounces. Alum powdered half an ounce. Honey about a good spoonful.

This gargle should be used two or three times of the day—if it is warm, it is so much the more powerful—a mouthful at a time is sufficient. If the palate is down,

No. XX

Take alum powdered, common salt, of each two drams, black pepper, ten grains, mix it into a powder.

Of this powder take a little on the tip of your spatula, and apply it to the palate, when it will immediately shrink up; and frequently make the person who before was hoarse, speak immediately quite clear again with a distinct voice; then proceed with the gargle as above directed.

But if the sore throat is of a more malignant nature, the parts inflamed and swelled, the patient seeming almost strangled, and attended with a high fever; it is certainly the quincey, and should be treated accordingly. In this case you must bleed freely and repeatedly, and give the fever powders, as will be directed in the treatment of the fever, and the management of the quincey.

The gargle notwithstanding should be plentifully used—a blistering plaister betwixt the shoulders, and if need requires, on the calves of the legs, the bigness of the palm of the hand, will also prove of singular service; and in every respect the treatment, as I shall observe under its proper head.

SECT. V.

Of the Venereal Disease

I have written and published a full treatise of this disease; divided in three parts; namely, I. An anatomical and physiological description of the genital parts of both sexes. II. An ætiological enquiry in the various stages of this disease. III. A true and rational method of cure. Nevertheless I shall here concisely give a safe method of the cure, in its various local degrees. But to those who would wish to know more circumstantial the true nature of that malady, and of the animal œconomy, I must refer to the above treatise.

The venereal disease may be considered in different degrees; namely the *first* and *second* infection.

To the first, I consider the gonorrhœa or clap, the phimosis, paraphimosis, chordee, priapism, shankers, dysuria, bubo, swelled testicles, and venereal excrescences, in their various degrees, in a local state. To the second infection, I consider those diseases that proceed from the venereal virus, infecting the whole mass of blood; namely cutaneous diseases; glandular diseases; diseases of the bones; dry pox; and the confirmed lues itself.

The first infection is what I shall here briefly consider, and give such remedies, which though simple and few, will prove efficacious. But the second I shall here only consider in the general.

Of the Gonorrhœa.

The gonorrhœa or clap, commences with an agreeable tickling in the urethra, attended with a thin limpid ouzing, which however soon degenerates into a disagreeable pain. A discharge of sharp fœtid matter, gradually commences with a scalding of urine; the præpuce and glans sometimes swells, with inflammation and a painful involuntary erection,

which adds considerable pain to the patient, and under which, the whole system seem disordered.

When all these symptoms appear, there is no room left to doubt a confirmed clap.

The first thing to be done, is to take some of the *fever powders* (L) in order to promote a perspiration, and to abate the scalding of the urine.

In the next place, and which is the principal object; have immediate recourse to my antacrid injection, which is simply this,

No. XXI

Take clean water one ounce, Calomel (N) one scruple, Mix it, and shake it when it is made use of.

Let this be injected up into the urethra, three times of the day, which if duely performed will stop the evil in its first bud; and prevent all the generally attending consequences. For it perfectly eradicates all the malignities of the venereal virus; prevents shankers, priapism, chordee, and any farther spreading of the infection. At the same time the calomel should be given internally; namely the pills, No. [X](#). to be taken every morning one.

Of the Phimosiſ, and paraphimosis.

A Phimosiſ is when the præpuce is so swelled, that the glans cannot be uncovered; and a Paraphimosis is when the præpuce is swelled behind the glans, so as not to cover the glans. Either way proceeds from the venereal virus having insinuated itself into the præpuce.

If the *injection*, No. [XXI](#). is made use of in time, all this will be prevented; but sometimes it is of a very stubborn nature. If the inflammation is very great, the poultice, No. [II](#). should immediately be applied, to which it will yield; being of a very powerful quality, particularly, if to the poultice *mercurial ointment* (P) two drachmes is added, by which it becomes an immediate antidote to the venereal virus.

Of the Chordee, and Priapism.

A priapism is an involuntary erection; and the chordee is a stricture in the urethra, by which the penis is bent either ways, according to the seat of the contraction, this however is generally downwards; whereby it makes the involuntary erection prodigiously painful.

It proceeds from some imprisoned virus in the urethra; which together with its contraction, irritates the penis to erection.

The *antacrid injection* No. [XXI](#). will generally, if timely applied, prevent this; but if by neglect, or the great virulence of the venom, it has actually commenced, recourse should be had to mercurial ointment; which should three or four times of the day be rubbed on the penis. Should the priapism be very violent, the poultice, No. [II](#). with the addition of *mercurial ointment* (P) must be applied.

Of the Dysuria.

The dysuria is a scalding of the urine in the urethra; this likewise is obviated by timely using the *injection*, No. [XXI](#). or No. [VIII](#). But when it is actually become troublesome, drink plentifully of diluting drinks, such as *balm tea*, &c. and take a few fever powders which will soon set this to right again; particularly as that is a symptom which generally vanishes, as the disease becomes milder.

Of Shankers.

These are little ulcers coming upon the glans, the frænum, and the præpuce of the yard. They are not only troublesome, but sometimes tedious in the cure; the *antacrid injection* No. [XXI](#). if timely applied, will generally prevent them.

But sometimes these very unwelcome visitors will appear even without a clap; and then they generally gain great ground, before timely remedies can be applied.

The *injection* No. [XXI](#). should in this case be injected betwixt the præpuce and glans; but if this is not sufficient, apply the following.

No. XXII

Take mercurial ointment, two drachmes; calomel (N), two scruples; precipitate (O), ten grains. Mix it into an ointment.

By the dressing with this, all the shankers will speedily heal up.

If warts sprout out, they should immediately be cut off, with a pair of good sharp scissars, and then dressed with the above ointment.

Shankers, when they appear, without a previous gonorrhœa or clap, ought always to be taken as a forerunner of the pox itself; and the patient must have immediately recourse to the *mercurial pills*, No. [X](#). one to be taken night and morning, in order to prevent the threatening evil consequences.

Of the Bubo.

A Bubo is a swelling that appears on a venereal infection, sometimes in one, sometimes in both groins. This swelling proceeds either from a transposed virus, or comes as the first messenger of a venereal infection.

A bubo appears with symptoms of a heavy dull pain in the groin, which on examination discovers itself with a little kernel that is painful to the touch: If left to itself it gradually encreases both in bulk as well as in pain, and at length forms a collection of matter.

The moment a bubo appears, take a purge, No. [IX](#). rub the part in the groin with *mercurial ointment*, and spread a *mercurial plaister* the bigness of a crown-piece, which, (the hair being removed,) lay on the part; let it remain on, till it separates from the skin, and the swelling is dispersed; the patient should be kept warm, and not have much exercise; at the same time the body should be kept open.

By this means the bubo will soon disperse, and the virus be destroyed in its first bud. But if it has already gained too much head to be dispersed, and matter is formed, it becomes larger and soft, and a fluctuation of matter is felt, then suppuration must be promoted; and when ripe be opened as soon as possible, for an outlet of the matter.

When it is come to its maturity, (that is the matter plainly perceived,) open it and dress it like an ulcer, first with the ointment, No. [I](#). and when it discharges pretty well, dress it with the ointment, No. [XXII](#).

By this management, though easy, you will not fail of success.

Of Swelled Testicles.

The venereal virus is liable to attack the testicles, which makes one of the most disagreeable circumstances of all venereal diseases; sometimes one, and sometimes both will be the seat of this complaint.

It begins with a heavy dull pain, attended with a disagreeable tightness in the spermatic vessels, leading from the testicle into the bowels. Gradually the testicle swells, becomes hard and grows very painful, and if not timely remedied, a suppuration, or a scirrhus, or even a gangrene and sphacelus, may be the consequences.

The moment the pain in the testicles is perceived, remove the hair all round the scrotum, and apply a mercurial plaister spread upon leather; over which put a suspensorium, and tie it up to the body as tight as it can be borne; which together with a purge, No. [IX](#). will not fail of giving immediate relief.

The patient should be kept as quiet as possible; for hard labour not only retards the cure, but becomes very painful. Should this not be sufficient, apply the poultice, No. [II](#). with the mercurial addition as warm as sufferable, and repeat it as often as it is necessary for keeping it warm to the part. By this method it will not fail of dispersing.

Should it however against all expectation come to a suppuration, it must be opened the same as another aposthume, and treated accordingly. But should a scirrhus, or a mortification ensue, extirpation is the only recourse. For which see my treatise.

Of Venereal Excrescences.

Besides all these disorders before mentioned that are the consequence of impure coition, there is one more to be mentioned, and that is a variety of disagreeable excrescences, which sometimes arise about the penis, the scrotum perinæum, and about the fundament.

When these appear, I generally take it for granted, that the pox is in the blood, and the patient ought to be treated accordingly.

If there is a possibility of extirpating them, either by a bistoury, or a pair of good sharp scissars, it ought to be done, and dressed with the following ointment.

No. XXII

Take mercurial ointment (P), and precipitate (O) equal parts: mix them.

The roots of these stubborn companions are frequently seated very deep; if this is particularly taken notice of, together with proper internal medicines, the excrescences taken away at the root, and the ointment applied, they may all of them soon be drove off from the field.

Of the second Infection.

In regard to the second infection, this is too tedious here to enter into, and to do properly justice, I must refer the reader to the treatise itself.

All I have at present to say, is, that my method of cure is particularly easy and simple, in itself; and I am happy to think at the same time, that it is more efficacious, than any former method, or practice, made use of.

The *antacrid injection* No. [XXI](#). is the most efficacious for subduing the gonorrhœa, and all the evil consequences in that part; and the other mercurial preparations in the manner I have recommended them, is equally powerful.

I am utterly averse to salivation; it is using mercury as a poison; very few have ever been cured by it, and those few have been more indebted to their good constitution, than the treatment which they have with so much danger undergone. But many, very many, have had their constitutions ruined by salivation, past the art of man to recover; and thousands have been sent miserable objects to the grave.

I am also averse to bleeding in venereal cases, unless the greatest necessity requires it; for in general it serves only to infect all the fluids, since the veins emptied, will naturally absorb from all parts of the body, and

thence liable to infect the whole mass of blood. Purging needs but be used very sparingly.

Mercury inwardly taken, should be used with the greatest moderation, and taken in very small doses; for with such judicious management it becomes one of the best medicines in the whole *Materia Medica*; but if abused, it is converted into a poison. The decoction of the sassafras should be plentifully made use of during the whole course of the venereal disease.

And lastly, from the real effect that the mercury has on the blood, I must add, that the same gentle method with respect to its internal use, as is recommended for the first infection, must also be pursued for the second infection; let the malignities be ever so bad, even in the most inveterate diseases of the bones themselves; with this difference, that in proportion as the disease is engrafted, time must be allowed for the cure.

Nature may be led by the gentle hand of assistance, and tenderness; but he who attempts to do good by main force, acts as a fool, and deceives not only himself, but those who are so unfortunate, as to trust to his pretended skill.

SECT. VI.

Of the Bite of Venomous Animals

It has been observed by all diligent inquirers into nature, that poisons are of two kinds, with respect to their effect in the animal œconomy. One kind will destroy if taken inwardly, but not have such ill consequences in wounded parts, externally. Another kind will be fatal in wounds; at the same time it does not affect the viscera.

The first which destroys internally, are poisons, either from corrosive minerals, or vegetables; but of the latter, namely what seems inoffensive taken in the mouth, is the poison that follows the bite of most venomous animals whose poison are natural; such is the scorpion, the adder, the rattle snake, &c. whose poison have by various experiments, been found to be attended with no ill consequences internally taken. This was known to the ancients, as well as to us; for we find Celsus,^[13] to have spoken of this in his medical works.

¹³. *Nam venenum serpentis, ut quaedam etiam venatoria venena, quibus Galli præcipue utuntur non gusto, sed in vulnere nocent. Ergo quisquis, exemplum Psylli secutus, it vulnus exsuxerit, et ipse tutus erit, et tutum hominem præstabit. Medicin. lib. v. c. 27.*

Without any farther preface, I shall proceed to give the immediate remedies for such unfortunate accidents as happen by the bite of these animals.

To Cure the Bite of a Rattle Snake.

The poison from this venomous beast, is more deadly than any other of the serpentine tribe. If its venom by a wound is infused into the blood, a mortification, and speedy death must follow.

The part bit, should immediately be sucked with the mouth, either of the patient itself, or one who will risk himself to be so much the friend of the wounded. Dr. Mead has recommended, that oil should be held in the mouth

all the time of sucking, but we have it from good authority, that the spittle itself is an antidote to that kind of poison.

It is needless to say, that the person who sucks the poison, should frequently spit out, as not to run the risk of swallowing it; and care should be taken, that his mouth is free from sores, or wounds.

When this is performed for a quarter of an hour, the part should be dressed with *mercurial ointment* (P). Sweet oil is much recommended, and may be plentifully used as an embrocation to the inflamed part; particularly if the oil is impregnated with camphor.

No. XXII

Take oil (C) two ounces; camphor, one scruple; mercurial ointment, one drachm; mix it.

The patient should take a vomit of *ipecacuanha* (S) *one scruple*; and a few drops of the essential *oil of peppermint* (U), on a lump of sugar, and dissolved in water, which should be repeated after the operation of the vomit as often as occasion requires. The fat of a rattle snake, is accounted a sovereign specific, if applied immediately to the wounded part, and the patient drinking a decoction made of *snakeroot*.

The same remedy here recommended to the bite of a rattle snake, (namely, sucking the fresh made wound) is also recommended to all other venomous bites. The *viper*, the *adder*, the *santapea* or *scolopandra*, the *scorpion*, *tarantula*, &c.

That oil is an antidote, (or as we may say a poison) to all these venomous animals is evident from their expiring in great agony, when oil is poured on them; but whether this effect is because of the penetration it has on their bodies, or annihilating their poison, is as yet undetermined.

I was once told a secret for the sting of a scorpion, avered to have been experienced, by a commander of a ship, trading to the bay of Honduras, viz. that the part stung by a scorpion should be touched with the glans of the penis, which would immediately draw out the poison, without effecting the glans that touched it; and it differs not, whether the patient himself or any other person does it.—If there is any virtue in this, I own it is beyond my

comprehension.—I thought proper however to mention it, though I cannot advise it as a safe experiment, unless farther confirmed by those who have actually experienced it.

I have however a great opinion of mercurials in such exigencies; and would advise recourse to be had to it immediately, both external as well as internal; this we are at least assured of, it cannot be attended with danger.

SECT. VII.

Of the Bite of a mad Dog

There are many surprizing things in nature, in so much, that ocular demonstration can only convince us of their existence. Amongst those wonderful phenomenons, the direful effect of the bite of a mad dog is equally astonishing as it is melancholy.

That dogs and wolves are more liable to madness of this kind, called the *Rabies*, is because of their deficiency of external perspiration; and are obliged to supply that defect with their open mouth and their lolling-out the tongue; whence their blood is subject to become overheated, and thence generate a malignant inflammation, which infects the saliva with this direful poison. But the astonishment is, why it should have so very powerful a contagion as to infect the mass of blood of every animal that receives the least atom of it? Yet so it is, confirmed by sad experience.

From frequent observation we are able however to communicate such signs, which attend the symptoms of madness in a dog, whereby we may the better avoid the danger of being plunged into the most deplorable misery by this wretched animal.

The Signs of a mad Dog.

The dog with symptoms of madness evidently discovers a malignant fever; he runs confusedly forwards; the tongue hangs out of the mouth with much slabber; his head hangs downwards; his eyes look dull and heavy; his tail is drawn up between his legs; and in every respect he has all the sign of confused wildness and pain; he runs confusedly forward, and snaps about him without distinction of persons or fear. He is always restless; will neither eat nor drink; and is terrified at the sight of water; the tongue hanging out of his mouth, grows black; and at last expires a spectacle of horror, unless the

poor beast is put out of his misery, and thereby many dreadful consequences prevented.

The Effect of the Bite of a mad Dog.

So subtle is this fatal poison, that the symptoms do not appear immediately; but the part bitten will sometimes even heal up, the same as any other wound; and two, four, or six months after the injury, the symptoms of the malady will appear and gradually reduce the patient to the most deplorable state in nature.

The first symptoms of the disease are, a pain in the part where he was bit, which gradually spreads all around; the whole body in general is attended with a dull heavy pain, a lassitude and weariness. The mind grows depressed and anxious, together with a loss of appetite and a nausea at the stomach; and, in particular, a difficulty of swallowing, sore throat, and accumulation of spittle.

Soon these symptoms encrease. The peculiar soreness of the throat, which renders the swallowing liquids very painful, begins to create the utmost horror and dread of every thing that is liquid; and this first gives rise to the *Hydrophobia*, or dread of water, which compleats the patient's misery in this terrible disorder.

The eyes become full and staring, the face bloated, florid intermixed with lividness; the mind is impaired; rage, lust and mischief, agitates the confused will; at length, convulsions, foaming of the mouth, priapisms, and seminal emissions, barking, howling, and endeavouring to bite every thing near, ends the wretched life of the most deplorable object in nature.

Of the Cure of the Bite of a mad Dog.

With respect to the remedies, we have had many delivered to us; but few indeed with success. Before I enter on the best method that I know, by experiment, to recommend, I must beg leave to mention a peculiar successful instance, in the course of my practice, in this deplorable case.

A mad dog on shore, in the West-Indies, bit two men, one belonged to my ship's crew, the other to a ship in the same harbour; both were bit in the calf

of the leg. It happened that the surgeon of the other ship and myself were immediately at hand at the time of the accident. I was for cutting the part bitten fairly out without delay; to which the man of my ship readily contented; and it was done accordingly upon the spot, having fortunately my pocket instruments about me. But this operation was not approved of by Mr. E——; nor suited it the timidity of the man belonging to his care; it was therefore omitted.

When I had performed the operation, viz. (as near as I can guess) I cut more than an ounce and a half of flesh out of the gastrocnemius muscle (every where round the bitten part) I spread a pledget with basilicon, mixed with precipitate, and laid it on the wound, having first let it bleed, perhaps six or eight ounces; over the pledget I laid plenty of dry lint; and when he was carried on board, I fomented it with spirits of wine and camphor, in which *sublimated mercury* was dissolved (about one grain to about four ounces of the spirit). Inwardly I gave him one grain of calomel every morning and night for at least a fortnight. The sore I continued to dress with mercurial ointment; which within a month healed up perfectly sound. From the mercurial treatment, the patient had a small ptyalism, which I checked by gentle cathartics. My patient, by this severe, though efficacious method, recovered perfectly well, and remained so at least three months after the accident, when we parted. The other unfortunate man, however, had not so good luck; for, as I was afterwards informed (the ship going soon to sea after the accident) the unhappy sufferer died a horrible object, raving mad in the *hydrophobia*.

Mercury is certainly the greatest antacid we have; and I am firmly of opinion, that in this case it is the most powerful of any medicine that is yet known.

The cold bath is strenuously recommended by Dr. Mead; and since his recommending it, is generally practised on suspicion of this misfortune; but in real cases, very seldom with the wished-for success. Yet, Reason says much in its favour, and should by all means be had recourse to. Others have recommended strong salt brine, which also is not to be rejected.

Before I lay down the method to be observed, I shall first give Dr. Mead's recipe in his own words.

“Let the patient be bled at the arm nine or ten ounces. Take of the herb called in Latin, *lichen cinereus terrestris*; in English *ash-coloured ground liverwort*, cleaned, dried, and powdered half an ounce. Of *black pepper powdered, two drachmes*, mix these well together, and divide the powder into four doses, one of which must be taken every morning fasting, for four mornings successively in half a pint of cow’s milk warm. After these four doses are taken, the patient must go into the cold bath, or a cold spring, or river, every morning fasting for a month: he must be dipt all over, but not stay in (with his head above water) longer than half a minute, if the water be very cold. After this he must go in three times a week a fortnight longer.”

But to speak my real sentiments, I have no great opinion of this recipe; yet as other methods may be pursued along with it, I thought proper to insert it, as in a malady of this consequence, nothing should be left undone that is practicable with propriety.

Celsus I think was the first who recommended cold bathing^[14]; but indeed to a degree of drowning and reviving the patient again alternately. In the mania itself, it carries much reason with it in this miserable extremity, for the poor wretch at the eve of expiring, a spectacle of horror, may as well run the risque of drowning; and besides, the shock that plunging gives to the whole system, may probably work an alteration in the malignant habit.—But alas! when the hydrophobia has actually commenced; Death, and death only—can give relief! Yet let it ever be a maxim, (and especially with a seaman) *never* to forsake *hope*; for *despair* is only a punishment reserved for the damned. I repeat it; no means should be left untried, that has the least prospect of recovery.

¹⁴. *Unicum remedium est, nec opinantem in piscinam non ante ei provisam projicere, et, si natandi scientiam non habet, modo mersum bibere pati, modo attollere; si habet, interdum deprimere, ut invitus quoque aqua satietur: sic enim simul et sitis, et aquæ metus tollitur. Lib. v. cap. 27.*

But to avoid being tedious, I will proceed to the method I would have pursued in an accident of that terrible kind.

If you by the before description find that the dog who gave the wound is actually mad, then without a moment’s delay let the piece bit, be cut fairly out; if that is not practicable, apply immediately the ointment made of *mercurial ointment* and *precipitate*, No. [XXI](#). Over which apply *mercurial*

plaister. Give one of the pills No. [X](#). every morning and night, and continue this course for a whole month. Let the *plaister* remain on the place a good while after the sore is healed up.

This method I pursued with a gentleman, and a little boy in New-York, who both were bit by the same dog, without provocation, the dog having never attempted the like before; upon which he was immediately shot, and was found to have all the signs of madness.—The accident happened at noon, and I was immediately sent for; but not being in the way, my assistant dressed it with basilicon, and laid over it a compress with *spirit of wine and Camphor*, supposing it of no consequence, as the wound was but very small. But on my hearing of it in the evening, I went that moment to their assistance, removed the dressing and applied what I have before mentioned.

I was not mistaken, for the sore and all around, after some time grew angry, and had an unfavourable appearance; but with diligent attendance gradually became better.— The little boy was bit deeper than the gentleman; His wounds were on both sides of the *tendo achillis*: But the gentleman's wound was a little above the *maleolus externus* on the right leg, inclining to the *tendo achillis*.

It was in the month of July—I ordered bathing every morning and night, which suited the season. In the mean time, during three months, I gave one grain of calomel every morning and night; and every thing succeeded to our wishes.

But to return; should the mercurial medicines go to the mouth, a gentle cathartic will divert this, for I am an utter enemy to salivation. If however these remedies are administered too late, so that, notwithstanding all those applications, the patient should grow melancholy, have a difficulty in breathing, and a sore throat; but especially a pain in the sore, or bitten part, recourse should be had to frequent cold bathing, or rather plunging; mercurials should be continued, and every method pursued as will be advised in the putrid malignant fevers.

Should the patient however be seized with madness, care should be taken that those about him have no wounds where the least of his spittle may be introduced; and that the deplorable patient is rendered incapable to make any wounds or mischief on himself, or those who attend him; as he will be apt to snap and bite about him like a dog. He should be sewed up in a

hammock, to which his arms should also be confined: in this method he may, by help of a tackle, be plunged up and down in the sea, but not his head under water. This is far superior to the unnatural and cruel way of smothering with beds, where the patient has no chance at all left.

The subject is melancholy; but, willing to contribute all the relief in my power to the distressed, I have been insensibly led on; and summed up all that might promise success. Innumerable are the remedies delivered to us, but few prove successful if the case be real.

Of all, I never knew any meet with better success than where mercury was concerned; and as I have been so fortunate as to have experienced the exalted virtue of that great medicine, in this and other deplorable cases, I stand so much upon the firmer grounds, to recommend it beyond any other medicine that I am acquainted with at present.

To conclude, I think it requisite to observe, that, dreadful as the bite of a mad dog is, when the poison has infected the unfortunate patient, yet this unlucky accident happens much seldomer, than mankind from the dread of the misfortune, are apt to apprehend it does. Many nostrums are published as specifics, and their virtue artfully confirmed, with cases of being bit by dogs, that had however not the least symptom of madness; and thence cures enumerated, that could have no foundation on reality; by which artifices much mischief has been done in real cases. Due observance should therefore be had to the state of the dog that has given the wound, and the circumstances attending; that either the patient may not be brought into unnecessary pain, or that proper and substantial remedies may not be neglected.

LECTURE V.

OF

FEVERS AND

INFLAMMATORY DISEASES,

AND

THEIR PROPER METHOD OF TREATMENT.

We now proceed to the more internal diseases; in which, the various steps that we intend to take of restoring health again, must be guided with the greatest caution, as the seat of the malady is hid from our external examination.

It is for this very reason I designed previously to prepare the reader with the two first lectures; particularly the second, I must beg the reader thoroughly to examine, and to understand; for on this will in a great measure depend the proper application of the various experienced remedies, which I here mean to unfold.

SECT. I.

Of Fevers in General

Hippocrates, the father of physic, was wont to say, “A fever is so universal, that no patient dies without it.” This is so true, that there is hardly an exception to it, even in the most momentary apoplexy itself.

To conceive this properly, we must observe, that every diminutive part in the animal fabric, like in a commonwealth, is endued with a sense of self-preservation; and the whole fabric is in so strict a conjunction with all its various particles, both solids and fluids, that nothing which may cause the least disturbance can pass unnoticed; but, let the injury be where it will, every individual particle, as a member of the whole animal state, feels the shock, and is immediately concerned for its own preservation.

This then is the cause of a fever in the general sense; and thence we shall be led to understand what this disturbance in its nature actually is; namely, an unnatural action of the solids upon the fluids.

Now as the fluids are the most yielding parts, that must submit to the pressing motion of the solids, we may without much difficulty, see one principle reason of its being set into a disturbed commotion, when the irritability of the nervous fibrillæ are acted upon.

Again, when we consider that the blood is liable to become acrimonious, and thence by its stimulus apt to act upon the coats of its containing vessels, we see demonstratively plain, how the solids are subject to be excited to act upon the very blood that stimulates them.

Moreover, as motion must naturally produce heat, and heat motion, and motion expansion, and expansion rarefaction, we at once bring to view the principal laws of mechanism, that governs the human automaton in the various degrees of health as well as disease. However, let me not forget to whom I write, but haste to the point I am at.

A fever may proceed from many different causes, and may affect many different constitutions, and also produce many different effects; it is therefore just that a proper distinction should be made.

Indeed, it is a subject that cannot be too minutely handled; because it is scarce possible that there is a disease void of a fever in some degree or other.—But preciseness and utility is the object I have in view.

SECT. II.

Of Accidental Fevers.

How uncertain is human life! One moment in a perfect state of health, the next perhaps no more; or confined under the painful sensations both of body and mind!

Every accident is liable, and unavoidably must bring on a fever; and this, like fire, should be extinguished in its earliest commencement. But in that, as I have before observed, the constitution should principally be consulted.

I have in the lecture of accidents, at every opportunity, recommended bleeding, but always with a proviso, *if requisite*; and, as this is of more consequence than generally is supposed, I find it necessary here to expatiate farther on it.

In the first place, no man should be bled who has lost a considerable quantity of blood, unless indeed he was known in his state of health to be very strong and robust; and that the fever sets on with such fury, that the system requires absolutely to be reduced; for the solids and fluids have such a strict dependance amongst themselves, that they keep each other in an equipoise. If therefore the blood is of an acrimonious nature, and at the same time impoverished by lessening its quantity, the coats of the arteries and veins will be the more stimulated; thence the fever will naturally encrease, and generate more acrimony; which at length will destroy the patient. I have been an eye witness more than once, that the fever has actually increased upon every repeated bleeding, till the whole supply of life was spent; and all my persuasions have not been listened to till too late. Nay, a fever is sometimes absolutely necessary; and proves to some constitutions a remedy on its own account, provided it is properly treated, and kept within due bounds.

Secondly, If the constitution of the patient before the accident was but slender, and of a phlegmatic nature, bleeding should be very little or cautiously performed; for the blood of itself being but poor, must naturally

be impoverished by being lessened; and whilst the tenseness of the fibres is lost, it is the more acted upon and rendered more acrimonious, and still unable to strengthen the tone of the irritated fibres; whence a relaxed body, together with a poor acrimonious blood, and all its evil tendencies must be the consequence.

Let it once for all be observed, that all the benefit we can possibly hope and expect from bleeding, is to lessen the whole system, to reduce the quantity of the fluids, and to relax the solids. This lessening of the system I must own is very often highly necessary; But, let us not fall into the ridiculous notion to suppose that bleeding will draw off the bad blood, and leave the good behind; far otherwise; the quantity of the blood is soon replenished, but the quality becomes impoverished.

The principle intention in fevers of whatever kind, should be to promote perspiration. This is the grand restorative of nature; and I cannot recommend it too much. Nature designed this evacuation in the formation of man; and by that, fevers were cured in time of yore, many ages before art contrived either lance or lancets.

But here let me not forget another great caution; namely, not to mistake profuse sweating for perspiration; for this is an extreme into which many have erroneously fallen. Nature frequently produces a sweat without force or compulsion; and if so, it should be favoured, but never brought on by stimulants, or strong forcing medicines.

Perspiration, sweat and urine, are the natural evacuations intended, for the purifying the system from animal acrimony; and when these excretions are in a just balance with the secretions of the fluids and motion in the system, without pain or fatigue; then nature is at peace with itself.

Lastly, I must observe, that the stomach is an organ which, by the nervous consent, carries on an immediate correspondence with every part in the human fabric; whence this viscera should likewise ever be consulted in fevers; that is, never to impose on it any aliment against its own inclination and appetite.

The want of appetite in a fever is a natural cause, and very often is in the patient's favour; but the mistaken notion, that a patient must eat, even against his inclination, in order to recover strength, is not only highly

absurd, but has been the destruction of many, who otherwise would have recovered, had they with patience waited the kind dictates of Nature.

Let the system be free from diseases, and the stomach will soon crave food, when there is a probability that the aliment may be converted into good blood; but whatever is forced into the stomach before that time, will instead of nourishing the body, only add fuel to the destructive fire, and encrease the already malignant blood.

Upon the principles of these observations, I shall lay down the rules that should be religiously observed in accidental fevers; and when they are well considered, we shall find that they will hold good also in all other kind of fevers, notwithstanding they arise from internal acrimony of the body; for the difference is barely this, whether the fire is kindled within or from without;—but to proceed.

Whatever then be the accident, a fall, a wound, a fracture, dislocation, bruise, &c. observe what the patient was before the accident.

If the patient was a stout hearty man, it naturally follows, that the fibres of his body were also in a rigid robust state; and the quantity and richness of the blood was in a due proportion; for this very reason, health being at such an *acmé*, a fever is the more to be feared, and ought studiously to be avoided.

If this patient has not lost any blood by the accident, by all means bleed him; but never exceed eight ounces at farthest; it is as much as the strongest man ought to lose at once.

In the next place, perspiration should be the principal object in view. To this end, let the patient immediately after bleeding go to rest, and take a dose of the *Fever Powder* (L); upon which let him drink plenty of *Balm Tea* (12); which will generally bring on a perspiration.

If however the fever increases, the pulse continues to beat hard and full, the head-ach, difficulty of breathing, a great draught and delirium should ensue, bleeding indeed may be repeated, provided it is within the same twenty-four hours; but if the symptoms come on beyond that time, then bleeding will seldom prevail, or do any good, but rather mischief; for by this time, the fever is generally agitated by acrimony, and the blood is of its own accord poor enough without being made more so by bleeding.

In the symptoms which I mentioned, regard should be had to the body being gently kept open, not by purges but by clysters; for purges in fevers are precarious, and liable to bring on heavy fluxes. If the patient under this high fever is costive, give him the following clyster.

No. XXIII

Take warm water one pint, or something more; in which scrape about one ounce of castile or hard soap, and let it perfectly dissolve; then add a spoonful of honey to it, and let it be as warm as that it may be easily borne to your cheek.

Let the patient keep this clyster within him as long as he can, till it comes away with a stool.

If the patient is sick at the stomach, a vomit of *ipecacuanha* (S) or No. [XXVI](#), will be very proper, and nothing in such a case can be given more safety and better advantage; for it is not only the stomach that gets relieved, but the whole body is brought thereby into perspiration, by the universal strain it occasions. After the vomit, or after the clyster, give the following draught.

No. XXIV

Take of fever powder (L) one scruple; common water or balm tea, one ounce or half a tea cup full; then take a lump of sugar, on which pour essential oil of peppermint (U) three drops; liquid laudanum (T) eight drops; sweet spirit of nitre (D) thirty drops; and mix them with a tea spoon.

This draught will not only promote perspiration, but it will comfort the stomach, and incline the patient to an agreeable rest; by which he will be greatly refreshed. If occasion requires, it may be repeated every six hours.

If rest comes of its own accord, the *liquid laudanum* (T) may be omitted; and if the stomach does not stand in need of any carminatives, the *oil of peppermint* (U) may also be omitted, &c. The constant drink when dry may

be balm tea; in which, if it suits the palate of the patient, a slice of lemon may be put, and then sweetened with sugar.

The diet, as I have before observed, should be sparing; and the patient's palate in this should principally be consulted. The following water gruel, or rather wine soup, or whatever other name it may acquire, I have often ordered to my patients at sea; and which has suited most stomachs, and proved an agreeable mess.

No. XXV

Take oatmeal, or pounded biscuit, a couple of spoonfuls; water, a quart; a small handful of raisins; a little allspice, a little mace, tied up in a fine rag; which boil together in a tin saucepan till consumed to a pint and a half; then add a gill of good wine, red or white, and sweeten it with sugar to suit the palate. You may put a little lemon peel in, to give it an agreeable flavour.

A little roasted fresh meat will hurt nobody; and broths most certainly are good, particularly if the body is costive; but when the body is inclined to looseness, I have known the best broth to produce a dangerous flux; and this should be taken notice of.

If the patient however who has met with the accident is of a puny weak constitution, or has lost a sufficient quantity of blood by the wounds, bleeding then is not so requisite as in the former case; though the treatment otherwise must be equally the same.

By this method of proceeding, there is little fear but that the patient will soon perfectly recover, even in the highest fever that may suddenly befall a man from an accident; provided the injury is not of a dangerous tendency. But as an acrimonious habit of body is liable on its own accord to diseases, it is easily imagined that an accident may become accessory, and heighten the malignity; hence often a fever of that kind is liable to degenerate into a malignant one, or even from the accident itself, when of a dangerous nature; and this I shall explain as I proceed. I shall therefore leave the accidental fever that proceeds from an external cause, and proceed to those which take their origin internally.

SECT. III.

Of a Cold.

The fever now to be treated of is the most common amongst mankind; especially seamen, who are ever exposed, to have the perspiration obstructed. It however would save many from the grave, and almost as many from being food for the fish, was it a little more regarded than what it actually is.

The sound of a cold is so gentle, that custom has almost made it effeminacy to pay any regard to it; and yet, if we maturely examine, we shall find the most malignant, acute, and mortal diseases, first make their appearances by the symptoms of a cold,—But let us proceed to particulars. A cold is an obstruction of perspiration; which may become destructive to the animal œconomy if neglected, in so much, that it may be considered as the first foundation of most diseases.

In my treatise on diseases of the lungs, I have enumerated the many causes of a cold to which mankind are liable, and their precautions to avoid them. To sea-faring men, I have one principal cause to mention whereby they make themselves subject to colds, and sow the seed of many dangerous diseases in their system, and that is their carelessness in shifting, after getting wet; in this respect, I have however a remedy to offer, which is as valuable and as efficacious to their preservation, as it is easily put in execution. The remedy is simply this:

A man who has been exposed on duty in the rain, and is become thorough wet, should not only shift himself in dry cloaths, but previously wash all over with sea water: if he carefully observes this, he will never get cold on that account. I have not only experienced this personally many times, but ever found it productive of salutary effects to those I have persuaded to it. —If we but make some observations on the accidental occurrences that happen at sea, we shall see the just foundation of this. When do we find a man who is continually washed by the sea to have a cold? while on the

contrary a foul rainy day, will almost breed a general sickness.—These observations first brought me on the notion of washing the body with salt water, and having verified it by experiment, I soon saw the propriety of it; namely, that the fresh water being of a sluggish nature, and having a tendency to putrefaction, is not only apt to infuse a putrescency into the system, but from its slimy nature apt to clog up the pores, and thus obstruct perspiration; whilst on the contrary, salt water is not only antiseptic, but stimulates the pores of the skin to their secretory and excretory functions. —Nay so perfectly have I experienced this remedy, that I have persuaded the people to salute one another by way of play with buckets of sea water on a foul rainy day, and with pleasure I have experienced that they have all remained brisk and hearty after it, and evidently seen the difference, that on a similar day the sea water play being omitted, scarce one who has been thorough wet, escaped a cold in some degree or other.—I must not forget, that a draught of grog is likewise no bad preservative. But this is best taken when the watch is out, and that rest and sleep can be indulged upon it, as that will then be apt to throw them into a gentle perspiration; whilst however their watch is upon deck, the best warmth is that which is produced by exercise; except liquors is taken with the greatest moderation.

But when a cold has actually commenced, care should be taken immediately to have it subdued in time. The term cold however is a very vague expression, that includes a great many disorders, occasioned by an obstruction of perspiration, which all in turn shall be taken notice of: But as I am now on the subject of fevers, I shall confine myself to that at present, and consider the cold as it affects the body universally; namely headaches, a weakness in all the limbs; intermittent fevers, intermixed sometimes with flushes of heat, and at other times shivering, chilling, &c.

These are not only the symptoms of a cold, but of almost all inflammatory fevers; which in fact, the cold is a commencement of. Now all that nature requires is a perspiration; for thereby the imprisoned acrimony is expelled, and which nature is always engaged to throw off from the system.—To this intention, if the patient is rich of blood, bleeding should be performed; but with such caution, as I have before recommended. Next give the *fever powder* ([I](#)), together with a good quantity of balm tea, or if the constitution is of a phlegmatic habit, sage tea, or even sage punch (No. [V](#).) will not be improper; but care should be taken, that if there is much

fever, all those things should be omitted, as it is liable to increase the fever. If there is a nausea at the stomach, give a vomit: take the following:

No. XXVI

Take Ipecacuanha (S), one scruple: sugar a little lump: water half a Tea cup full, and make it up into a draught.

This vomit should be well worked off with chamomile tea, or even common luke warm water; first letting the vomit operate, and then drinking plenty of the water after every operation. When this is over, the patient should keep his bed or hammock, and take either the *fever powder* (L), or the before mentioned draught, No. [XXIV](#).

The catarrhal fever, and all the other symptoms of cold, viz. cold in the head, cough, hoarseness, &c. generally keep one another company, leave the patient generally at one time; and in fact, one method of cure, namely, promoting a perspiration, serves for all; for thereby the obstructions become removed, and nature is restored to its primitive tranquillity again.

SECT. IV.

Of Intermitting fevers, and Agues.

There is scarce a disease that makes a patient more wretched, and unhappy, than the fever and ague. The causes of these kind of fevers is principally from a pituitous, or slimy blood, together with its being loaded with acrimony; and a flacidity of the moving fibres.

They are divided into various classes; though in fact all of one nature. When the ague with its succeeding fever comes on regularly every day, it is called *quotidian*, or every day's ague; when every second day, *tertian*, or third day's ague, and when every third day; a *quartan*, or fourth day's ague. There are many other classes as subdivisions of these, but they are the mere speculations of physical pedants, who are ever sworn enemies to plain understanding, and delight in perplexities; which therefore, we shall neither spend paper nor time upon to recount.

Suffice it, let their paroxysms return at what stated period of time they will, either regularly or irregularly, they differ in nothing but this, that the farther the fits are distanced, the severer they generally are, when they come on; and the more regular, the more obstinate in their cure; though in this even there are exceptions.

It is surprizing to imagine, how many different opinions have been sent into the world concerning the returns of the periodical fits of the ague; and equally astonishing, how few with any probability of truth or reason.

I shall not here enter into an enquiry of this nature; certain it is, that at the stated times of these returning fits of the ague, there is an obstruction in some particular part of the system, that corresponds with the dated time of the circulation of fluids in such parts—which when disturbed, effects by the nervous consent, the whole animal structure.

The principal intention in all intermitting fevers, should be to correct the petuity and acrimony of the blood; to remove the obstructions; and to

strengthen the solids.

With respect to the blood, it is an inevitable consequence, that when it is loaded with a vicidity and lentor, that it will ever occasion a depraved nutrition; and hence we see, that people labouring under agues, and intermittent fevers of any kind, have a depraved appetite, and many disagreeablenesses in the first passages.

The intermittent fever and ague approach with the following symptoms, viz. a heavy dull head-ach, a shivering chill all over the body, with a yawning, &c. pain in all the limbs attended with weariness, a nausea at the stomach, a depressed pulse; which after it has lasted for half an hour, changes into a burning heat, a throbbing head-ach, pain in the loins, and in every limb, a great draught, a *full hard pulse*, and which at last terminates in a profuse sweat; this seems to give the patient ease, for one, two, or three days; and then attacks the patient with the same symptom, increasing every time, and reduces the patient both in body and spirit.

The first thing to be done is to give a vomit, like No. [XV](#). and XXVI. This is best taken at the approach of the cold fit, and worked well off with plenty of luke warm water. After the operation of the vomit is over, let the patient take the following draught.

No. XXVII

Take fever powder (L), thirty grains; calomel (N), one grain; elixir of vitriol (G), sixty, or seventy drops; essential oil of peppermint (U), four drops upon a little lump of sugar; to which add water, half a tea cup full.

This draught will not only promote perspiration, but help to dissolve the siziness in the blood; and may therefore be repeated every six hours, for a few days. But the patient should always be kept moderately warm upon it, that it may work by perspiration.—This simple method I have known put the patient to rights again, after a severe onset.

But sometimes this wretched disorder will require more powerful medicines to keep it in subjection. When the fibres are debilitated, and the blood still retains its viscid corruptive quality, the ague is apt to hang on the

patient a long time, the succeeding fever apt to melt the patient down to a mere skeleton, and even destroy him. In this it is requisite to brace up the solids, and to infuse an astringency into the blood.

To this intent, the Peruvian bark is now frequently prescribed, and if judiciously administered it is an excellent medicine; but as it is liable however, to be improperly given, and then productive of irreparable mischief, I have purposely omitted it in this catalogue; nevertheless, I shall give in its stead a medicine, superior both in virtue, and by far more safe in the application; not only in this kind of complaint, but in scorbutic cases, as I shall hereafter take notice of. The medicine is this,

No. XXVIII

Elixir of Iron.

Take a quantity of rusty iron, the smaller the pieces are, the better, for the less quantity will do; pour on it good strong vinegar, sufficient to cover it; let them steep till the vinegar becomes of a deep brown colour, which it soon will, and at length acquire a consistence of a black elixir. The vessel should be either stone or glass; but what is yet better is an iron pot, which of itself will contribute to the medicine.

If the patient therefore, after the vomit, and the above draught, is attacked again with the fit of the ague, and succeeding fever, have immediate recourse to your elixir of iron.—Let him take after the fever is off, every three hours, a table spoonful of the elixir, and should it seem to come up again at first (which it sometimes will, particularly if the stomach is yet foul) give immediately another spoonful, and repeat this at least three times of the day.

A little gentle exercise upon taking the elixir is highly beneficial; and if it throws a gentle warmth over the body, and brings it into a gentle perspiration, you are sure of success.

The quantity of it should be adapted to the condition of the patient, and the strength of the stomach; if it sits easy, a larger quantity may be taken, and seldomer; and if it is more nauseous, a smaller dose and oftener.—

Should the fits seem obstinate, the following electuary may be used; and is a powerful specific.

No. XXIX

Take stomach powder (M), two drachmes; nutmeg powdered, one drachm; powdered alum (17), half a drachm; essential oil of peppermint (U), twenty drops; calomel (N), ten grains; honey (E), one ounce: the whole well mixed.

Of this let the patient take every eight, or twelve hours, the bigness of a nutmeg, taking his elixir as before observed; and this will ever prove successful in the most obstinate case.

As patients of that kind at sea, have ever a taint of the scurvy in their composition, or at least have a tendency towards it, exercise and freshness of food, will prove one of the restorative means. The malt decoction should also be their drink, as well as in the scurvy.

To conclude, I shall only observe, that bleeding in agues must at all events be omitted, as being absolutely pernicious, as it cannot fail of impoverishing the blood; which is the principle source of the disorder. Also, in the intermitting time, exercise should be observed, and the diet should, when the stomach craves, be good. Sometimes indeed, the stomach is too ravenous and greedy; but that is a default which proceeds from the same cause as that which makes it loath its victuals; a vomit in either case is the most effectual medicine, and may be repeated as often as it is requisite, without the least fear of injury, particularly in this disease.

SECT. V.

Of Putrid and Malignant Fevers.

I Shall on this head restrain myself from saying more than what is absolutely requisite to the cure. It seldom happens to infect a ship's crew at sea, unless the ship has carried it from some place where the disease was contagious.

Though notwithstanding, great sickness on board may give rise to it, if care is not taken to prevent it. I would therefore ever advise, in case of sickness, to steam the birth with vinegar; namely, taking a vessel with vinegar, in which put a hot iron, and thereby raise a steam, which is a powerful preservative.

A putrid fever may be generated from the contaminated atmosphere of a sick person, though it rarely is the case; yet prudence demands every precaution that is in our power.

The fever that I now speak of is of the malignant kind, and peculiar to the hotter climates.

The patient is seized first with a shivering chilliness, which is soon succeeded with a weak unequal fluttering pulse; the body is seized with a universal pain, together with a great debility; the mouth is hot and dry; the respiration depressed; and the mind labours under a heavy anxiety.—The fever continues, though not very high; and the pulse not very full;—a delirium or else a stupor ensues; the urine becomes deep coloured, and has an oily skim at top; and the patient is in so imminent a state of danger, that, unless a favourable crisis happens soon, twenty-four hours rapid progress is liable to put him beyond all help. The first and immediate remedy should be a vomit, No. [XXVI](#), which is the grand step to be taken in all fevers; not so much for the sake of cleansing the stomach, as for the universal ease it gives to nature.

The next step is to correct the putrescent acrimony in the system; for which purpose make the following draught.

No. XXX

Take fever powder (L) one scruple; elixir of vitriol (G) thirty drops; sweet spirit of nitre (D) thirty drops; calomel (N) two grains; essential oil of peppermint (U) four drops; upon a lump of sugar; and half a tea cup full of water.

Let the patient take this draught every six hours; and after four have been taken, the calomel should be omitted. If the nerves seem to be affected, the essential oil of peppermint may be increased. If a delirium should come on, the *liquid laudanum* (T) from ten to twelve drops and upwards may be given in the draught.

I know many will think it odd, that I should give calomel in a fever; but all this is merely the effect of blind prejudice; a better medicine than calomel is not in all Nature, if judiciously applied; and so fortunate have I been in my success with this exalted medicine in contagious fevers, that I have preserved myself and my patients in the midst of Death's spreading horror around us. But to proceed; if the patient continues to be dry on the surface of the skin, and perspiration seems backward, a blister betwixt the shoulders, the bigness of the palm of the hand^[15], will greatly facilitate the crisis or turn of the fever; but let it not be erroneously supposed, that a blister will draw off the bad humour, or that it does good by such an operation; for this notion has proved destructive to many patients. All the good a blister does is to spur on the animal function; by which sometimes the acrimony is thrown off from the system; but if thus irritating the system be overdone, or wrong applied, it is productive of many additional evils to the disease; causes spasms, strictures, and other acute disorders, that of themselves may become mortal. Therefore blisters should never be used but when the body is defective of perspiration, or when the debilitated system is defective in its excretory functions.

¹⁵. The method how to apply a blister is described under the article of Blister Plaister, in the second section, p. 75.

When the patient is rid of the fever, a general weakness will be left behind; this is best assisted with the *elixir of iron*; and now and then as a stomachic, a tea spoonful of *Turlington's balsam* (W), or some few drops of *oil of peppermint* (U), as also the malt decoction, and fresh diet.

Malignant fevers are generally attended with eruptions on the skin, which is sometimes in favour of the patient, and therefore by no means to be hindered; but if there appears purple spots on the skin, and that they incline to turn black, it is rather a dangerous symptom, and indicates very little hope of the patient's recovery.

The same method as above must be observed; which if medicine can avail, will afford benefit.

On the whole, I have this to observe, that all acute fevers are liable to become malignant and putrid; and in my practice I never found, that there was any real distinction to be made in the method of cure, whether the acrimony was generated in the habit by disease, or introduced by contagion; and if I may be allowed to judge by comparison, it must appear that my method is superior to the more general doctrine; because I have ever found it to be attended with much more success.

LECTURE VI.
OF THE
VARIOUS INTERNAL DISEASES,
AND
THEIR METHOD OF CURE.

Having in the foregoing said what I thought most material to be observed in fevers, as the universal disorder that attends almost every ailment in the human body, I shall for regularity sake, treat on the most material diseases, as they may seem to follow in order in the various divisions of the human body; and be as concise in their description and cure, as their nature will admit of.

SECT. I.

Of Diseases of the Head, Of the Apoplexy.

The apoplexy is a disease, wherein the patient falls down motionless, void of sense and sensation, and is, in fact, a temporary death. It will scarce however admit of a strict definition; for there are not only various degrees of this disorder, but even different disorders under the same denomination, that have quite distinct different causes for their effect. Many indeed may be accounted for, but many more cannot.

The brain is the root of the nerves, and consequently the fountain of all sensation, and life itself. That part of the human fabric is extremely delicate, and blood vessels plentifully distributed. When therefore these vessels become either so distended with blood, so as to press the substance of the brain, that it is hindered from its motion, confusion to the animal œconomy must naturally ensue; thence an apoplexy in its various gradations is liable to be the consequence; but as the vessels in such a distention are liable to break, and the extravasated blood pressing the brain more forcibly without a chance of dispersing, we plainly see one reason, why such a malady is apt to turn out fatal, as we find it sometimes does.

Such causes may either proceed from external, or internal injuries; thence we see that a heavy fall, or contusion on the head, may occasion an apoplexy, or at other times, a congestion of blood to the head may produce the same malady.

Sometimes the patient is motionless; and to all appearance dead, and nothing but a faint pulsation remains; at other times there is indeed more signs of life, but more terrible in appearance,—namely a convulsive breathing and foaming at the mouth, without the least signs of sensation or perception; in which case there is much to be feared, that vessels in the brain are burst.

However in all these extremities, nothing can contradict the indication of bleeding, except extreme weakness, and debility of the system; which here is very rarely the case.—After bleeding, a large blister between the shoulders should immediately be applied, and treated as directed in the second lecture, page 75.—A draught like No. [XXX](#). will also be of singular service, till the patient comes to; after which he will require such treatment as is directed under fevers; for that disorder will always attend the shock that the apoplexy gives to the animal system.

Of the Frenzy and Delirium.

When the root of the nerves are stimulated by acrimony, so as to heighten their sensibility, or set their regulated perception in confusion, the ideas become bewildered, and the sensations act at random. This is generally the case with patients in acute and malignant fevers; and this conception differs from dreaming in this respect, that the phantoms of the confused imagination may be somewhat guided by the objects around them; as they may be said to dream with their eyes open and full awake.

When the frenzy attends acute fevers, the blood is generally in a great confusion, and the pulse high and fluctuating. Yet bleeding should by all means be avoided, unless indeed the disorder has come on remarkably sudden. Blisters here are requisite; and in order to take off the stimulus of the nerves, give the patient the following draught.

No. XXXI

Take Fever Powder (L) one Scruple; Liquid Laudanum, (T) ten Drops; Spirit of Wine and Camphor (A) twenty Drops; Essential Oil of Peppermint (U), four Drops upon a lump of Sugar; Water half a Tea Cup full.

This draught may be taken every fourth or sixth hour; and now and then a grain of calomel may with great success be added to it. Cooling clysters in this case are also of singular service; for example:

No. XXXII

Take cold water, eight ounces; Nitre, or instead of this^[16] Fever Powder (L) two drachms; spirits of Camphor (A) thirty drops, mix them.

- ¹⁶. The fever powder may be steeped in a little water before hand, and the clear poured off, because nothing but the nitre here is useful.

If the patient is very costive, a little honey and soap may be added, previously dissolved; which will both open the body, and promote urine.

For the rest, every method must be taken as is already observed in fevers.

Of a Stupor, and Lethargy.

This is a symptomatic affliction, that attends acute and malignant fevers, in its nature quite opposite to the former; in which the nervous system is benumbed, and have lost their sensation; this is very apt to succeed, but seldom precede the Frenzy. When a patient lays in that kind of stupefaction in a continual dosing, there is in general little hopes of recovery; nevertheless remedies must be applied.

Blisters here are very proper, and these should be pretty large. Horse radish should be frequently held to the nose, which has a reviving quality, and is very innocent with respect of adding acrimony to the system.

A vomit has often in this extremity proved very beneficial; which the patient's weakness should not retard, unless very weak and near the end indeed. The following draught may also prove highly beneficial.

No. XXXIII

Take Sweet Spirit of Nitre (D) one drachm; Essential Oil of Peppermint (U) ten drops upon a lump of sugar; water one ounce, mix them.

Let the patient take this every four or six hours, till he comes to himself again:

But care should be taken that a good sound sleep, is not mistaken for a stupor; for it frequently happens that a patient upon a favourable crisis will fall into a long sound deep, that terminates all his danger; if therefore the

patient breathes freely, and has a moistness all over the body, together with a tolerable regular pulse, disturb him not; but let nature finish her gracious work.—This kind of sleep is materially different from the stupor, wherein the patient neither sleeps nor wakes, but is equally insensible when the eyes are open, as when they are shut.

Of a Violent Head-Ach.

A Violent head-ach often attends fevers, or comes spontaneously without; it generally precedes a delirium, as the irritation of the nerves is the principal cause thereof.

The first intention must be to draw the humours from the head. A compress with vinegar, in which is dropt a little spirits of wine and camphor, is of great service, and will often have the defined effect to carry it off. Warm foot-baths also is of great use; a blister may be applied, and the draught No. [XXXIV](#). will also prove of great service.

The head-ach may proceed from many other different causes, and in the general is only a symptomatic affection; therefore particular attention should be had to the primitive seat of the disorder.—Frequently a foul stomach will occasion the head-ach, and a vomit most times prove the best application.

SECT. II.

Of Diseases of the Neck

The diseases in that part, are indeed very numerous; but as they have so strict a connexion amongst themselves, the classes may be considered but very few; of which some have been already taken notice of.

Of the malignant Sore Throat.

Every body who is liable to catch cold, is frequently apt to have a hoarseness and consequently in some degree, more or less, a sore throat. If a fever attends it, methods should be used accordingly, as is mentioned under the head of colds: with respect to the throat, use the gargle, No. [XVIII.](#) pretty frequently, and the disorder will soon go off again.

But when an ulcerated sore throat is attended with a malignant fever and inflammation, it is called a quincey; this case is alarming, and requires very careful treatment; for first we must observe, that the larynx is of a very complicate and curious structure; and secondly, it is composed of various, and delicate materials; namely, cartilages, membranes, muscles, and glands.

The glands are very numerous in the various parts of the composition, whence the inflammation becomes not only painful, but is apt to swell up and obstruct respiration; wherefore an inflammation in such parts becomes highly dangerous to the life of the patient.

When the patient perceives a burning pain in the throat, and the parts all round swell up, respiration becoming difficult, add to this a hoarseness, great difficulty in swallowing, there is no time to lose, for immediate assistance is wanting.

Bleeding is indispensably necessary, particularly if the inflammation is sudden; warm foot-bathing also is highly requisite; to remove the humours downwards; a gargle should also immediately be applied like the foregoing No. [XVIII.](#) with the addition of a little *camphor spirits*. Externally apply the poultice, No. [II.](#) and give the following draught.

No. XXXIV

Take fever powder (L), one scruple; calomel (N), two grains; liquid laudanum (T), twelve drops; water two spoonfuls; mix it into a draught.

This may be given every six hours, for two days successively; and this generally will break the disorder. The patient should be made to drink as freely as he possibly can; and perspiration should ever be promoted. A blister as the case requires, may also be used with much advantage. If the body is bound, clysters may tend to incline the humours to be drawn down from the part.

No. XXXV

Take Glauber's salts (19), one ounce and a half; honey (E), one spoonful; hot water, a pint and a half.

Sometimes the inflammation is so very great, as to choak up both the passage of respiration, as well as that of deglutition; so that the poor afflicted patient can neither breathe, nor swallow; and is not only liable to be suffocated, but even starved to death, besides the danger from the malignant inflammation.

In this exigency, have recourse to the following simple but excellent remedy.

No. XXXVI

Take a pipkin, or a tin sauce-pan, with a quart of water, a pint of vinegar, and two spoonfuls of honey; then take a funnel that will pretty well cover the vessel; then take a piece of iron, or some old nails, or a little stone, or what you have handy; heat it red hot, and put it into the pipkin, and put on the funnel so that steam may ascend through it. Over this let the patient hold his mouth, so that he breathes nothing but the steam; and to secure it the better, put something over his head, that he may reap the more the benefit of

it. When the steam ceases, heat it again as before. Instead of heating it with the iron, it may be shifted with boiling, but the former is generally more handy.

In the mean time, if the patient should by the inflammation, be deprived of all nutriment, recourse must be had to nourishing clysters: milk in this respect is the best; but in want of that, fowl broth may be used; and if that is wanting, take water gruel. Milk however is superior to all: I shall give a receipt of each kind.

No. XXXVII

Take milk and water, of each a pint; cinamon (16), one drachm; boil them and strain the liquor; then dissolve one ounce and an half of good loaf sugar; add six drops, of the essential oil of peppermint (U), upon a little lump of sugar.

No. XXXVIII

Take half a fowl, bruise the bones, and cut the flesh small; this boil in three pints of fresh water, with about one drachm of cinamon; strain it clear, and add as before, an ounce of sugar, with a few drops of the oil of peppermint.—A tea cupful of good white wine may be added, if the patient is very weak, and not feverish.

No. XXXIX

Take oatmeal, two spoonfuls; malt, one spoonful; cinamon, two drachmes. Boil it in two quarts of water for some time, till when it is strained, it becomes one quart; put to it fine sugar, six ounces; a few drops of essential oil of peppermint, and a tea cupful of good wine.

These clysters are perfectly safe, and very nourishing. Some have added salt to the animal broth, perhaps through a notion to make it the more palatable, and so of course more nourishing; but probably they forget that

the part they are now about to feed, differs widely from the depraved taste of the palate.

Salt is stimulating; besides, it indurates the aliment, that it never yields so much nourishment, because it always opposes assimilation; and it is universally known, that salt victuals is by far, less nourishing than fresh.

The quincey is sometimes so desperate, as to strangle the afflicted patient, particularly if assistance is given too late; in so much that *bronchotomy*, or opening the wind-pipe, has been the last resource in this deplorable dilemma. I have mentioned this operation, in the third lecture, and have only to observe, that in this case little hope is left after that operation, on account of the great inflammation of the parts; though we have instances of its success. If however the method which I have here laid down, together with strict observance of managing fevers is duly observed, there can scarce be a failure of success; for if I may be allowed to judge from my own experience and success, it is merely neglect, in the first setting out of these diseases, that for the most part, renders them so very desperate and mortal.

SECT. III.

Of Diseases in the Breast

Under this head, I mean principally to treat of diseases of the wind-pipe, the lungs, and the pleura, or membrane that lines the cavity of the breast, and encompasses the lungs.

Of the Cough.

A Cough is generally occasioned by a cold; unless it is symptomatic with other inflammatory diseases, or the relict of a heavy sickness; in which case, it is either near a-kin to, or else the consumption itself.

If from a cold, it seldom is deep founded, and generally has its seat in the larynx, or in the wind-pipe; though indeed sometimes in the lungs.

It generally begins first with a tickling in the throat, and occasions a short dry cough; which gradually becomes strong, and is attended with a discharge.

It is surprizing what absurd remedies are sometimes given for the cough; as if they supposed that every thing taken should immediately march at the word of command into the wind-pipe, and there set to work accordingly. Amongst the many celebrated nostrums, spermaceti, or whales fat is remarkably celebrated; it softens the phlegm say they; but in fact, a little fat pork, or hog's lard will do equally as much good.—The truth of the matter is, the cough proceeds from an obstructed perspiration. When a person gets cold, he generally first feels a snuffling, and obstruction in the nose, then a sore throat or hoarseness, and lastly a cough; all which is a gradual defluxion from the pituitous membrane, which reaches from the cavity of the nose, all down the lungs. Whatever medicines therefore, are capable of yielding benefit, must be calculated to open and remove the obstruction, and promote universal perspiration; whence therefore we find it is impossible to cure one, without the other.—Medicines neither do, nor is it

fit they should come immediately farther, than the top of the larynx; for if they happen to come into the wind-pipe, the most innocent medicine would certainly act the part of a poison.

The first indication therefore in a cough, is to promote perspiration; for then nature will repair the respirative organs with the rest.

Let the patient take a draught or two of good sage punch, No. [V](#). and thereby promote perspiration; and in order to continue this principal intention, and to assuage the slight inflammation in the throat, which the heavy cough occasions; make the following electuary.

No. XL

Take honey, or sugar dissolved to syrup, two ounces; lemon juice, half an ounce; or elixir of vitriol, two drachmes; sweet oil, one ounce; mix them into a linctus.

Take of this, every half hour, but half a tea spoonful; and swallow it gradually, so that it may as it slides by, do some service to the larynx; and keeping the body all the time in a gentle perspiration. This method is superior to all your nonsensical pectorals, which have nothing but quackery for their foundation; for whatever you may take for the cough, it cannot possibly have any other effect on the part, till the real cause is removed, than barely to touch the larynx; for all goes to the stomach, and undergoes the same chance as your victuals. Therefore strike at the root of the evil; namely, to promote perspiration; and all the rest of the complaints will vanish, in consequence of this principal intention.

Of the Pleurisy, and Peripneumony.

The *pleurisy* is an inflammation of the lining of the inside of the pectoral cavity and lungs; and the *peripneumony* an inflammation of the lungs themselves.

These two principal kinds of inflammation have given birth to the names of many other inflammations contiguous to those parts, which I hope to be excused from entering into, as it is of very little signification to the proper treatment of those disorders; for I have often known the most skilful practitioner, to be mistaken of these disorders in his diagnostics, and yet

been successful in his cure. The signs of this inflammatory disease, is an acute fixt pain in the breast; on whatever side the disorder is in, with a difficulty of breathing, and frequently attended with a small short dry cough, which however stings the patient like so many daggers. A constant fever attends with all its symptoms; which adds greatly to the patient's misery.

On the immediate signs of these symptoms, bleeding must be had recourse to; and if the inflammation is great, and the pain pungent, it must be repeated; though with skilful management. Bleeding may be repeated after the first twenty-four hours; yet in general it is needless, and sometimes dangerous; for my own part, I never approved of it; nor have I seen any necessity for it. Perspiration however should be promoted.

If the pain is violent, let the patient use the following medicine.

No. XLI

Take sweet Spirit of Nitre (D) twenty drops; Spirit of Wine and Camphor (A) ten drops; Fever Powder (L) one scruple; mix it up with sugar and water into a draught.

Let the patient take this every six hours; if there is a great nausea at the stomach, a small vomit of twelve grains of *Ipecacuanha*, (S) may be given first, notwithstanding it is great pain to the patient; and I have often known it productive of great good, and laid the first foundation to a successful cure.

If the pain still continues, twelve drops of *liquid laudanum* (T) may be added to the above draught; No. [XLI](#). If the spirits of the patient is low, give the *essential oil of Peppermint* (U) to the quantity of three or four, or six drops upon a lump of sugar, either put, into the draught above-mentioned, or give it by itself occasionally.

To the inflamed side, warm cloths may be put wrapt up with a hot made trencher, brick, a bag of sand, or whatever will retain heat; as that will greatly facilitate the internal resolution.

Perspiration is the principal object to be considered in this acute disease, and should therefore be promoted; but every thing forcing that produces

profuse sweats should be avoided, as that only encreases the fever.

If the patient is very restless, the fever depressed, and greatly oppressed in respiration, blisters may be applied; but if the patient can do without them, it is much better; as they very often are apt to encrease the inflammation.

Cooling clysters, No. [XXXII](#). which at the same time promote urine, is of singular service; and in every respect methods must be taken, as have been directed in fevers, and the ulcerated sore throat; as all these inflammatory diseases are of one nature, and require almost one method of treatment. For farther particulars on this head, see my treatise on diseases of the lungs.

SECT. IV.

Of Diseases of the Bowels, and the Alimentary Canal

Diseases of mankind under this head are very numerous indeed; which are for the most part owing to their deviating too much from the common path of nature. But besides all this, seamen are particularly liable; not so much indeed on account of luxury as the confined diet, and the different climates they visit.

I shall in this be as concise as the subject will admit of, and confine myself to such diseases as occur most commonly; and by which others that are here omitted, may be understood.

Of a Depraved Appetite.

There is nothing that people in general are more apt to fly to, when appetite fails them, than bitters; but if we consider what bitters really are, and by what means they act in the alimentary canal, we shall find them, either only to stimulate the coat of the stomach, or else add to the strength of the bile.

But generally the default of a bad appetite is a pituitous slime in the stomach, that prevents this viscera from performing its part towards assimilation; and therefore we generally find that such stimulants only spur it on for a little while longer, till at length it loses its tone, so as to be wholly unable to prepare the food for a good nutrition; or else the stomach becomes so used to these forcing stimulants, as not to be able to do without them.

But this is not all; for in bilious complaints, when the stomach is most disordered, bitters never fail of making evil worse. The stomach is a delicate viscera, and should ever be the most cautiously dealt with. Whatever part of the human body is in exquisite pain, the stomach from the nervous consent always partakes, and is apt to be depraved in its functions;

whence a nausea is liable to ensue; and in such a case any stimulants, particularly where bitters are concerned, is ever of evil tendency.

In general however, a vomit is what nature points out; and which gives not only relief to the stomach, but assists in giving a general effort to the whole animal fabric. In short, there are so very few cases where bitters are of any benefit at all, except where the stomach has injuriously been used to those stimulants, that I have entirely exploded them from my catalogue; being persuaded that every body will do better without them.

When the stomach loaths victuals, the mouth is slimy, has a bitterish taste, and attended with sour belching, and the victuals siting heavy on the stomach, nothing is preferable to a vomit.

This I know has also been carried into abuse, as well as many other good medicines.—Yet that should be no rule; for if a vomit is properly and timely given, it will ever be attended with success. Take here the following direction for a foul stomach.

First take a vomit, and work it well off with bare luke warm water; then take every night, or when going to rest, a dose of *stomach powder* (M). If the stomach seems very weak, add a few drops of the *essential oil of peppermint* (U), which will soon convince you of the good effects.

If you require a stimulant in the morning, take *Turlington's balsam* (W), about a tea spoonful in a glass of wine, previously put on a lump of sugar, that it may the better incorporate: which is a good stomachic, and ever preferable to the customary bitters.

Dram drinking is another pernicious custom, whereby many have been sent to eternity long before the period of life prescribed by nature; and of which I would advise every man, who has the least regard for his health, to beware, as of a deadly poison.

On the whole, the want of appetite is what many complain of without reason. Some fancy themselves very bad, because they cannot eat their breakfast; those I would advise to stay till dinner time; others again have no appetite for their dinner, unless they wet their stomach with a glass of Stoughton, or some other bitters: Those should eat nothing in the morning, or else stay till supper time.—In short, nature should be consulted, both

in the state of health, as well as in diseases, and her salutary dictates should be followed.

The custom of seasoning victuals with strong stimulating condiments is often prejudicial, and nothing should be used with more moderation; for though it may warm and stimulate the stomach, yet, it frequently gives cause to great weaknesses, and generally are the primitive causes of chronic distempers; as it hurries on the chylefaction, and thence is apt to feed the blood with many acrimonious humours.

This little dissertation on depraved appetite I hope will suffice, and doubt not if attended to, that health and longevity will become the intimate acquaintances of the man who wishes to enjoy their blessings.

Of the Colic.

If we consider the length of the alimentary canal from the mouth to the anus, namely six times the length of the subject they belong to; and if we consider the delicacy of its substance, as also, the office for which it is ordained; we shall not at all wonder, that so many disorders are incident to mankind in that part.

The colic however is the general term for various painful sensations in the intestines; though the causes as well as the seat, and the manner of its affecting the system, is materially different.

The general cause however, is a stimulant, which irritating the inner coat of the intestines to constriction; and whence a spasmodic contraction, and a perverted peristaltic motion is produced.

If the stomach is affected with the pain of the colic, so that a nausea and reaching attends, or has been previously felt, a vomit is the first, and most expedient method to be had recourse to; for by this, not only the crudities in the stomach gets disburdened, but from the joint action on the internal canal, a stool or two is generally produced, by which frequently the offending acrimony is carried off.

When however this avails not sufficiently, and the pain rather increases, particularly if the body at the same time is costive, recourse should be had to clysters; for purges in this case, do not operate speedily enough, and in

cases of costiveness, is not always safe. The clyster No. [XXIII](#). may be of great benefit, and if necessity requires it, occasionally repeated.

But more frequently, it is from a nervous affection, and then carminatives prove the most effectual; in such cases give the following,

No. XLII

Take essential oil of peppermint (U), six drops on a lump of sugar; elixir proprietatis (H), five large tea spoonfuls; wine or water, half a tea cupful.

Let the patient take this immediately; a little *Turlington's balsam* (W) about a tea spoonful, also is very beneficial. If the pain still continues violent; let ten or twelve drops of *liquid laudanum* (T), be added. Clysters should not be neglected, but plentifully administered; the same medicines as is given to the patient, may also be added to the clyster.

The dry belly-ach, is a spasmodic contraction of the intestines, and throws the patient into the greatest misery imaginable; clysters is the only recourse and anodynes, together with gentle mercurials will likewise prove beneficial; warm clothes and something that can retain the warmth should constantly be put to the bowels; and in great extremities, cold foot baths have often given immediate relief.

The bilious cholic is of a dangerous tendency, and always attended with the bilious belching and vomiting of bile. In this case vomits is the only method to be used with safety; and if properly continued, as long as the bile is brought up, will never fail of success. But bleeding in bilious complaints is ever pernicious. If a fever attends, method must be used accordingly.

Of Looseness, and the Bloody Flux.

A Cholic is often the forerunner of a looseness, and if the looseness is not too violent very often is salutary, and carries off many kind of disorders in the body. But if a looseness continues after twenty-four hours, it is time something should be done, in order to prevent a bloody flux.

If the flux is attended with a nausea at the stomach, and an inclination to vomit, give without delay the following draught:

No. XLIII

Take rhubarb (O) one scruple; ipecacuanha, ten grains; stomach powder (M) fifteen grains; calomel (N) three grains; mix them and make it into a draught or bolus, with sugar and water.

This probably will vomit a few times, and then occasion a few stools; but as the rhubarb has an astringency as well as cathartic quality, it will in general, after the purging has gently carried off the acrimony, act as an astringent; particularly if after the above draught, the following paragoric is taken.

No. XLIV

Take oil of peppermint (U) four drops upon a lump of sugar; liquid laudanum (T) eight drops; sweet spirits of nitre (D) twenty drops; mixt in a glass of wine.

But if the patient still continues after these medicines, with great griping, a succession of stools, together with a *tenesmus*, (that is a perpetual inclination to go to stool) then give the following:

No. XLV

Take rhubarb (O) ten grains; stomachic powder (M) fifteen grains; essential oil of peppermint (U) three drops; calomel (N) one grain; and make it into a bolus or draught.

Let the patient take this every six hours, for the first two days. If the stools are bloody, slimy, and attended with great pain; recourse must immediately be had to clysters, such as No. [XXXVII](#), [XXXVIII](#), [XXXIX](#). mentioned in the quincey; for the more nourishing they can be made the better; as thereby they will lubricate the bowels, and strengthen them. In the

mean time the *white decoction* should be used as a constant drink, which make as follows:

No. XLVI

Take chalk (18) pounded small, three ounces; cinamon (16) bruised, two drachms; boil them in two quarts of water for one hour, and strain off the decoction.

Of this let the patient drink for his constant drink; a few drops of oil of peppermint may be dropped upon a lump of sugar, and dissolved in it.

The same decoction may also be given as a clyster when the flux is very violent. After the two first days the rhubarb should be given but sparingly; for if the flux is bloody, we may justly suppose the intestines are in a state of inflammation, and even ulceration. In such extremity, a little calomel mixed with the clyster, as also liquid laudanum, will not only correct the acrimony, but give ease to the emaciated stamina of the intestines.

By this treatment the most inveterate bloody flux may be subdued, especially with the help of patience, cautious diet, and cleanliness.

SECT. V.

Of the Scurvy

If I was to write as much on that subject, as I have already written in this treatise, the whole would only amount to this, that the scurvy is a state of putrifying corruption in the living human body.

The causes of this disorder is a defective diet, contaminated air, and want of proper exercise.

In regard to the diet, salt provisions and foul water are the accessory means, which not only impoverishes the blood, but gives it a tendency to a putrescency, whereby it loses its spirit, the lymph its gelatinousness, and the crasamentous part its resinous consistency. Thence the serum becomes sharp and acrimonious; the lymphatic part in the blood unable to nourish the stamina of the solids; and the red crasamentous part becomes crumous, and void of its vivacity: and as the solids must in consequence suffer in proportion to the state of the blood, from whence they have their existence, and become maintained, it must follow also, that they degenerate into the same state of corruption: thence we find not only a universal lassitude and debility in the limbs, but gradually the smaller vessels become unable to circulate the corrupt humours, break, bleed, and even mortify of their own accord, without any other cause than the putrescent nature of their containing fluids.

Though the symptoms are in general too well known, not to be easily mistaken; yet I shall here describe this destructive disease in its various gradations.

When a patient first begins to be tainted with the scurvy, he feels a lassitude all over his body, and a peculiar proneness to indulgence and idleness; work becomes irksome; and neither has he strength, nor inclination, to do his work with the same chearfulness as before. He sleeps more in general, but his rest is for ever disturbed with heavy and

melancholy dreams; and particularly of that nature, as for ever to be engaged in escaping from some sad danger, but unfortunately that his limbs are so remarkably heavy, that he cannot move them. With these and the like labour in vain phantoms his sleep is disturbed, till at length he awakes, and finds himself more fatigued than when he first went to rest. The mouth is always foul and slimy, and frequently attended with a disagreeable taint. The appetite is depraved, and the victuals sits very heavy on the stomach.

This is the first stage of the scurvy; and when a patient is sure of those symptoms, it is time some remedy should be used, in order to prevent the approaching evils; which gradually come on thus.

The gums begin to swell, though not very sore; and frequently bleed either on their own accord, or at the least touch of any thing that comes near them. The breath becomes very disagreeable, even to the patient himself. The teeth grow dark, brown or black, become loose, and the very jaw bone, together with the roots of the teeth are exposed and seem uncovered from the decayed gum.

Sometimes a scurfy itch ensues, and sometimes little spots appear. The body becomes remarkably weak, and sensibly falls away, and the colour of the skin gets a disagreeable tawny hew; which is most remarkable, together with other emaciated features in the patient's countenance.

This is the second state, and now dissolution actually begins to take place.

The patient loses his strength sensibly, so as not to be able to be supported by his own limbs. Over his body variously appear red, scarlet, yellow, green, livid and black spots, of various magnitude; some of which will bleed on their own accord; others will break into malignant ulcers, and others again will mortify. The appetite fails; a flux frequently ensues; the urine will be thick and stinking; and the poor wretch gradually dies merely by putrefaction.

The scurvy has had many definitions, and has been divided into various classes. All the difference of the disorder is, that sometimes it is attended with a fever, only in some degrees, more in one habit, than in another; which may be owing either to the nature of the constitution itself, but principally depends on the climate the patient is in: to which I must not

forget to add, that frequently the scurvy is complicated with other disorders. I could make many observations on this head, but, this perhaps would serve more to indulge myself than to benefit my reader.

I shall therefore suppose that I have said enough, to proceed to the cure; which I do with the more chearfulness, as I am happy in knowing it by experience to be not only easy, but efficacious.

The first time I ever experienced, and had it in my power to examine into the nature of the *true* scurvy, I was fortunate enough to discover so much of its corruptive quality, as to hit upon the true antidote; namely, the saccharine quality and acidity of vegetables; and upon this foundation I established my method of cure with success.

I shall not here enumerate the many experiments that I made in the course of these inquiries, but proceed to the result thereof, and the method to be observed.

In the first place, sugar is an article of more virtue, than what is in general known; and of so great a value to seamen, that he ought to esteem it as one of the greatest preservatives of his health. Next to this is vinegar, which is not only a resister of putrefaction, but has a coagulating, and condensing quality to the animal juices; whereby it not only consolidates the fluids without adding acrimony, but on the contrary, absorbs animal acrimony, and braces up the solids.

On the same principle, I have found *dried malt*, to be of great advantage in scorbutic cases; in which, both the saccharine and acidity of vegetables is contained even in a heightened degree,—This being premised, I proceed in the following manner:

When a man is found to have evident signs of the scurvy, let his drink be acidulated with one fourth, or third, &c. of good vinegar, or what is yet better, lemon juice, and well sweetened with sugar.

If there is malt on board, let a malt decoction, or temporary beer be made in the following manner:

Take malt bruised, a pound, or three handfuls; sassafrass chips, a small handful; boil it in a gallon of water for an hour, and strain it off for use.

Let this be the constant drink, which in a short time will revive the already tainted habit. Onions, horse-raddish, and mustard, &c. such as will keep on board, should plentifully be made use of, and the pills No. [X](#). should be taken every twenty-four hours, at a time when going to rest.

This method will soon bring the patient to health again, particularly if in the first stage; but when it is come to the second degree, or in a state of rapid decay, more effectual and immediate means must be taken.

Here then I shall present the method, which by experience I know to be efficacious.

No. XLIX

Take a large tub, (a butt sawed in two) in which place a stool, or something to sit on; Then let the patient be stript naked and seated in the tub; pour into the bottom of the vessel about one gallon of good vinegar; put a blanket all over the patient, and the tub, so that the whole is covered like a hut. Then heat a large poker; which put into the vinegar in order to rise a steam; which continue for about a quarter, or half an hour, or longer, according to the circumstances.

If the patient is very scorbutic, it should be repeated three times a day, or even oftener, for the effect of this steam-bath is so powerful, that I have with pleasure seen the patient when not able to crawl to the tub; he has been so revived by the operation, as to come out of his own accord quite refreshed.

In the mean time the elixir of iron as is described No. [XXVIII](#). should be taken; the mouth often washed with vinegar, lemon juice, or rather with the elixir of the iron, and sugar, should be both plentifully used in the drink, and with the victuals.

Salt provisions should be avoided as much as possible, or else previously well deprived of its salt.

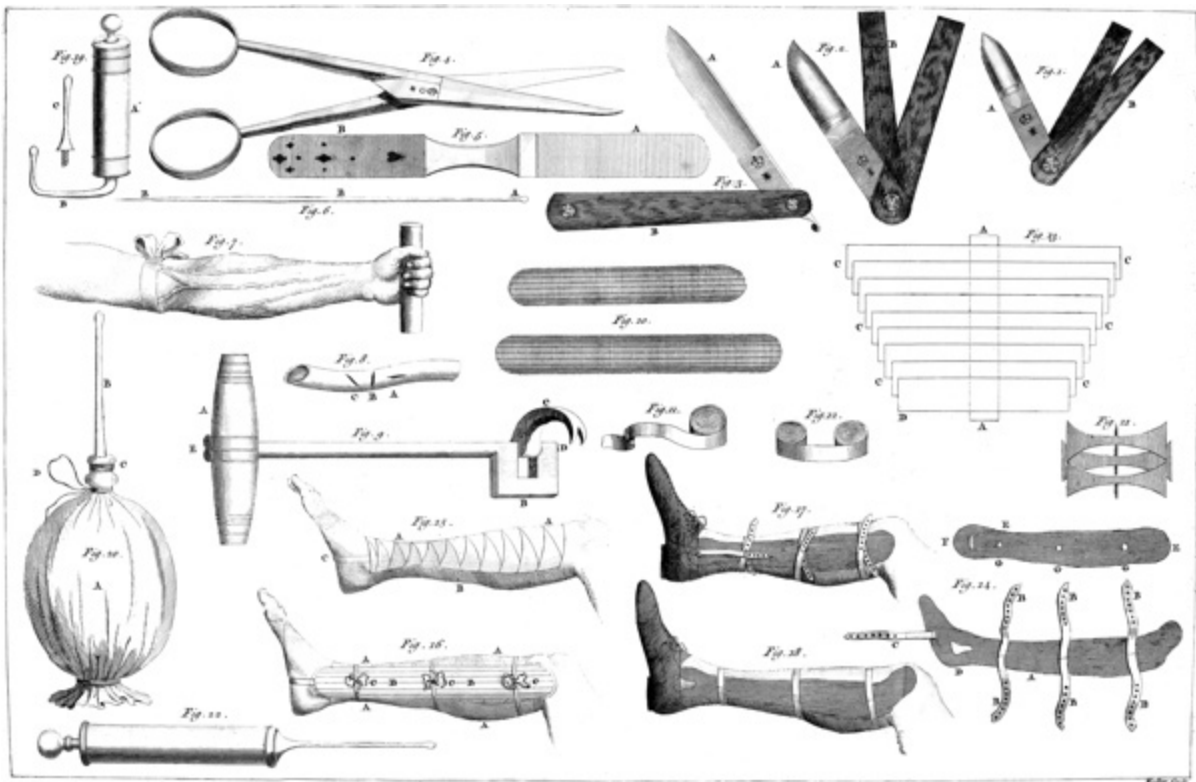
By this method the scurvy may not only be prevented at sea, but even cured in the most inveterate degree.

If however there is an opportunity for the patient to come on shore, it will greatly facilitate his cure; though he should reap no other benefit than the exhalations of the earth in the warmth of the day: and I have with astonishment seen patients devour vegetables in large quantities, and mended surprizingly, even in a few hours; so powerful an antidote is the vegetable effluvia to an animal putrescency.

To conclude, I have only this to observe, that the scurvy is principally owing to defaults with respect to provisions, and the customary way of living in ships in general; and such as might be much amended, both with respect to conveniency, wholesomeness, and expence.

I speak this from long experience, from the many observations I have made, and the good success that have attended my improvements, when it has been in my power to put them in execution. I have many things to offer with respect to provisions at sea; but as that is too extensive to be inserted in this volume, I shall treat of it distinctly. In the mean time give me leave to close these lectures: and being conscious that truth and utility has guided my endeavours through the whole, I lay down my pen for this time, with the pleasing reflection, that I have advanced one step farther in the performance of my duty, and in answering the end of my existence.

FINIS.



EXPLANATION OF THE PLATE.

Fig. 1. A common bleeding lancet. A. The blade; which, when it is wiped after bleeding, should always be laid upon the handle B. and wiped but one way, to save the point.

Fig. 2. An incision lancet. A. The point of the blade, which ought to incline one way most, in order to make it the better fit for opening of aposthumes.

Fig. 3. A bistoury for opening of sores, &c. in general use. A. The blade. B. The handle in which the blade folds.

Fig. 4. A pair of strong scissars.

Fig. 5. A spatula. A. The part for spreading of plaisters. B. The part for pressing down the tongue; for which reason it is made full of holes that it may the better press down the tongue.

Fig. 6. A probe. A. the probe. B. A triangular point on which lint may be wrapped to wipe the sore with, as it is easy taken off again.

Fig. 7. A arm tied up for bleeding, with the three veins as they generally appear when swelled up.

Fig. 8. A vein of a kind of three orifices. A. Is a longitudinal orifice, which will not let the blood run freely. B. Transverse; in making such orifice the vein is apt to be divided; besides it is not very easily stopped, and will not admit of an orifice large enough to let the blood run out with freedom. C. An oblique opening, the medium between the two former; and always the best orifice.

Fig. 9. A key-tooth instrument. A. The handle, made of ivory. B. The comb of the instrument, which should be wrapped round with a little rag, and is that part which rests against the jaw bone. C. The hook which is brought over the tooth. D. The screw at the end of the key, by which the hook is shifted as occasion requires. E. The screw by which the handle is fixed to the instrument.

Fig. 10. Common splints used for fractures; these splints are made of thin boards of pine or cedar, glewed on to leather, and then ridged so, that they may easily bend and suit to the limb which they are applied to.

Fig. 11. A single headed rowler, or bandage, this bandage is generally used for a fractured arm or leg.

Fig. 12. A double-headed rowler, generally for dislocations.

Fig. 13. A many-tailed bandage. A. The back part slip, that keeps the bandage together. B. The stitches coming thro' the bandage. C. The slips of the bandage. D. The part next the ancle, from whence the parts upwards have a gradual increase.

Fig. 14. Mr. Sharp's invented splints, made of paste-board. A. The external splint. B. The leather straps. C. The strap, that comes under the foot. E. The internal splint. F. The loop through which the strap that comes under the foot is put. G. Small iron buttons, on which the straps are fastened.

Fig. 15. A leg with the many-tailed bandages. A. The slips laid over each other. B. The under slip that secures the whole. C. A slip that secures the bandage to the foot.

Fig. 16. A leg dressed with the splints. A. The bandage. B. The splint. C. The tie knots of the tape that secure the splint to the leg.

Fig. 17. The inner view of a leg with Mr. Sharp's splints.

Fig. 18. The external view of the same. These two are copied from Mr. Sharp's pamphlet.

Fig. 19. A clyster syringe. A. the syringe made of pewter. B. A pipe made in such manner, as to enable a man to administer a clyster to himself. C. A common clyster pipe, that occasionally may be screwed on to the syringe.

Fig. 20. A clyster pipe and bladder. A. The bladder. B. The pipe. C. The manner of fixing the bladder to the pipe. D. The tying of the bladder when it is filled, with a slip knot; which is handier than the customary way of the cork.

Fig. 21. The manner of closing a wound with plaisters, called the dry suture.

Fig. 22. A common syringe made of ivory.

- Transcriber's Notes:

- In Lecture VI, Sect. VI. has been corrected to Sect. V. to match the table of contents.
- In SECT. VI. on page [118](#) there is a reference to the authors description of the skin in Lect. I., Sect. VII. The discussion of the skin is actually in Lect. I., Sect. IX. on page [40](#).
- There are two SECT. VIII. in LECTURE I. They have been left with
 geh duplicate number.
- The “recipes” for various medicines are numbered somewhat sequentially with Roman numerals across all Lessons and Sections. However there are “recipes” with duplicate numbers (XXII appears three times in Section V of Lect. IV.) and some recipe numbers are missing altogether.
- Missing or obscured punctuation was silently corrected.
- Typographical errors were silently corrected.
- Inconsistent spelling and hyphenation were made consistent only when a predominant form was found in this book.

*** END OF THE PROJECT GUTENBERG EBOOK THE SEAMAN'S
MEDICAL INSTRUCTOR ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.